Progressive Fines and Tax Evasion*

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Abstract

We ask whether making fines progressive raises tax revenue. In a laboratory tax-evasion experiment, subjects face either a progressive fine schedule or a linear benchmark of similar average intensity, under both a between- and a within-subject design. A simple model predicts that progressivity weakly increases compliance and total revenue among high evaders, while effects elsewhere in the evasion distribution are ambiguous.

We find that progressive fines raise total revenue, with the clearest gains concentrated among high-evasion types. The effect is robust across both designs and is driven by total revenue rather than by its decomposition into declarations and fines. These results show that the shape of the penalty schedule—not just its level—matters for revenue, primarily through differential incidence in the upper tail of the evasion distribution**[Note:** would be great to have a behavioural explanation for the asymmetry in T3 vs T4 results, ATM our interpretation is all in the standard EU framework**]**.

Keywords: progressive fine, progressivity, tax evasion, dishonesty.

JEL Classification: C9; H26; H83.

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1 Introduction

Most real-world deterrence systems adopt progressive penalties where the punishment increases more than proportionally than the offence. In most countries, errors and omissions are lightly punished – or not punished at all –, whereas more serious offences, characterized by intentionality are harshly punished. This is observed, e.g., in speeding violations, murders (first- and second-degree, voluntary- and involuntary-manslaughter), or tax crimes, the topic that interests us here. Even though this is so central for any deterrence system, both theoretical and experimental research has largely neglected this progressivity in penalties.

There are many examples of real-world fine progressivity in tax crimes. In the German tax system, monetary fines can range from 5 to 360 daily rates, which amounts to one to thirty thousand euros each. Tax evaders may also face a minimum of one year imprisonment. In France, simple omissions do not result in a fine, but penalties for tax evasion can range from 10 to 80% of the underpaid tax. In Italy, not paying taxes may result in a penalty ranging from 120% to 240% of the taxes due. In the USA, simple mistakes are overcharged with a 20% fine, while proven tax evasion incurs a 75% fine.

The study of the role of the fine function in the optimal evasion decision runs deep into the theoretical literature on tax compliance. Right after the seminal Allingham and Sandmo (1972) paper, Srinivasan (1973) first considered a general (convex and increasing) fine function in its analysis. In the paper that developed the theoretical framework still used today to model the evasion decision, Yitzhaki (1974) showed that adopting a more realistic penalty structure, based on evaded taxes (instead than on undeclared income) has crucial implications for the model predictions. Further works have then investigated different aspects of the fine function: the use of monetary fines proportional to evaded tax liabilities vs imprisonment and steep fines when evasion configures as a felony (Pencavel 1979), the use of retroactive penalties (Rickard et al. 1982), the interaction between tax rate progressivity and the penalty function on evasion (Koskela 1983), the limitations of the use of maximal penalties for evasion deterrence (Cowell 1985) and the possibility to mix penalties with rewards to increase the taxpayers' utility while leaving revenues unaffected (Falkinger and Walther 1991).

Starting from Friedland et al. (1978), a large experimental literature has studied tax compliance in the lab, mainly by varying audit probabilities and the overall level of fines. By contrast, much less is known about the shape of the sanction schedule itself. Our focus is therefore not on whether harsher enforcement matters in general, but on whether reallocating sanction severity across the evasion distribution changes revenue collection when average sanction intensity is held broadly comparable.

To do so, we compare two penalty structures: a progressive piecewise-linear schedule and a linear benchmark. To the best of our knowledge, this is the first paper to study theoretically and experimentally a progressive fine schedule that depends on fraud severity.[1] The experiment combines two between-subject treatments and two within-subject switch treatments. The progressive schedule ranges from 1 to 5 times the unpaid tax, so low evasion is penalized relatively lightly while high evasion faces substantially steeper sanctions. The linear benchmark uses a multiplier of 3.25, chosen ex ante to provide a pragmatically comparable flat schedule relative to the average sanction intensity generated by the progressive schedule while remaining close to standard values in the experimental literature. Throughout, the linear regime should therefore be read as a pragmatic flat benchmark matched to the progressive schedule at the level of broad sanction intensity, not as a schedule that equalizes expected burden point-by-point across the evasion space.

The paper makes three contributions. First, it isolates the shape of the sanction schedule as the policy object of interest: rather than varying audit probability or overall fine level, we compare a progressive penalty schedule with a pragmatically comparable flat benchmark. Second, the paper shows that the effect of progressivity is not uniform across the evasion distribution. The clearest gains in total revenue are concentrated among predicted high evaders, while effects for the other strata are either ambiguous ex ante or more design-sensitive empirically. Third, the paper combines a between-subject benchmark with a within-subject switch design and an external predictive partition of evasion propensity. We use this architecture not to claim multiple equally clean causal estimates, but to compare how the same substantive pattern travels across designs with different strengths and vulnerabilities.

Empirically, progressivity raises total revenue most clearly among predicted high evaders, and this result appears in both the between-subject comparison and the pooled within-subject design. Aggregate total revenue is also higher under progressivity. By contrast, the evidence for predicted low evaders is more design-sensitive: it appears in the pooled within-subject evidence, but it is not confirmed in the between-subject comparison and is driven mainly by one sequence. The most robust takeaway is therefore not that progressivity improves compliance uniformly, but that sanction-shape design changes the incidence of revenue effects, with the clearest gains concentrated among predicted high evaders and in the aggregate.

The paper is structured as follows: Section 2 presents our tax evasion experiment on progressive vs. linear fine structure. Section 3 provides a theoretical model with predictions. Section 4 presents the empirical identification strategy. Section 5 describes

[1] The only mention of a progressive fine in the experimental literature comes from Samid and Suleiman (2008) where participants play a Public Good Game and fines for shirkers depend on the number of people opting-in an anti-shirker policy.

the results. Section 6 reports robustness checks and Section 7 concludes.

2 Experiment

The experiment consists of two between-subject treatments: progressive fine rate (T1) and linear fine rate (T2), along with two within-subject treatments: Linear-Progressive (T3) and Progressive-Linear (T4). All treatments involve two parts: the first part consists of the Tax Evasion Game, the second one is a risk aversion elicitation task (Gneezy and Potters, 1997). Finally, participants are asked to complete a socio-demographics questionnaire.

2.1 Tax Evasion Game

The first task is a standard Tax Evasion Game (Malézieux, 2018; Alm and Malézieux, 2021), with 20 periods and identical course of action.[2] Instructions are framed in tax-related terms (see Appendix A.1). In each period participants receive an initial income equal to 125 ECU (with constant exchange rate of 1,00€ per 25 ECUs) and are asked to declare it for tax purposes. Alm and Malézieux (2021) show that earning or receiving an income does not materially alter evasion decisions. The income level is fixed because we are not interested here in wealth effects. There is a uniform tax rate τ equal to 20% applied only to reported income. Participants can declare any whole-number amount between 0 and 125 ECU. Collected taxes are invested in a research fund, and participants do not receive anything back from those taxes. In each period, an audit occurs with probability 20%; realized audited counts therefore vary across periods, but the governing audit probability is constant. The fine is determined as q times the unpaid tax and varies according to the treatment implemented (see below), with q representing the experimental counterpart of the model multiplier. Two periods are randomly drawn for the final payment. We also monitor the decision time of each compliance decision (Jacquemet et al., 2020).

2.2 Risk task

The second task is a risk aversion elicitation task (Gneezy and Potters, 1997). Herein, participants receive a 10 € sum that they can choose to invest fully, partially or not at all in a heads or tails game. If they win, they get 3 times the amount invested. One participant out of ten is selected randomly to receive a payment for the risk task (this type

[2]With this respect, we closely follow Heinemann and Kocher (2013) which studies the impact of tax-rate progressivity in a 20-period experiment.

of incentive is known as between-subjects random incentivized system and is increasingly used in experimental social sciences).

2.3 Socio-demographic questionnaire

In the post-experimental questionnaires, participants are asked questions about their age, gender, year of study, if they were French, their field of study, if it was their first experiment, how willing they were generally to take risks (from 1, completely unwilling to 10, completely willing), how much they spend each month in leisure/recreational activities and how much they saved each month.

2.4 Experimental treatments

2.4.1 Treatment 1: Progressive fine rate

In T1, we implement a progressive penalty structure that depends on the severity of the evasion. The fine rate q varies between 1 to 5 times the unpaid tax (0 for full compliance). This structure has been chosen so as to expand at the maximum the fine rate, given the fiscal parameters and the necessity for the participants to end up with positive or null earnings. Participants are thus informed that in case of an audit and discovered evasion, they will face a payment of a fine with a progressive structure.

2.4.2 Treatment 2: Linear fine rate

In T2, we implement a linear penalty structure. The fine rate q is fixed at 3.25 in each period. This benchmark was chosen before collecting data in T2–T4 to provide a pragmatically comparable flat schedule relative to the sanction intensity generated by the progressive regime. The purpose of this benchmark is comparability at the level of overall sanction intensity, not exact equality of expected burden for every evasion level or subject type. It is also close to the average fine implemented in the literature, i.e., 3.01 in Alm and Malézieux (2021). Participants are thus informed that in case of an audit and discovered evasion, they will face a fixed fine.

2.4.3 Treatment 3: Linear - Progressive

In T3, we implement first a linear penalty structure for the first 10 rounds, as described above. At the 11th round, participants shift to a progressive penalty structure for the last 10 rounds.

2.4.4 Treatment 4: Progressive - Linear

In T4, we implement first a progressive penalty structure for the first 10 rounds, as described above. At the 11th round, participants shift to a linear penalty structure for the last 10 rounds.

2.5 Experimental protocol

Overall, 490 subjects participated in the experiment (131 in T1, 127 in T2, 109 in T3, 123 in T4). They earned on average 9,33€ (SD: 5€). The experiment took place between January and December 2022. As France was still under varying social-distancing rules in late 2021, the experiment was run exclusively online. Participants are students from the Burgundy School of Business (Dijon, France) and were recruited through ORSEE (Greiner, 2015). They were paid through Lydia, an electronic payment service that allows payment to be transferred using a cellular phone number, with a delay of about a week after participation.

3 Model

This section develops a simple, one-period model of evasion choices under linear and progressive fines, tailored to the experimental setting . The objective is to investigate how moving from a linear sanction to a progressive one (and vice-versa) changes the incentives to under-report, holding fixed income, the tax rate, and the audit probability. The model is therefore used to organize the treatment comparisons and the heterogeneity predictions. The use of a one-period model is supported by the lack of evidence of learning over rounds.

3.1 Formal Setup

We consider an individual with income Y, proportional tax rate τ, and audit probability p. The individual declares $X \in [0, Y]$ and evades

$$E = Y - X \in [0, Y].$$

In the experiment, declarations are integer-valued. For the theoretical characterization, we work with the continuous interval $[0, Y]$ and use the resulting cutoffs to organize the discrete choice problem.

Conditional disposable income is

$$C^{na}(E) = Y - \tau(Y - E), \qquad C^{a}(E; f) = Y - \tau(Y - E) - \tau f(E)E, \tag{1}$$

where C^{na} and C^{a} denote non-audit and audit states, respectively, and $f(E)$ is the fine schedule. The experimental fine multiplier q is the empirical counterpart of $f(E)$ in the model.

The taxpayer chooses E to solve

$$\max_{E \in [0,Y]} \mathcal{U}(E; \rho, f) = (1 - p)\, U\big(C^{na}(E); \rho\big) + p\, U\big(C^{a}(E; f); \rho\big). \tag{2}$$

Utility is increasing in consumption ($U_c > 0$), with curvature allowed to vary by type (U_{cc} can be negative or positive). Evasion raises consumption when no audit occurs, but it also enlarges the sanction base under audit. The key heterogeneity object is therefore the latent type ρ, which governs how strongly a subject trades off the non-audit gain against the audit-state loss.

We focus on two fine systems. The linear schedule f_{lin} and the progressive schedule f_{prog} are given by

$$f_{\text{lin}}(E) = f_{avg} = 3.25,$$

$$f_{\text{prog}}(E) = \begin{cases} f_1 = 1, & 0 = B_0 < E \leq B_1 = 25, \\ f_2 = 2, & B_1 < E \leq B_2 = 50, \\ f_3 = 3, & B_2 < E \leq B_3 = 75, \\ f_4 = 4, & B_3 < E \leq B_4 = 100, \\ f_5 = 5, & B_4 < E \leq B_5 = Y = 125. \end{cases} \tag{3}$$

3.2 Solution Characterization and Predictions

On any flat segment with multiplier $\bar{f}$, audited consumption can be written as

$$C^{a}(E; \bar{f}) = Y - \tau(Y - E) - \tau \bar{f} E,$$

and the marginal return to evasion is

$$M(E; \rho, \bar{f}) := \tau \Big[(1 - p)\, U_c\big(C^{na}(E); \rho\big) + p\,(1 - \bar{f})\, U_c\big(C^{a}(E; \bar{f}); \rho\big) \Big].$$

Within any flat segment, an interior optimum satisfies

$$M(E; \rho, \bar{f}) = 0. \tag{4}$$

Equation (4) summarizes the local incentive problem: evasion increases when $M > 0$, decreases when $M < 0$, and is locally optimal when $M = 0$.

For any flat-rate segment with fine level $\bar{f}$, define

$$\Gamma(E;\rho,\bar{f}) := \tau^2\Big[(1-p)U_{cc}\big(C^{na}(E);\rho\big) + p(1-\bar{f})^2 U_{cc}\big(C^{a}(E;\bar{f});\rho\big)\Big]. \tag{5}$$

The condition $\Gamma < 0$ is a strict-concavity restriction delivering uniqueness on each flat segment. For the linear regime, it implies a single-valued mapping between type ρ and optimal evasion $E^{lin}(\rho)$ on $[0, Y]$. Under the progressive regime, the global choice is obtained by comparing the finitely many interior and threshold candidates generated by the piecewise schedule.

We impose $M_\rho(E;\rho,\bar{f}) < 0$ on the relevant domain, so higher risk aversion lowers the marginal material gain from evasion at any given E. This is the standard single-crossing condition in common utility classes because audits are the low-consumption state. We also use, on flat segments, the deterrence restrictions

$$M_E(E;\rho,\bar{f}) < 0, \qquad M_f(E;\rho,\bar{f}) \le 0.$$

The first condition requires diminishing marginal incentives in evasion. The second requires that stronger fines weakly reduce those incentives at any given E. Together with increasing bracket thresholds and fine rates, these restrictions deliver an ordered mapping from latent types into progressive brackets. Under standard risk-averse expected utility on flat segments (with $U_{cc} < 0$ and fines $\bar{f} \ge 1$ in our setting), these signs are satisfied in the usual way; under the richer baseline that also allows risk-loving types, we maintain them as shape restrictions to preserve a coherent type partition.

Proposition 1 *Assume $U_c > 0$ and, on each flat segment, maintain $\Gamma < 0$, $M_\rho < 0$, and $M_f \le 0$. Under these conditions:*

1. ***Linear schedule: uniqueness and monotone comparative statics.*** *Under the linear schedule $f(E) \equiv f_{avg}$, the problem has a unique optimizer $E^{lin}(\rho) \in [0, Y]$. Define the corner cutoffs by*

$$M\big(0;\rho^0_{lin}, f_{avg}\big) = 0, \qquad M\big(Y;\rho^Y_{lin}, f_{avg}\big) = 0.$$

Then $\rho^Y_{lin} \le \rho^0_{lin}$ *and*

$$E^{lin}(\rho) = \begin{cases} Y, & \rho \le \rho^Y_{lin}, \\ \text{the unique interior solution to } M(E;\rho,f_{avg}) = 0, & \rho^Y_{lin} < \rho < \rho^0_{lin}, \\ 0, & \rho \ge \rho^0_{lin}. \end{cases}$$

On the interior, comparative statics satisfy $E^{lin}_\rho < 0$. *For flat-rate comparisons in the fine level,* $E^{lin}_f \le 0$.

2. ***Progressive schedule: candidate set, global selection, and local threshold attraction.*** *For the progressive schedule, define for each bracket* k *the within-bracket candidate* E^I_k *as the unique solution to* $M(E;\rho,f_k) = 0$ *when it lies in* (B_{k-1}, B_k). *Define the finite candidate set*

$$\mathcal{C}(\rho) = \{B_0,\dots,B_5\} \cup \{E^I_k : \ E^I_k \text{ exists for } k = 1,\dots,5\}.$$

The progressive optimizer correspondence is

$$\mathcal{E}^{prog}(\rho) := \arg\max_{E\in\mathcal{C}(\rho)} \mathcal{U}(E;\rho,f_{prog}).$$

If, at threshold B_k, *the one-sided marginal conditions satisfy*

$$M(B^-_k;\rho,f_k) \ge 0, \qquad M(B^+_k;\rho,f_{k+1}) \le 0,$$

then $B_k \in \mathcal{E}^{prog}(\rho)$ *and the threshold is locally optimal. If the distribution of types has positive density on a neighborhood of such types, this local threshold selection appears as observable bunching at* B_k. *The weak sign predictions below do not depend on how ties inside* $\mathcal{C}(\rho)$ *are broken.*

3. ***Progressive type-space partition by bracket.*** *Define the corner cutoffs*

$$M\big(0;\rho^0_{prog},f_1\big) = 0, \qquad M\big(Y;\rho^Y_{prog},f_5\big) = 0,$$

and, for $k = 1,\dots,4$, *define the interior-threshold cutoffs by*

$$M\big(B_k;\rho^{B_k}_{prog},f_k\big) = 0. \tag{6}$$

Whenever the cutoffs exist, they are unique and ordered as

$$\rho^0_{prog} \ge \rho^{B_1}_{prog} \ge \rho^{B_2}_{prog} \ge \rho^{B_3}_{prog} \ge \rho^{B_4}_{prog} \ge \rho^Y_{prog}.$$

These cutoffs induce the exhaustive bracket partition

$$\begin{aligned}
\mathcal{S}_1 &:= \{\rho : \rho > \rho_{prog}^{B_1}\}, \\
\mathcal{S}_k &:= \{\rho : \rho_{prog}^{B_k} < \rho \leq \rho_{prog}^{B_{k-1}}\}, \quad k = 2, 3, 4, \\
\mathcal{S}_5 &:= \{\rho : \rho \leq \rho_{prog}^{B_4}\},
\end{aligned}$$

such that

$$\begin{aligned}
\rho \in \mathcal{S}_1 &\Longrightarrow E^{prog}(\rho) \in [0, B_1], \\
\rho \in \mathcal{S}_k &\Longrightarrow E^{prog}(\rho) \in (B_{k-1}, B_k], \quad k = 2, 3, 4, \\
\rho \in \mathcal{S}_5 &\Longrightarrow E^{prog}(\rho) \in (B_4, Y].
\end{aligned}$$

Within the outer sets, the corner choices are selected when $\rho \geq \rho_{prog}^0$ (full compliance) and when $\rho \leq \rho_{prog}^Y$ (full evasion).

Proof. See Appendix A.2.3.

Proposition 1 delivers the behavioral mapping used in the rest of the paper. Under linear fines, type heterogeneity produces a unique evasion choice, with sufficiently risk-averse subjects selecting full compliance and sufficiently low-ρ subjects selecting full evasion. Under progressivity, the key object is the ordered sequence of bracket cutoffs. Kinks make threshold choices locally attractive; when a positive density of types lies near a cutoff, that local force translates into observable bunching. The upper tail is therefore admissible only for sufficiently low-ρ types satisfying the corner condition at Y. Primitive sufficient conditions for cutoff existence are collected in Appendix A.2.1, and Appendix A.2.2 records the local threshold argument used below.

3.3 Switch of Fine Regime: Effects on Revenue by Bracket/Group

We now turn from choice characterization to the policy comparison that matches the experiment. The key institutional change is a switch between the flat sanction $f_{avg} = 3.25$ and the progressive schedule in (3). That switch affects evasion through two channels. First, within a given progressive bracket, the relevant fine level becomes f_k instead of f_{avg}. Second, the kinks at the bracket thresholds create local threshold attraction. For upper brackets, both forces weakly reduce evasion under progressivity. For lower and middle brackets, they work in opposite directions, so the sign is not pinned down globally.

Expected revenue is the sum of revenue from declared taxes $R^d(E) := \tau(Y - E)$ and expected penalty revenue $R^f(E; f) := p\tau f(E)E$:

$$R(E; f) = R^d(E) + R^f(E; f) = \tau(Y - E) + p\tau f(E)E. \tag{7}$$

Fix a type $\rho \in \mathcal{S}_k$ and define

$$\Delta E_k^{lin \to prog}(\rho) := E^{prog}(\rho) - E^{lin}(\rho).$$

The revenue change under $lin \to prog$ is

$$\Delta R_k^{lin \to prog} := R\big(E^{prog}; f_k\big) - R\big(E^{lin}; f_{avg}\big) = \tau\Big[(pf_k - 1)\Delta E_k^{lin \to prog} + p(f_k - f_{avg})E^{lin}\Big]. \tag{8}$$

Conditional on selecting bracket k, the progressive problem coincides inside that bracket with a flat schedule at rate f_k. By Proposition 1(i), the within-bracket candidate therefore moves monotonically with the fine level: it is weakly lower than the linear benchmark when $f_k > f_{avg}$ and weakly higher when $f_k < f_{avg}$. Kinks add a second, local force. When the one-sided conditions in Proposition 1(ii) hold, the threshold B_k becomes locally attractive and can replace nearby interior choices with a boundary choice. This local threshold force weakly reduces evasion near the threshold, but it does not globally sign the response away from the threshold in lower brackets.

Type-specific predictions. To translate the bracket mapping into empirically tractable groups, define

$$\mathcal{S}^L \text{ (Low evasion)} := \mathcal{S}_1 \cup \mathcal{S}_2,$$
$$\mathcal{S}^M \text{ (Mid evasion)} := \mathcal{S}_3,$$
$$\mathcal{S}^H \text{ (High evasion)} := \mathcal{S}_4 \cup \mathcal{S}_5.$$

Types in $\mathcal{S}^L$ choose declarations that map into the lower progressive brackets. Types in $\mathcal{S}^M$ select the intermediate bracket. Types in $\mathcal{S}^H$ select the upper progressive brackets, where the relevant multipliers are $f_4 = 4$ and $f_5 = 5$.

Then we have:

Hypothesis 1a *Under $lin \to prog$, high-evasion types in $\mathcal{S}^H$ weakly increase compliance, weakly increase declared-tax revenue, and weakly increase total expected revenue; the effect on expected penalty revenue is ambiguous. For low- and mid-evasion types $\mathcal{S}^{L,M}$, all revenue effects are ambiguous.*

Hypothesis 1b *Under $prog \to lin$, the previous signs reverse for compliance, declared-tax revenue, and total expected revenue among high-evasion types $\mathcal{S}^H$; the effect on expected penalty revenue remains ambiguous. For low- and mid-evasion types $\mathcal{S}^{L,M}$, effects remain ambiguous.*

Consider first high-evasion types $\mathcal{S}^H$ $(k = 4, 5)$. In these brackets, $f_k > f_{avg} = 3.25$. The within-bracket candidate under progressivity is therefore weakly below the

linear benchmark, and threshold selection can only weakly lower evasion further. Hence $\Delta E_k^{lin\to prog} \leq 0$, so compliance weakly increases. It follows immediately that declared-tax revenue weakly increases:

$$\Delta R_k^d = \tau\big[(Y - E^{prog}) - (Y - E^{lin})\big] = -\tau \Delta E_k^{lin\to prog} \geq 0.$$

For penalty revenue, however,

$$\Delta R_k^f = p\tau\big(f_k E^{prog} - f_{avg} E^{lin}\big) = p\tau\Big[f_k \Delta E_k^{lin\to prog} + (f_k - f_{avg})E^{lin}\Big].$$

For high-evasion types, the second term is strictly positive when $E^{lin} > 0$ because $f_k - f_{avg} > 0$, whereas the first term is weakly non-positive because $\Delta E_k^{lin\to prog} \leq 0$. The sign of ΔR_k^f is therefore ambiguous: higher marginal sanctions tend to raise penalty revenue, but the induced reduction in evasion shrinks the penalty base.

Total expected revenue is still weakly higher for high-evasion types. Using (8),

$$\Delta R_k^{lin\to prog} = \tau\Big[(pf_k - 1)\Delta E_k^{lin\to prog} + p(f_k - f_{avg})E^{lin}\Big],$$

and noting that under the experimental range ($p = 0.2$, $f_k \leq 5$) we have $pf_k - 1 \leq 0$, the first term is weakly non-negative because $\Delta E_k^{lin\to prog} \leq 0$, while the second term is strictly positive when $E^{lin} > 0$. Hence $\Delta R_k^{lin\to prog}$ is weakly positive in the upper brackets. In the knife-edge case $k = 5$ ($pf_5 - 1 = 0$), only the strictly positive second term remains.

Low- and mid-evasion types $\mathcal{S}^{L,M}$ ($k = 1, 2, 3$) remain ambiguous. In those brackets, the within-bracket fine level under progressivity is weakly below the linear benchmark, which tends to raise evasion, whereas local threshold attraction can pull choices downward toward the bracket cutoffs. Because the model does not rank these forces globally, $\Delta E_k^{lin\to prog}$ is not signed. The same ambiguity therefore carries over to ΔR_k^d, ΔR_k^f, and ΔR_k^{tot}. This ambiguity is most pronounced for $k = 3$, where f_3 is close to f_{avg}.

For the reverse switch, define

$$\Delta E_k^{prog\to lin} := E^{lin} - E^{prog} = -\Delta E_k^{lin\to prog}, \qquad \Delta R_k^{prog\to lin} := -\Delta R_k^{lin\to prog}.$$

The signs for compliance, declared-tax revenue, and total revenue therefore reverse. For penalty revenue, the sign remains ambiguous because it continues to reflect the opposing effects of the fine-rate change and the evasion-base change.

Aggregate predictions. Let G_ρ be the distribution of ρ. Aggregate revenue change under the switch is

$$\Delta R^{tot}_{lin\to prog} = \sum_{k=1}^{5} \int_{\mathcal{S}_k} \Delta R_k^{lin\to prog}(\rho)\, dG_\rho(\rho), \tag{9}$$

with $\Delta R_k^{lin\to prog}$ given by (8). Substituting yields

$$\Delta R^{tot}_{lin\to prog} = \tau \sum_{k=1}^{5} \left[(pf_k - 1) \int_{\mathcal{S}_k} \Delta E_k^{lin\to prog}\, dG_\rho + p(f_k - f_{avg}) \int_{\mathcal{S}_k} E^{lin}\, dG_\rho \right].$$

Hence, aggregate gains from progressivity are larger when the type distribution puts more mass on high-evasion brackets $\mathcal{S}^H$. More mass in low- and mid-evasion brackets $\mathcal{S}^{L,M}$ makes the aggregate effect more ambiguous by placing greater weight on types for which the sign is not pinned down. Under the opposite switch, signs reverse at the bracket level, and the aggregate effect remains composition-dependent.

Hypothesis 2 *The aggregate effect of switching from linear to progressive on total revenue is ambiguous ex ante and depends on the mass of subjects in each bracket/type.*

Equation (9) also clarifies why aggregate treatment effects can differ across samples: even with exogenous assignment, overall effects remain composition-dependent through G_ρ.

Prediction-to-estimand mapping. The empirical estimands are directly linked to model objects through $\Delta R_k^d = -t\, \Delta E_k^{lin\to prog}$ and (8) for ΔR_k^{tot}.

4 Empirical strategy and econometric implementation

We study four observed revenue outcomes:

$$R_{it}^d = \tau\,(125 - E_{it}) \in [0, 25], \tag{10}$$

$$R_{it}^f = \mathbf{1}\{\text{audit}_{it}\}\, \bar{f}_{it}\, \tau\, E_{it} \in [0, 125], \tag{11}$$

$$R_{it}^{tot} = R_{it}^d + R_{it}^f \in [0, 125], \tag{12}$$

$$R_t^{tot} = \sum_{i=1}^{N} R_{it}^{tot} \in [0, 125N]. \tag{13}$$

These outcomes are empirical counterparts of R_k^d, R_k^f, R_k, and R in the model. Here, i indexes subjects, t indexes rounds, and N is the number of subjects in the relevant group.

The indicator $\mathbf{1}\{\text{audit}_{it}\} \in \{0, 1\}$ records whether subject i is audited in round t, and $\bar{f}_{it}$ denotes the realized fine multiplier in that round. All individual outcomes we observe are bounded, so treatment effects can operate both through interior shifts and through changes in mass at the bounds.

4.1 Grouping strategy

Across grouped analyses, each subject is assigned to

$$\widehat{S}_i \in \{\text{Low}, \text{Mid}, \text{High}\},$$

which we call the participant's *predicted evasion stratum*. Consistent with our model, we treat each participant's evasion stratum as invariant across rounds. The central identification concern is that grouping directly on realized progressive-round declarations can induce mechanical endogeneity in heterogeneity analysis. Our baseline design therefore uses externally predicted strata, constructed from participant-level characteristics and elicited risk measures via supervised multiclass ML.

This design reduces, but does not eliminate, endogeneity concerns. The resulting labels are predictive rather than structural: they are measured with error, depend on observed predictors and model choice, and need not coincide exactly with latent evasion types. For this reason, we treat the first-stage classification as a structured sensitivity exercise and, to ensure stability of results, we perturb it along three dimensions: predictor set, training-target specification, and model class.

We estimate four model families to limit functional-form dependence: penalized multinomial logit (logit), feed-forward neural network (nnet), probability forest (forest), and gradient-boosted trees (boost). We then vary predictors using three feature sets:

- **Risk set**: incentivized risk investment and self-reported risk willingness.
- **Socio-demographic exogenous set**: age and a female indicator.
- **Risk + socio-demographic exogenous set**.

The risk set is the baseline because it is the closest empirical proxy to the model's heterogeneity channel, while still being interpreted as an elicited post-experiment measure rather than a pre-treatment type classifier **[Note:** it was a post-experiment questionnaire or not? would be cool if pre-experimental**]**. The socio-demographic set provides a more exogenous but potentially less informative partition; the combined set tests whether conclusions are sensitive to richer observables.

For the training sample, we estimate two target specifications. The *baseline specification* uses progressive observations from T1 and the first ten rounds of T4, avoiding the late-switch progressive block where adaptation, learning, and session dynamics are most likely to contaminate the target. The *extended depurated specification* adds the progressive block of T3 (rounds 11–20), increasing sample size, and depurates declarations before aggregation by removing sequence and progressive-block event-time components, recentering, clipping to $[0, 125]$, and mapping back to evasion units.

For the three main feature sets and each target variant, weighted classification grids contain a sizeable set of configurations: 1230 for penalized multinomial logit, 1280 for the feed-forward neural network, 1152 for gradient-boosted trees, and either 720 or 1440 for the probability forest depending on feature-set dimensionality.[3] Each model grid is evaluated using repeated stratified out-of-fold prediction (10 folds $\times$ 10 repeats). In the inference run used for the main tables, specifications are ranked lexicographically by macro-F1, then multiclass log-loss, and then classification error rate, so minority-class performance is prioritized before overall-fit tie-breakers. In the main text we report the top-ranked specification, while robustness analyses also examine the 10 highest-ranked alternative hyperparameter configurations; all 10 satisfy the support requirements.

We also report a transparent benchmark based on realized progressive-round behavior:

$$\overline{E}_i^{\text{prog}} := \frac{\sum_t Prog_{it}\, E_{it}}{\sum_t Prog_{it}}.$$

For T3/T4, benchmark groups use bracket-consistent evasion cutoffs (< 50, $[50, 75]$, > 75). For T1–T2, progressive-arm group shares are transported to the linear arm by rank-order assignment, so quantile shares are aligned across arms.[4] Because this benchmark uses realized behavior to define groups, it is more vulnerable to endogenous partitioning and is therefore used as a sensitivity benchmark, not as the preferred causal grouping design.

4.2 Within-subject regime switches: T3 and T4

In treatments 3 and 4, each subject faces both fine regimes in two blocks of ten rounds (T3: linear then progressive; T4: progressive then linear). Identification is therefore within-subject, with regime variation induced by the within-sequence switch.

Grouped analysis In the baseline specification, we use the external ML partition from Subsection 4.1. For each outcome $y_{it} \in \{R_{it}^d, R_{it}^f, R_{it}^{tot}\}$, define $Post_{it} = \mathbf{1}\{round \geq 11\}$.

[3] Full hyperparameter grids are reported in Appendix Table 6.

[4] More details are provided in Section 4.3.

The pooled baseline model is a random-intercept (RE) Tobit estimated separately by predicted stratum $g \in \{\text{Low}, \text{Mid}, \text{High}\}$, with censoring bounds $(a, b) = (0, 25)$ for R^d and $(a, b) = (0, 125)$ for R^f and R^{tot}:

$$y_{it}^* = \alpha + \beta_1 \mathbf{1}\{T3_i\} + \beta_3 \big(\mathbf{1}\{T3_i\} \times Post_{it}\big) + \sum_r \delta_r \mathbf{1}\{round = r\} + \boldsymbol{\gamma}' \mathbf{Z}_i + u_i + \varepsilon_{it}. \quad (14)$$

Here $\mathbf{Z}_i$ includes age and study year, plus Female, French-nationality, economics-student, and experience indicators. The term $\sum_r \delta_r \mathbf{1}\{round = r\}$ is a saturated set of round fixed effects (one omitted category), allowing flexible control for common round dynamics in the pooled sample.

For observed-scale inference, define the population-average censored mean under RE-Tobit as

$$\tilde{m}(\mu, \sigma, \theta_u; a, b) = \mathbb{E}_u[m(\mu + u, \sigma; a, b)], \qquad u \sim \mathcal{N}(0, \theta_u^2). \quad (15)$$

Let $\mu_{it,g}^{\text{pool}}(s, p)$ denote the pooled-model latent index after setting $(\mathbf{1}\{T3_i\}, Post_{it}) = (s, p)$, and let $m_{sp,g}^{\text{pool}}$ be the corresponding integrated observed-scale mean. The grouped-pooled policy-relevant target is

$$\Delta_g^{\text{T3/T4}} := (m_{11,g}^{\text{pool}} - m_{10,g}^{\text{pool}}) - (m_{01,g}^{\text{pool}} - m_{00,g}^{\text{pool}}) = (Prog - Lin)_{T3,g} + (Prog - Lin)_{T4,g},$$

with plug-in estimator

$$\widehat{\Delta}_g^{\text{T3/T4}} := (\widehat{m}_{11,g}^{\text{pool}} - \widehat{m}_{10,g}^{\text{pool}}) - (\widehat{m}_{01,g}^{\text{pool}} - \widehat{m}_{00,g}^{\text{pool}}). \quad (16)$$

We use superscript T3/T4 to denote this pooled-switch differential estimand. Positive $\widehat{\Delta}_g^{\text{T3/T4}}$ means higher expected revenue under progressive fines in the within-subject T3/T4 design. This is a difference-in-differences (DiD) estimand in a sequence×post layout: it first computes the within-sequence pre/post contrast in T3 and in T4, then differences those two contrasts across sequences. We therefore refer to $\widehat{\Delta}_g^{\text{T3/T4}}$ as the grouped T3/T4 DiD estimator (equivalently, $\widehat{\text{DiD}}_g$; and in aggregate, $\widehat{\Delta}^{\text{T3/T4}} = \widehat{\text{DiD}}$). Round fixed effects absorb sequence-level level differences, so this differential contrast is the identified target. We also estimate sequence-specific Prog.−Lin. contrasts as heterogeneity diagnostics. So, for each outcome and separately for T3 and T4, the grouped sequence-specific RE-Tobit is

$$\begin{aligned} y_{it}^* = \beta_0 &+ \beta_1 \, Prog_{it} + \beta_2 \, \mathbf{1}\{\widehat{S}_i = \text{High}\} + \beta_3 \, \mathbf{1}\{\widehat{S}_i = \text{Low}\} + \beta_4 \left(Prog_{it} \times \mathbf{1}\{\widehat{S}_i = \text{High}\}\right) \\ &+ \beta_5 \left(Prog_{it} \times \mathbf{1}\{\widehat{S}_i = \text{Low}\}\right) + \lambda \, xround_{it} + \boldsymbol{\gamma}' \mathbf{Z}_i + u_i + \varepsilon_{it}, \end{aligned} \quad (17)$$

where $Prog_{it} := \mathbf{1}\{f(E_{it}) = f_{\text{prog}}(E_{it})\}$, $xround_{it} := round_{it}/10$ is a linear round trend, and the Mid stratum is omitted. Let $\mu^{\text{seq}}_{it,g,T}(p)$ denote the sequence-specific latent index after setting $Prog_{it} = p \in \{0, 1\}$. The grouped sequence-specific observed-scale target is

$$\Delta^T_g := \mathbb{E}\Big[\tilde{m}\big(\mu^{\text{seq}}_{it,g,T}(1), \sigma, \theta_u; a, b\big) - \tilde{m}\big(\mu^{\text{seq}}_{it,g,T}(0), \sigma, \theta_u; a, b\big) \mid \widehat{S}_i = g,\, T\Big],$$

with plug-in estimator using estimated quantities:

$$\widehat{\Delta}^T_g = \frac{1}{N^{(T)}_g} \sum_{(i,t)\in(g,T)} \Big[\tilde{m}\Big(\widehat{\mu}^{\text{seq}}_{it,g,T}(1), \widehat{\sigma}, \widehat{\theta}_u; a, b\Big) - \tilde{m}\Big(\widehat{\mu}^{\text{seq}}_{it,g,T}(0), \widehat{\sigma}, \widehat{\theta}_u; a, b\Big)\Big].$$

We use superscript $T \in \text{T3}, \text{T4}$, to denote sequence-specific estimands.

Causal interpretation of the pooled differential estimands $\widehat{\Delta}^{\text{T3/T4}}_g$ and $\widehat{\Delta}^{\text{T3/T4}}$ requires that, conditional on round fixed effects, covariates, and subject random intercepts, residual sequence-specific shocks are not differentially concentrated at the switch. Because learning, fatigue, and end-of-block dynamics are first-order threats in this crossover design, the pooled baseline uses the most saturated round controls (full round fixed effects).

By contrast, causal interpretation of the sequence-specific contrasts $\widehat{\Delta}^T_g$ and $\widehat{\Delta}^T$ requires stronger assumptions: within each sequence, $Prog_{it}$ is tightly linked to timing, so identification relies on the adequacy of the sequence-specific time control (here, the linear trend $xround_{it}$) to absorb residual dynamic confounding. We therefore use sequence-specific contrasts as heterogeneity diagnostics rather than as baseline causal estimands.

In this assigned-sequence design, $\Delta^{\text{T3/T4}}_g$ is best interpreted as an assignment-effect contrast within an externally fixed predictive partition. A stronger latent-type interpretation would require additional transport assumptions: predicted strata must be stable and informative proxies for time-invariant heterogeneity, the mapping from latent type to predicted stratum must be comparable across sequences, and support must hold in the relevant sequence-by-stratum cells. We assess those conditions descriptively, but throughout the paper we treat the grouped estimates primarily as predictive-partition heterogeneity and report elicited risk measures and covariate means by predicted evasion stratum as partition-quality diagnostics.

Under the average-evasion grouping rule, the estimating equations and within-subject contrasts are mechanically unchanged, but causal interpretation is weaker because strata are endogenous: group labels are built from realized progressive-round behavior that is mechanically linked to outcomes. This benchmark therefore, remains informative as a robustness comparison, but it is not our preferred design for causal heterogeneity interpretation.

Aggregate analysis Without group splits, the baseline pooled aggregate target uses the same within-subject structure. Let $\mu_{it}^{\text{pool}}(s,p)$ denote the pooled-model latent index and m_{sp}^{pool} the corresponding integrated observed-scale means. The aggregate pooled target is

$$\Delta^{\text{T3/T4}} := (m_{11}^{\text{pool}} - m_{10}^{\text{pool}}) - (m_{01}^{\text{pool}} - m_{00}^{\text{pool}}),$$

with plug-in estimator $\widehat{\Delta}^{\text{T3/T4}}$.

As in the grouped analysis, we also report sequence-specific aggregate contrasts. The sequence-specific aggregate RE-Tobit is the same specification without stratum indicators and $Prog_{it}\times$stratum interactions.

Let $\mu_{it,T}^{\text{seq}}(p)$ denote the latent index from the sequence-specific aggregate model after setting $Prog_{it} = p \in \{0,1\}$. The corresponding sequence-specific aggregate observed-scale target is

$$\Delta^{T} := \mathbb{E}\big[\tilde{m}\big(\mu_{it,T}^{\text{seq}}(1),\sigma,\theta_u;a,b\big) - \tilde{m}\big(\mu_{it,T}^{\text{seq}}(0),\sigma,\theta_u;a,b\big) \mid T\big]\,,$$

with plug-in estimator $\widehat{\Delta}^{T}$.

4.3 Between-subject comparison: T1 vs. T2

In T1, subjects face the progressive schedule in all 20 rounds while, in T2, subjects face the linear schedule in all 20 rounds. This design provides between-subject variation across regimes.

Grouped analysis In this setting we use the same external ML partition from Subsection 4.1 in both arms. The resulting group-specific contrasts are

$$\widehat{\Delta}_g^{T1-T2} = \frac{1}{N_g^{T1}} \sum_{i\in g,\,T1} \overline{y}_i - \frac{1}{N_g^{T2}} \sum_{i\in g,\,T2} \overline{y}_i, \tag{18}$$

which provide an unadjusted benchmark and should be read as group-specific treatment contrasts conditional on the external partition. In the baseline covariate-adjusted specification used for the main grouped T1–T2 results, we estimate for each outcome and within each stratum g the subject-level ANCOVA

$$\overline{y}_i = \alpha_g + \beta_g\,\mathbf{1}\{T1_i\} + \boldsymbol{\delta}_g'\mathbf{Z}_i + \varepsilon_i, \qquad i\in g, \tag{19}$$

unlike T3/T4, no round trend enters because outcomes are subject-level means over rounds. We use ANCOVA as the baseline specification for the T1–T2 comparison because this contrast comes from a between-subject randomized design. The relevant grouped es-

timand is therefore a stratum-specific contrast in subject-level mean outcomes, and in this specification $\widehat{\beta}_g$ targets $\Delta_g^{\text{ITT-part}} := \mathbb{E}\Big[\overline{y}_i(1) - \overline{y}_i(0) \mid \widehat{S}_i = g\Big]$. Stratum-specific ANCOVA on subject-level mean outcomes thus provides a direct estimate of the randomized grouped ITT contrast under weak functional-form assumptions and with transparent interpretation.[5] The unadjusted contrast $\widehat{\Delta}_g^{T1-T2}$ in (18) is also computed for robustness.

As in T3/T4, the default interpretation of (18) and (19) is partition-conditional (grouped ITT-style) heterogeneity, not latent-type heterogeneity. A stronger latent-type reading would require cross-arm comparability of the predicted partition, stable mapping from latent heterogeneity into predicted strata, and non-zero treatment-by-group support in both T1 and T2. We verify support directly from treatment-by-group cell counts and assess comparability by reporting elicited risk measures and covariate means for each arm-by-group cell. Tables 1 and 2 report the corresponding descriptives and T1–T2 Fisher randomization p-values (overall and by predicted stratum) as balance and comparability diagnostics. Robustness across alternative feature sets in the prediction step is interpreted as partition-sensitivity analysis.

For the average observed evasion grouping benchmark, a structural reading of percentile-share transport in T1–T2 requires rank invariance (single crossing) in a one-dimensional latent index. It also requires single-valued optimal choices (or a common deterministic tie-breaking rule). In the model of Section 3, these are natural regularity conditions. Under these assumptions, aligned quantile shares identify comparable latent-index groups, so (18) converges to latent-quantile-specific treatment effects. Without these assumptions, the same contrasts remain informative descriptive heterogeneity comparisons, but they need not recover latent-type-specific causal parameters.

Aggregate analysis Finally, we complement the group-specific estimates with an overall (no-group) ANCOVA under the same baseline covariates as grouped T1–T2:

$$\overline{y}_i = \alpha + \beta\, \mathbf{1}\{T1_i\} + \boldsymbol{\delta}'\mathbf{Z}_i + \varepsilon_i. \tag{20}$$

We report $\widehat{\beta}$ as the aggregate covariate-adjusted T1–T2 contrast (Prog.−Lin.). Random assignment identifies the underlying intent-to-treat (ITT) target:

$$\Delta^{\text{ITT}} = \mathbb{E}[\overline{y}_i \mid T1] - \mathbb{E}[\overline{y}_i \mid T2], \tag{21}$$

[5] A randomization-inference Tobit specification with marginal effects is also feasible. However, it introduces additional distributional and latent-index assumptions and therefore increases specification risk without improving identification beyond random assignment.

where $\overline{y}_i$ is subject i's mean outcome across rounds. Causal interpretation of aggregate T1–T2 magnitudes is therefore design-based and direct: $\widehat{\beta}$ identifies the overall ITT under random assignment. Unlike grouped T1–T2 contrasts, this aggregate interpretation does not rely on additional cross-arm partition comparability assumptions for latent-type transport.

4.3.1 Inference protocol.

For reported treatment-effect estimators (T3/T4 pooled contrasts, T3/T4 sequence-specific contrasts, and T1–T2 ANCOVA contrasts), uncertainty is computed by subject-level bootstrap resampling with re-estimation of the same model in each replication. For grouped analyses, predicted strata are treated as externally fixed during resampling. In T3/T4 (within-subject regime switches), the same sequence-stratified bootstrap is used for both pooled and sequence-specific contrasts: subjects are resampled within treatment sequence. In T1–T2 (between-subject comparisons), subjects are resampled within treatment arm. For these treatment-effect objects, we report bootstrap standard errors, two-sided plus-one p-values, and percentile confidence intervals (99%, 95%, and 90% in the tables). By contrast, latent RE-Tobit coefficient tables (grouped and aggregate T3/T4) use model-based standard errors from the fitted Tobit and two-sided normal-approximation p-values for significance markers.

Estimand recap. The first-stage ML grouping is predictive and has no causal estimand. In the causal stage, effects are interpreted by default as partition-conditional contrasts under externally fixed strata. In T3/T4, the baseline grouped and aggregate targets are pooled within-subject differential (DiD) contrasts, $\Delta_g^{\mathrm{T3/T4}} \equiv \mathrm{DiD}_g$ and $\Delta^{\mathrm{T3/T4}} \equiv \mathrm{DiD}$. Sequence-specific T3 and T4 RE-Tobit contrasts (Δ_g^T, Δ^T, $T \in \{\mathrm{T3}, \mathrm{T4}\}$) are reported as heterogeneity diagnostics rather than as baseline causal targets. In T1/T2, grouped and aggregate ANCOVA contrasts ($\widehat{\beta}_g$, $\widehat{\beta}$) target partition-conditional ITT contrasts; with random assignment and no noncompliance, assignment-effect ITT and assignment-effect ATE coincide within group and in aggregate.

A latent-type interpretation (effects by true latent evasion type) requires assumptions beyond assignment: predicted strata must be stable, informative proxies for time-invariant latent heterogeneity; the mapping from latent type to predicted stratum must be comparable across arms/sequences; and support must be non-zero in the relevant treatment-by-stratum and sequence-by-stratum cells. Under these conditions, grouped effects can be read as latent-type ATE heterogeneity; otherwise they should be read as predictive-partition heterogeneity. For the average-observed-evasion benchmark, this interpretation additionally requires rank invariance/single-crossing and single-valued choices

(or a common deterministic tie-breaking rule).

5 Results

This section presents the evidence in order of evidentiary strength. We begin with descriptive patterns, then turn to grouped and aggregate treatment effects. Throughout, we treat the between-subject T1–T2 comparison as the cleaner benchmark, the pooled T3/T4 differential as corroborative within-subject evidence under round fixed effects, and the sequence-specific T3 and T4 contrasts as diagnostics about sequence asymmetry rather than as standalone causal estimates.

5.1 Descriptive diagnostics

The regime-level ECDF in Figure 1 reports the distribution of evasion E (equivalently, declarations $X = 125 - E$). In both regimes, choices cluster at the boundaries. The linear regime places slightly more mass at the extremes than the progressive regime (53.5% vs. 49.8%), largely because full compliance is more frequent under linear fines (42.0% vs. 38.4%). This evidence is in line with the narrower range of variation under the linear regime. At progressive-schedule thresholds $X \in \{25, 50, 75, 100\}$, masses are higher under progressivity. The corresponding shares are 1.17%, 1.92%, 2.49%, and 7.06% under progressivity, versus 1.09%, 1.46%, 1.30%, and 4.40% under linear fines. This pattern is consistent with focal-value behavior in both regimes and with additional bunching at progressive kinks driven by economic incentives.

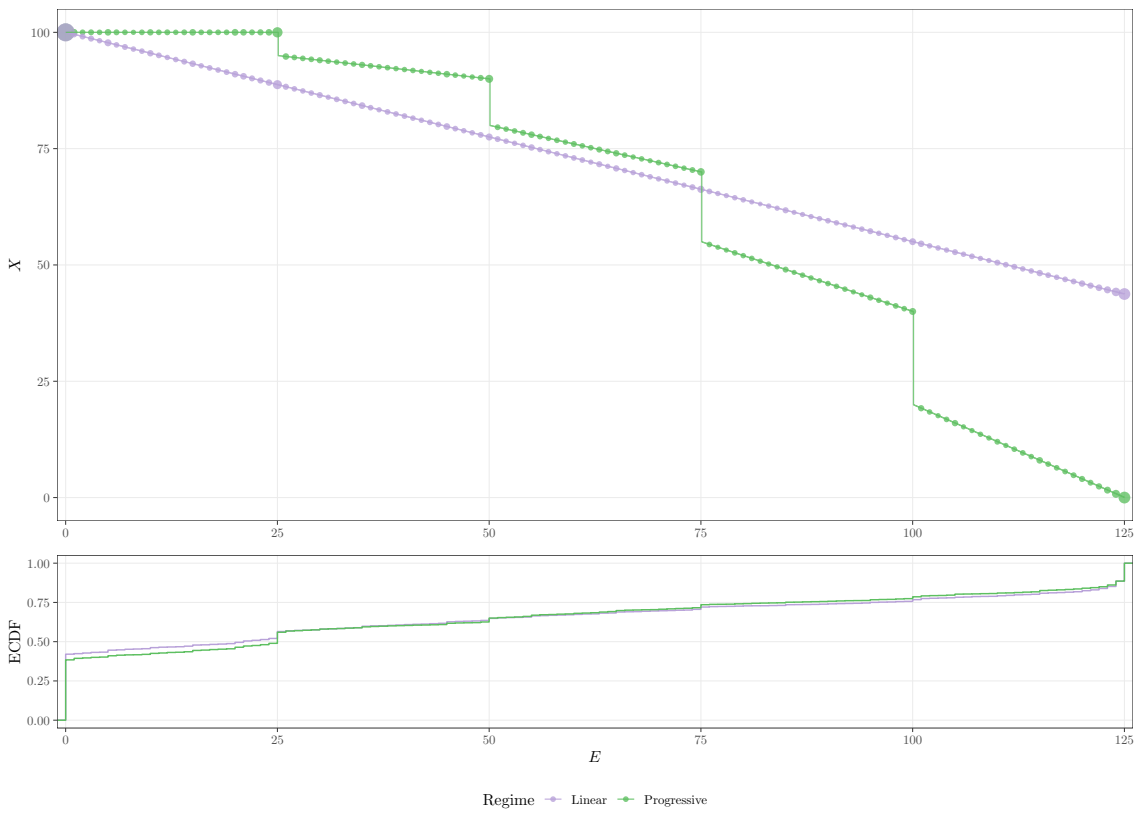

Figure 1: ECDF of evasion and schedule-overlaid scatter diagnostics by regime.

At the treatment level, descriptives in Table 1 show comparable support across arms (T1: 131 subjects; T2: 127; T3: 109; T4: 123), with generally strong aggregate covariate balance. In T1–T2, the only statistically meaningful differences are age and prior experimental experience, with a small age gap of less than one year. At the outcome level, clearer differences appear for fine and total revenues, both higher under progressivity ($R^f = 8.11$ vs. 5.29, $R^{tot} = 24.68$ vs. 22.25).

Table 1: Descriptive statistics by treatment arm with Fisher randomization p-values (T1–T2 contrast).

	T1 (Progressive)		**T2 (Linear)**		**Fisher RI**	**T3 (Lin. → Prog.)**		**T4 (Prog. → Lin.)**	
Variable	Mean	SE	Mean	SE	p(T1-T2)	Mean	SE	Mean	SE
N subjects	131		127			109		123	
N rounds obs	2,620		2,540			2,180		2,460	
Age	20.34	0.16	21.06	0.15	0.002^{***}	22.34	0.50	20.84	0.21
Female	0.695	0.040	0.661	0.042	0.623	0.532	0.048	0.626	0.044
French	0.939	0.021	0.953	0.019	0.777	0.963	0.018	0.976	0.014
Experienced	0.832	0.033	0.583	0.044	$<0.001^{***}$	0.706	0.044	0.667	0.043
Risk preference (self)	5.60	0.18	5.81	0.19	0.414	5.94	0.21	6.37	0.17
Risk preference (task)	5.71	0.28	5.98	0.27	0.477	5.84	0.33	5.82	0.28
Spending	159.28	15.56	174.93	13.26	0.443	192.66	19.51	165.24	16.09
Saving	132.92	15.54	369.09	235.67	0.419	222.66	23.74	172.03	27.14
Full compliance ($X = Y$)	0.413	0.010	0.449	0.010	0.357	0.397	0.010	0.347	0.010
Full evasion ($X = 0$)	0.132	0.007	0.106	0.006	0.315	0.100	0.006	0.116	0.006
R^d (declaration)	16.57	0.19	16.96	0.20	0.585	16.78	0.20	15.46	0.20
R^f (fines)	8.11	0.56	5.29	0.35	0.002^{***}	5.77	0.45	6.46	0.44
R^{tot} (total)	24.68	0.51	22.25	0.34	$<0.001^{***}$	22.54	0.43	21.93	0.42
Decision time	6.31	0.33	6.10	0.24	0.704	6.10	0.25	6.23	0.22

Notes. p(T1-T2) is the two-sided Monte Carlo Fisher randomization p-value for the T1–T2 contrast (B=5000, seed=20260323). For decision-level outcomes, randomization tests are performed at the subject level using each subject's mean across rounds. Stars on p(T1-T2): $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (unadjusted). Rows *N subjects* and *N rounds obs* are counts and therefore have no SE or p-value.

Table 2: Descriptive statistics by predicted evasion stratum for T1–T2 and T3–T4 treatment arms/sequences (single integrated table; Fisher randomization p-values reported for T1–T2).

	Between variation					Within variation			
	Progressive (T1)		**Linear (T2)**		**Fisher RI test**	**Linear → Progressive (T3)**		**Progressive → Linear (T4)**	
Variable	Mean	SE	Mean	SE	Fisher p-value	Mean	SE	Mean	SE
Panel A: Low predicted evasion									
N subjects	75		59			50		48	
N rounds obs	1,500		1,180			1,000		960	
Age	20.25	0.21	21.02	0.24	0.020**	21.32	0.30	21.17	0.43
Female	0.747	0.051	0.746	0.057	1.000	0.640	0.069	0.667	0.069
French	0.973	0.019	0.932	0.033	0.400	0.960	0.028	0.979	0.021
Experienced	0.840	0.043	0.576	0.065	<0.001***	0.760	0.061	0.688	0.068
Risk preference (self)	4.35	0.16	4.32	0.19	0.953	4.40	0.21	4.58	0.20
Risk preference (task)	3.76	0.29	4.15	0.28	0.369	3.66	0.34	3.54	0.28
Spending	125.17	12.53	175.42	21.85	0.036**	155.60	20.20	128.44	18.96
Saving	140.99	22.01	636.44	506.84	0.277	157.80	24.67	121.35	18.63
Full compliance ($X = Y$)	0.415	0.013	0.515	0.015	0.063*	0.409	0.016	0.374	0.016
Full evasion ($X = 0$)	0.097	0.008	0.097	0.009	1.000	0.074	0.008	0.095	0.009
R^d (declaration)	17.47	0.24	18.55	0.27	0.267	17.51	0.29	16.62	0.30
R^f (fines)	7.35	0.70	4.57	0.47	0.036**	5.51	0.66	5.68	0.64
R^{tot} (total)	24.82	0.63	23.12	0.44	0.047**	23.02	0.62	22.30	0.61
Decision time	6.39	0.31	6.36	0.40	0.973	6.22	0.31	6.43	0.30
Panel B: Mid predicted evasion									
N subjects	9		20			18		21	
N rounds obs	180		400			360		420	
Age	20.78	0.89	20.70	0.33	1.000	22.11	0.77	20.76	0.43
Female	0.556	0.176	0.500	0.115	1.000	0.444	0.121	0.619	0.109
French	1.000	0.000	1.000	0.000	1.000	0.944	0.056	1.000	0.000
Experienced	0.778	0.147	0.500	0.115	0.229	0.556	0.121	0.571	0.111
Risk preference (self)	7.22	0.22	7.40	0.13	0.539	7.61	0.18	7.67	0.21
Risk preference (task)	5.00	0.17	4.00	0.38	0.123	3.67	0.46	3.76	0.38
Spending	339.44	118.12	165.30	26.38	0.048**	256.67	35.45	207.62	46.94
Saving	88.89	40.64	107.00	28.77	0.753	262.22	63.21	163.33	32.38
Full compliance ($X = Y$)	0.417	0.037	0.390	0.024	0.794	0.325	0.025	0.312	0.023
Full evasion ($X = 0$)	0.100	0.022	0.140	0.017	0.705	0.061	0.013	0.067	0.012
R^d (declaration)	16.04	0.73	15.90	0.52	0.943	16.83	0.46	16.29	0.42
R^f (fines)	7.26	2.02	6.03	0.96	0.534	4.35	0.90	5.48	0.88
R^{tot} (total)	23.31	1.88	21.93	0.92	0.411	21.18	0.87	21.78	0.85
Decision time	5.87	0.54	6.34	0.67	0.851	6.23	0.49	6.42	0.86
Panel C: High predicted evasion									
N subjects	47		48			41		54	
N rounds obs	940		960			820		1,080	
Age	20.40	0.25	21.25	0.23	0.017**	23.68	1.22	20.57	0.23
Female	0.638	0.071	0.625	0.071	1.000	0.439	0.078	0.593	0.067
French	0.872	0.049	0.958	0.029	0.160	0.976	0.024	0.963	0.026
Experienced	0.830	0.055	0.625	0.071	0.037**	0.707	0.072	0.685	0.064
Risk preference (self)	7.28	0.23	6.98	0.28	0.422	7.07	0.32	7.46	0.16
Risk preference (task)	8.96	0.23	9.06	0.19	0.780	9.46	0.17	8.65	0.26
Spending	179.21	29.26	178.33	20.12	0.980	209.76	42.35	181.48	26.65
Saving	128.49	24.34	149.69	25.41	0.555	284.39	46.37	220.46	57.86
Full compliance ($X = Y$)	0.409	0.016	0.393	0.016	0.813	0.413	0.017	0.336	0.014
Full evasion ($X = 0$)	0.196	0.013	0.103	0.010	0.043**	0.150	0.012	0.155	0.011
R^d (declaration)	15.22	0.35	15.43	0.33	0.870	15.86	0.35	14.11	0.31
R^f (fines)	9.49	1.02	5.88	0.62	0.013**	6.69	0.81	7.54	0.74
R^{tot} (total)	24.70	0.95	21.32	0.60	0.002***	22.56	0.77	21.65	0.70
Decision time	6.27	0.77	5.69	0.27	0.569	5.90	0.50	5.98	0.27

Notes. Predicted evasion strata are external ML strata merged by idparticipant from strata_rank_001_glmnet_cfg1205.csv (group_ml_main). T1–T2 p-values are two-sided Monte Carlo Fisher randomization p-values (complete random assignment, B=5000 draws, seed=20260323); for decision-level outcomes, randomization is evaluated at the subject level using each subject's mean across rounds. Stars on p-values use $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (unadjusted). T3/T4 identification in the baseline uses within-subject regime-switch variation, so no between-sequence p-values are reported for T3/T4 columns. Rows *N subjects* and *N rounds obs* are counts and therefore have no SE or p-value.

For the baseline predicted-strata assignment, macro-F1 is 0.403. This indicates informative but moderate separation across Low, Mid, and High strata. We therefore interpret grouped effects as heterogeneity across predicted evasion strata, not as exact recovery of latent structural types.

Over rounds, average declarations are broadly stable, with a mild downward drift in both treatment (Figure 2) and treatment-by-stratum aggregation (Figure 10). This pattern is consistent with the round-related threats discussed in Subsection 4.2 and motivates the round control in the baseline specification.

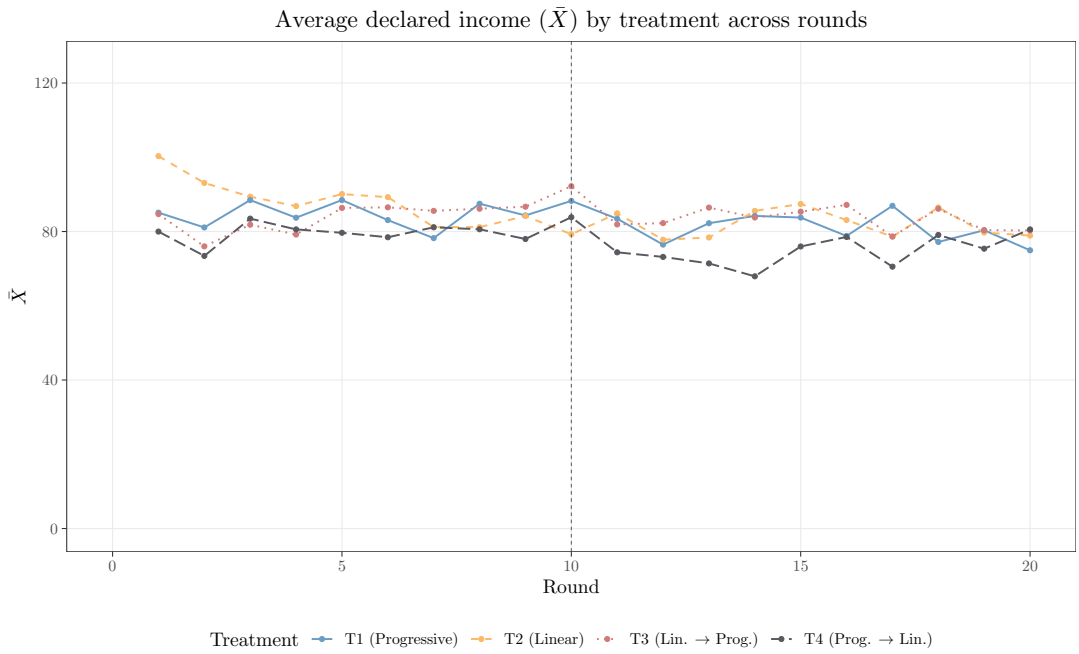

Figure 2: Average declaration by round and treatment.

5.2 Group-level effects

Our grouped results use the baseline covariate-adjusted specification from Section 4, with external ML-predicted evasion strata from Subsection 4.1. We start by reporting the coefficients in the latent space for T3/T4 pooled within-subject DiD specification with round fixed effects in Table 7.[6] Heterogeneity appears mainly in the fine-revenue interaction for High (DiD × High = 23.300, $p = 0.044$). Covariate patterns are selective rather than broad: female is positive in declaration revenues (3.211, $p = 0.056$) and in total revenues (1.819, $p = 0.026$); French nationality is negative in declaration revenues (-9.891, $p = 0.039$) and in total revenues (-4.848, $p = 0.033$); and experience is negative

[6] We report all two-sided p-values as p. Latent RE-Tobit coefficients use normal approximation, while observed-scale T1–T2 ANCOVA contrasts and T3/T4 treatment-effect contrasts use bootstrap plus-one.

in declaration revenues (-3.955, $p = 0.029$). These patterns are broadly consistent with the sequence-specific latent estimates reported in Table 8.

Figure 3, with details in Table 9, juxtaposes the between-subject benchmark with the pooled T3/T4 within-subject differential and the two sequence-specific diagnostics. For the group of high evaders, total revenues are higher under progressivity in both designs and the magnitudes are sizeable: $\widehat{\beta}_{\text{High},R^{tot}} = 3.066$ ($p = 0.005$) in T1–T2, which corresponds to about a 14%[7] increase, and $\widehat{\Delta}^{\text{T3/T4}}_{\text{High},R^{tot}} = 3.984$ ($p = 0.027$) in pooled T3/T4, which corresponds to about 19%. This is the pattern predicted by the model: progressivity reallocates the strongest revenue gains toward the upper tail, where steeper sanctions make large evasion episodes more costly. The common message across designs is therefore about total revenue, not about a precisely identified reallocation between declaration and fine revenues.

A second positive estimate appears for predicted low evaders in the pooled within-subject design, where $\widehat{\Delta}^{\text{T3/T4}}_{\text{Low},R^{tot}} = 3.310$ ($p = 0.026$), corresponding to a 15% increase. However, we interpret this result more cautiously than the corresponding High-stratum effect. First, it is not confirmed in the between-subject comparison, where $\widehat{\beta}_{\text{Low},R^{tot}} = 1.137$ ($p = 0.202$); there, a small negative declaration component (-0.404, $p = 0.658$) partly offsets the positive fine component (1.541, $p = 0.171$). Second, the pooled Low-group estimate is driven mainly by one sequence, as discussed below. The Low result is therefore best read as a sequence-sensitive within-subject pattern, not as a co-equal headline effect of progressive sanctions. Because the model does not sign effects for low-evasion types, this pattern remains consistent with model ambiguity, but it does not by itself support a strong mechanism claim**[Note:** Antoine has ideas on the interpretation of this?**]**. Estimates for the mid-evaders group are close to zero and imprecise in both T1–T2 and T3/T4 settings.

[7]Percent effects are computed as $100 \times \widehat{\theta}/\bar{y}^{lin}$, where $\widehat{\theta}$ is the reported contrast and $\bar{y}^{lin}$ is the corresponding linear-regime mean in the same outcome and cell (same stratum/treatment; for aggregate T1–T2, the T2 linear benchmark mean).

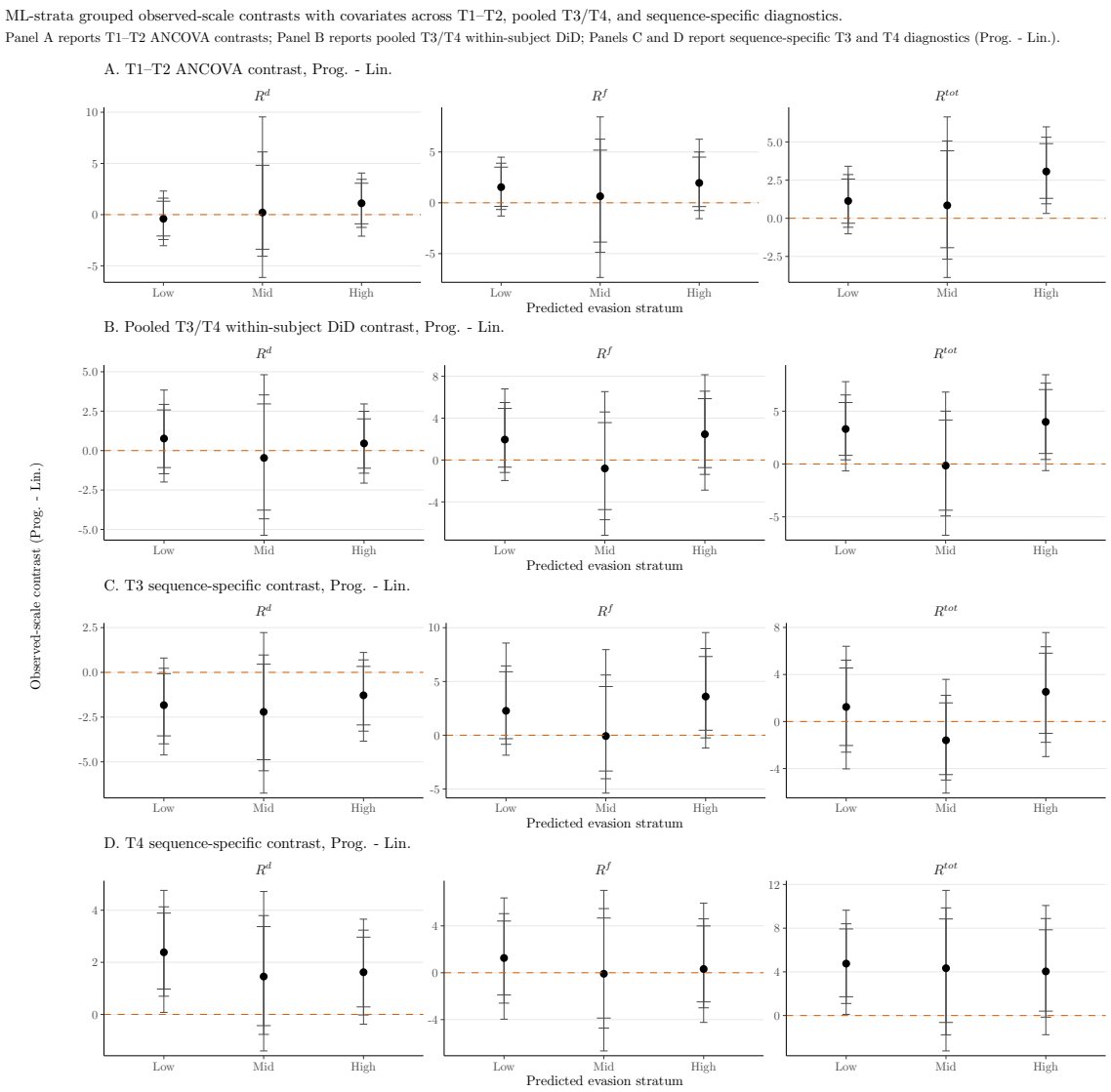

Figure 3: Grouped observed-scale contrasts with covariates (Prog. − Lin.). Panel A reports T1–T2 between-subject ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject DiD contrast; Panels C and D report sequence-specific T3 and T4 diagnostics. Error bars show nested 90%, 95%, and 99% confidence intervals.

Panels C and D of Figure 3 report sequence-specific T3 and T4 diagnostics, which reveal marked cross-sequence asymmetry. In the group of high evaders, the T3 profile is weaker and internally mixed: declarations decrease ($\widehat{\Delta}^{T3}_{\text{High},R^d} = -1.282$, -8%, $p = 0.187$), fine revenues increase ($\widehat{\Delta}^{T3}_{\text{High},R^f} = 3.599$, 66%, $p = 0.072$), and the net effect on total revenues remains imprecise ($\widehat{\Delta}^{T3}_{\text{High},R^{tot}} = 2.527$, 12%, $p = 0.222$). By contrast, T4 is aligned with the pooled pattern, though still less precisely estimated. The Low-group asymmetry is more consequential: the pooled positive estimate is mainly driven by T4, where declaration, fine, and total revenues are all positive ($\widehat{\Delta}^{T4}_{\text{Low},R^d} = 2.384$, $p = 0.008$; $\widehat{\Delta}^{T4}_{\text{Low},R^f} = 1.257$, $p = 0.467$; $\widehat{\Delta}^{T4}_{\text{Low},R^{tot}} = 4.755$, $p = 0.010$). In T3, the Low profile is

weaker and less stable, with no statistically informative revenue impact. We therefore use the sequence-specific contrasts as diagnostics about where the pooled within-subject signal comes from, not as standalone causal evidence.

Tables 13–15 report pooled T3/T4 covariate-ladder results, progressively adding controls from age alone through the full Risk set. For R^{tot}, the Low and High effects are positive and statistically informative across all ten rungs: $\widehat{\Delta}^{\text{T3/T4}}_{\text{Low},R^{tot}}$ ranges from 3.13 to 3.34 ($p \in [0.023, 0.038]$) and $\widehat{\Delta}^{\text{T3/T4}}_{\text{High},R^{tot}}$ from 3.87 to 3.93 ($p \in [0.022, 0.032]$), while Mid remains near zero and uninformative. The component margins R^d and R^f are directionally stable but individually imprecise throughout the ladder, consistent with the baseline specification. Treatment-specific covariate-ladder diagnostics—sequence-specific T3 and T4 contrasts (Prog. − Lin.) and T1–T2 ANCOVA contrasts—confirm sign and magnitude stability across rungs and are reported in Tables 16, 17, and 18.

5.3 Aggregate

We next report aggregate effects using the same covariate-adjusted baseline specification as in the grouped analysis. Figure 4, with details in Table 11, reports the aggregate between-subject contrast together with the pooled T3/T4 differential and the sequence-specific diagnostics. As in the grouped analysis, we treat T1–T2 as the cleaner benchmark, pooled T3/T4 as corroborative within-subject evidence, and T3/T4 separately as diagnostics about sequence dependence. The results display a coherent pattern across designs, with differences mainly in precision. For declaration revenues, both contrasts are small (about 2%) and imprecise: $\widehat{\beta}_{R^d} = 0.355$ ($p = 0.632$) in T1–T2 and $\widehat{\Delta}^{\text{T3/T4}}_{R^d} = 0.369$ ($p = 0.597$) in pooled T3/T4. For fine revenues (R^f), both contrasts are positive and similar in size (about 33%), but only T1–T2 is statistically informative: $\widehat{\beta}_{R^f} = 1.701$ ($p = 0.037$) versus $\widehat{\Delta}^{\text{T3/T4}}_{R^f} = 1.778$ ($p = 0.132$). For total revenues (R^{tot}), both designs show positive and statistically informative effects, with a larger pooled within-subject magnitude: $\widehat{\beta}_{R^{tot}} = 2.055$ (9.2%, $p = 0.001$) and $\widehat{\Delta}^{\text{T3/T4}}_{R^{tot}} = 3.043$ (about 14%, $p = 0.010$). The larger pooled R^{tot} estimate is consistent with separate nonlinear RE-Tobit OS-APE estimation by outcome, where R^d and R^f effects are not mechanically additive. Again, sequence-specific diagnostics in Panels C and D of Figure 4 reveal the same directional asymmetry observed in the grouped analysis.

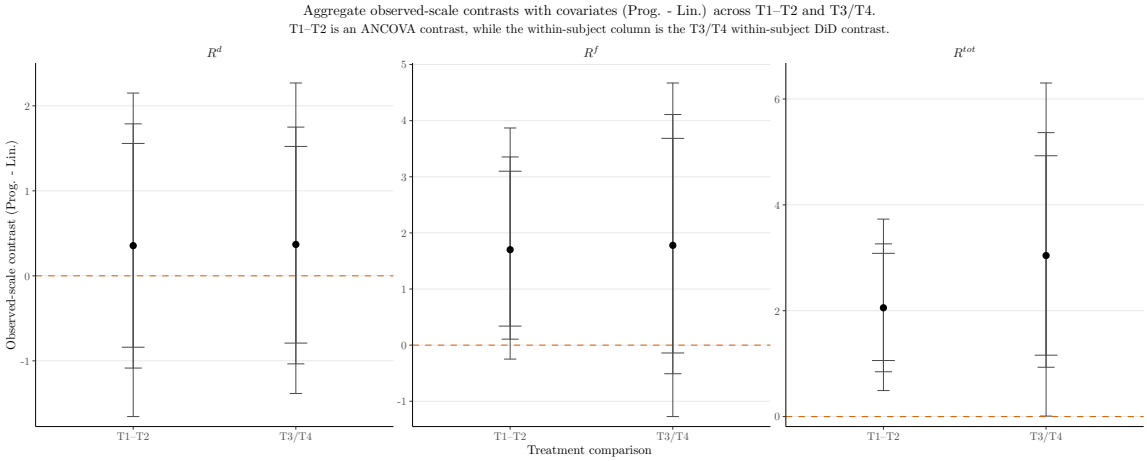

Figure 4: Aggregate observed-scale contrasts with covariates (Prog. − Lin.) across T1–T2, pooled T3/T4, and sequence-specific T3 and T4. T1–T2 is an ANCOVA contrast, pooled T3/T4 is the within-subject DiD contrast under round fixed effects, and T3/T4 are sequence-specific diagnostics. Error bars show nested 90%, 95%, and 99% confidence intervals.

Because aggregate effects reflect both within-stratum treatment responses and stratum composition, we examine each treatment arm separately. In T1–T2, the High share is similar across arms (35.9% in T1 and 37.8% in T2), but T1 has more Low (57.3% vs. 46.5%) and fewer Mid (6.9% vs. 15.7%). Because grouped T1–T2 total-revenue gains are strongest in High and smaller in Low/Mid, this composition pattern does not mechanically inflate the progressive aggregate effect; if anything, it attenuates it. The positive aggregate T1–T2 $\widehat{\beta}_{R^{tot}}$ is therefore mainly a within-stratum result, with the largest contribution from High.

In T3–T4, composition and within-stratum responses move in the same direction. Under the baseline external-ML partition, T3 is 45.9% Low, 16.5% Mid, and 37.6% High, whereas T4 is 39.0% Low, 17.1% Mid, and 43.9% High. Relative to T3, T4 shifts mass away from Low and toward High, and grouped total-revenue effects are also stronger in T4 in the sequence-specific diagnostics. This alignment makes the larger aggregate $\widehat{\Delta}^{T4}_{R^{tot}}$ relative to $\widehat{\Delta}^{T3}_{R^{tot}}$ internally consistent: the T3–T4 gap reflects sequence-specific heterogeneity, with composition differences reinforcing that gap under composition-sensitive aggregation.

6 Robustness

We organize robustness checks along two dimensions. The first examines the prediction and partition stage, asking whether grouped effects depend on how strata are constructed. The second examines treatment-specific inferential threats, checking whether the main

contrasts survive tests tailored to each experimental design.

6.1 Prediction and partition robustness

Grouped effects may be driven by first-stage classification choices rather than by stable behavioral heterogeneity. We therefore evaluate robustness to: i) alternative first-stage specifications, ii) alternative hyperparameter selection, and iii) assignment uncertainty near classification boundaries.

6.1.1 Alternative first-stage specifications

We begin by varying the specifications of the predictive stage along the dimensions introduced in Subsection 4.1. Relative to the baseline ML predictions, we compare three alternatives that employ: i) the socio-demographic exogenous set, ii) the Risk plus socio-demographic exogenous set, and iii) the Risk set under the extended depurated specification. Finally, we also consider iv) the average-observed-evasion grouping as a non-ML benchmark. The corresponding results are shown in Figures 11, 12, 13, and 14.[8] The reliability of results should take into account the different classification quality across predictor sets: out-of-fold macro-F1 is 0.403 for the baseline Risk-only partition, 0.395 for Risk plus socio-demographic exogenous, 0.383 for the extended depurated Risk specification, and only 0.318 for the socio-demographic-only classifier.

In T1–T2, the main result is stable across ML-based partitions: the High-stratum total-revenue effect remains positive and informative, with $\widehat{\beta}_{\text{High},R^{tot}} \in [2.780, 3.066]$ and every 95% confidence interval excluding zero. The declaration and fine components remain positive but individually imprecise, so the stable result is the composite revenue effect rather than its decomposition. The average-observed-evasion benchmark delivers a larger but noisier High estimate, $\widehat{\beta}_{\text{High},R^{tot}} = 8.173$ ($p = 0.018$).

For pooled T3/T4, the same ordering holds in the specifications that retain risk information among the predictors. The High-stratum total-revenue effect is tightly clustered, ranging from 3.715 to 4.079 ($p \in [0.026, 0.058]$), and the Low-stratum effect is likewise stable, ranging from 3.184 to 3.592 ($p \in [0.025, 0.036]$). Mid remains near zero in these risk-based specifications. By contrast, the lower-accuracy socio-demographic-only partition attenuates Low to 1.219 ($p = 0.459$) and shifts Mid upward to 3.293 ($p = 0.078$). The average-observed-evasion benchmark reshuffles the pooled pattern more sharply: High rises to 5.778 but becomes imprecise ($p = 0.138$), Mid turns positive and significant at 6.139 ($p = 0.019$), and Low becomes small and uninformative.

[8] Details are reported in Tables 19, 20, 21, and 22.

Sequence-specific diagnostics add little to the risk-based ML message beyond locating the heterogeneity mainly in T4: T4 High remains positive and fairly stable ($\widehat{\Delta}^{T4}_{\text{High},R^{tot}} =$ 4.040 to 4.562, $p \in [0.042, 0.062]$), whereas T3 High stays weaker and statistically uninformative (2.095 to 2.527, $p \in [0.222, 0.304]$). The model-ambiguous T4 Low cell also remains positive, with precision strongest in the risk-based specifications. The substantive difference again comes from the non-ML benchmark, which concentrates the High effect almost entirely in T4 (8.337, $p = 0.022$) rather than T3 (0.031, $p = 0.987$) and generates positive Mid effects in both sequences (5.014, $p = 0.077$; 3.588, $p = 0.094$). Thus, the alternative ML groupings that retain risk information largely confirm the baseline partition, whereas the average-observed benchmark reorganizes the heterogeneity profile.

6.1.2 Alternative hyperparameter selection

We next ask whether the grouped results depend on a single tuning choice in the first-stage classifier. Holding the causal pipeline fixed, we compare the baseline classifier with the 10 highest-ranked alternative hyperparameter configurations by out-of-fold macro-F1; all 10 satisfy the support requirements. Because all 10 are penalized multinomial logit specifications, this exercise varies regularization hyperparameters within a fixed model family rather than across learner classes. Appendix Tables 24 and 6 report the selected configurations and the full pre-specified search spaces, respectively. Figure 5 reports the resulting effect envelope, while Appendix Table 23 reports the corresponding cell-level summaries together with partition similarity across the 10 alternatives.

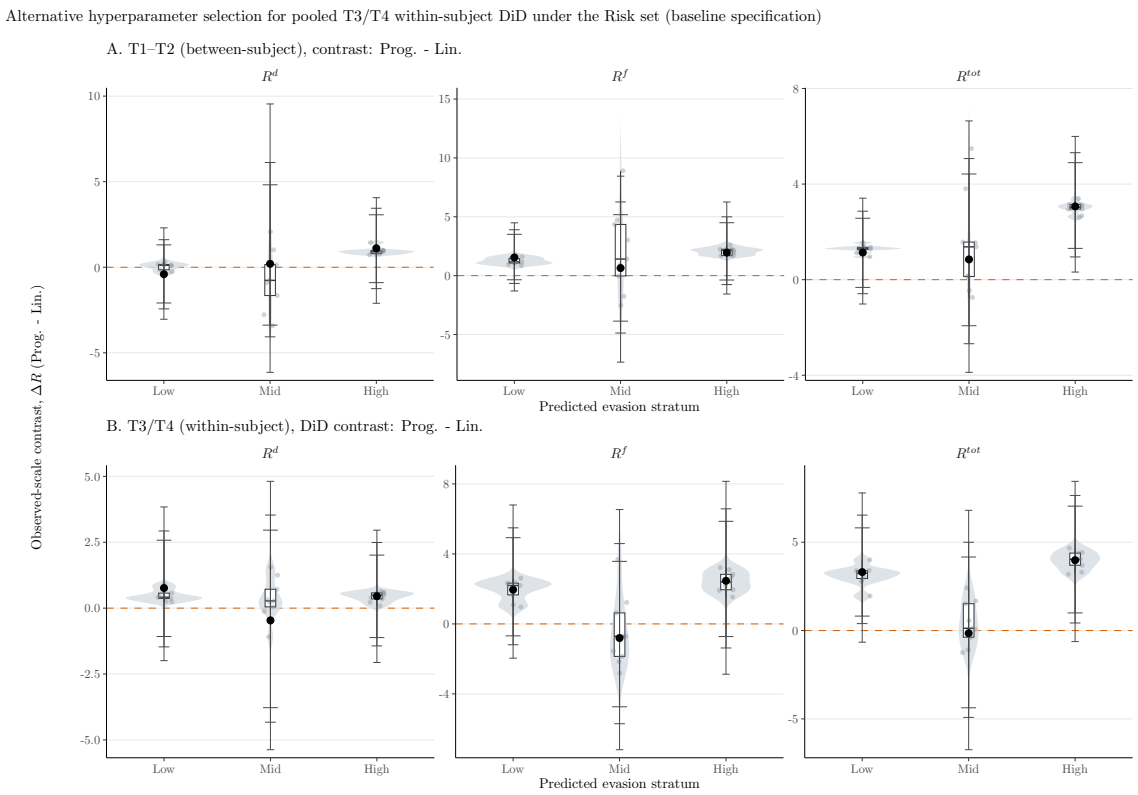

Figure 5: Alternative hyperparameter selection for the Risk set (baseline specification). Panel A reports T1–T2 between-subject contrasts; Panel B reports pooled T3/T4 within-subject DiD contrasts (Prog. − Lin.). In each panel, the black marker with nested 90%, 95%, and 99% confidence intervals reports the baseline estimate. Violins with embedded boxplots and gray points summarize the 10 alternative hyperparameter configurations.

Main cells are stable across the 10 alternative hyperparameter configurations. For T1–T2 High total revenue, the baseline estimate 3.07 ($[0.95, 5.31]$) is close to the median across alternatives, 3.05, with range $[2.59, 3.39]$; all 10 alternatives retain the baseline sign. Pooled T3/T4 High total revenue is similarly robust: the baseline estimate is 3.98 ($[0.43, 7.65]$), the median across alternatives is 4.01, the range is $[3.19, 4.66]$, and all 10 alternatives retain the baseline sign. Pooled T3/T4 Low also preserves sign and direction (baseline 3.31, $[0.39, 6.53]$; median 3.26, range $[1.96, 4.00]$), while pooled T3/T4 Mid remains near zero and directionally unstable, with the baseline sign retained in only 40% of alternatives. Classification agreement is moderate to high (median ARI 0.77). This pattern indicates that the baseline specification is not an outlier within this range of nearby hyperparameter choices and that the main total-revenue conclusions in the High and Low strata are not driven by a single tuning choice.

6.1.3 Assignment uncertainty

Because the first-stage classifier assigns each subject to a single stratum, grouped effects may be sensitive to hard class boundaries. We test this by holding the downstream causal estimators fixed and perturbing only assignment through three complementary checks: *low-margin exclusion*, *boundary flip*, and *stochastic reassignment based on posterior probabilities*.

Low-margin exclusion ranks subjects by the top-two posterior gap (the difference between the two highest posterior probabilities), and drops the bottom 10% or 20% of the sample, so estimates are recomputed on a smaller but higher-confidence subset; the perturbation design and affected sample shares are reported in Table 3.

Table 3: Assignment-uncertainty perturbation design for the Risk set (baseline specification)

Perturbation	Rule	Perturbed	Retained	Changed
Baseline specification	None	0%	100%	0%
Low-margin exclusion (10%)	Drop low-margin units	10%	90%	10%
Low-margin exclusion (20%)	Drop low-margin units	20%	80%	20%
Boundary flip (10%)	Flip to second class	10%	100%	10%
Boundary flip (20%)	Flip to second class	20%	100%	20%
Stochastic reassignment (50)	Random class draw	100%	100%	67%

Notes. Shares are percentages of the baseline sample. *Perturbed* is the share directly affected by each perturbation rule; *Retained* is the estimation sample share after perturbation; *Changed* is the share whose assigned group differs from baseline (equal to the dropped share in exclusion scenarios).

Boundary flip is run on exactly the subjects selected in the low-margin step (bottom 10% or 20% by the top-two posterior gap). For each selected subject, we replace the top-posterior class with the second-highest-posterior class, providing an adversarial boundary-misclassification check; the resulting baseline-to-scenario reassignment shares are summarized in Table 4.

Table 4: Assignment-uncertainty reassignment matrix by baseline group for the Risk set (baseline specification). Low-margin exclusion and boundary-flip entries average the 10% and 20% perturbations within each family; stochastic-reassignment entries average across 50 stochastic reassignments.

	Low-margin exclusion				*Boundary flip*			*Stochastic reassignment*		
Base	To Low	To Mid	To High	Dropped	To Low	To Mid	To High	To Low	To Mid	To High
Low	86%	–	–	14%	86%	12%	2%	35%	33%	32%
Mid	–	54%	–	46%	33%	54%	13%	34%	33%	32%
High	–	–	94%	6%	1%	5%	94%	35%	34%	32%

Notes. Entries are mean reassignment shares by baseline-specification group. The "Dropped" destination arises only under low-margin exclusion.

Stochastic reassignment based on posterior probabilities performs fifty replications; in each, a single class is drawn per subject from the subject-specific posterior distribution

$(p_{\text{Low}}, p_{\text{Mid}}, p_{\text{High}})$, and the grouped causal pipeline is rerun on the resulting partition. Unlike the exclusion and boundary-flip checks, which perturb only the most ambiguous units, this exercise perturbs assignments throughout the sample, producing a distribution of grouped effects that directly tests whether the main qualitative conclusions survive global classification uncertainty.

Figure 6 visualizes the full assignment-uncertainty effect envelope, while Appendix Tables 25 and 26 report the corresponding cell-level summaries for the deterministic perturbations and the stochastic reassignment exercise.

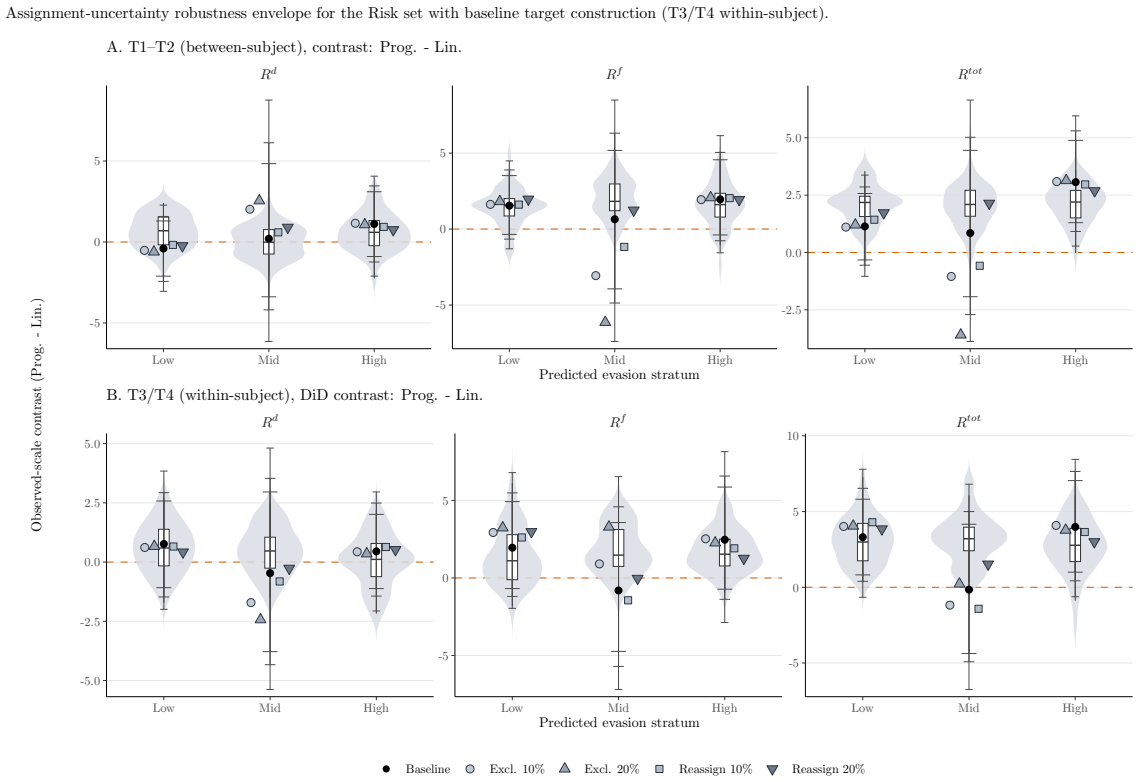

Figure 6: Assignment-uncertainty robustness envelope for the Risk set (baseline specification). The upper row reports T1–T2 ANCOVA contrasts and the lower row reports pooled T3/T4 within-subject DiD contrasts; columns report outcomes (R^d, R^f, R^{tot}). In each panel, the black marker reports the baseline estimate with nested 90%, 95%, and 99% confidence intervals. The violins summarize stochastic reassignment based on posterior probabilities, while the remaining markers report the exclusion and boundary-flip checks.

The key total-revenue cells remain directionally stable. In T1–T2 High, the baseline estimate is 3.07 ($CI_{95} = [0.92, 5.30]$), and 98% of stochastic reassignments retain the baseline sign. In pooled T3/T4 High and Low, the baseline estimates are 3.98 ($[0.43, 7.65]$) and 3.31 ($[0.39, 6.53]$); the corresponding shares are 92% and 100%. By contrast, pooled T3/T4 Mid is near zero and directionally unstable. These results indicate that the main

High- and Low-stratum total-revenue findings remain intact when classification uncertainty is propagated through the full sample. Overall, assignment perturbations do not overturn the positive total-revenue findings in Section 5, but they identify the Mid stratum and the R^d/R^f decomposition as the main loci of sensitivity.

Alternative first-stage specifications, assignment perturbations, and alternative hyperparameter configurations leave the main economic conclusion unchanged. In the High and Low strata, total-revenue gains in both T1–T2 and pooled T3/T4 remain robust in sign, magnitude, and statistical informativeness across all partition checks; the Mid stratum is uninformative throughout. The main sensitivity concerns the internal R^d/R^f decomposition and the survival of marginal significance in a few component-level cells. Sequence-specific envelopes across alternative hyperparameter configurations (T3/T4 split) are reported as supplementary diagnostics and confirm that the pooled signal is not driven by a single sequence.

6.2 Treatment-specific inferential robustness

This second set of robustness checks addresses design-specific inferential risks in the treatment contrasts themselves. For T1–T2, the main concern is that treatment is assigned at the session-day level, inducing within-cluster dependence that standard errors may understate; for T3/T4, the primary concern is dynamic confounding around the regime switch.

6.2.1 T1–T2 session-day RI robustness

In T1–T2, treatment is assigned between subjects at the session-day level, so within-cluster dependence is a first-order concern. We therefore keep the baseline grouped ANCOVA specification unchanged and re-compute uncertainty with session-day randomization inference (RI). For each subgroup-outcome cell (Low/Mid/High $\times$ R^d, R^f, R^{tot}), treatment is reassigned across session-day clusters while holding fixed the treated-cluster count. The contrast is then re-estimated under each reassignment and p_{RI} is computed from the resulting randomization distribution. We use 9,999 RI draws per cell and report subject-bootstrap percentile confidence intervals to summarize effect-size precision. Holm and BH adjustments across the grouped-cell RI tests are also reported to separate cellwise informativeness from familywise claims.

Table 5: T1–T2 RI robustness with session-day clustering.

		Grouped		Aggregate
Stat	Low	Mid	High	
Panel A. R^d				
$\widehat{\beta}$	-0.404	0.202	1.111	0.355
p_{RI}	0.608	0.931	0.479	0.629
CI_{99}	[-3.174, 2.194]	[-6.766, 9.837]	[-1.958, 4.036]	[-1.542, 2.226]
CI_{95}	[-2.503, 1.634]	[-4.428, 6.513]	[-1.262, 3.362]	[-1.062, 1.852]
CI_{90}	[-2.082, 1.326]	[-3.564, 4.802]	[-0.825, 2.930]	[-0.868, 1.656]
N_{subj}	134	29	95	258
$N_{cl}(t/c)$	13 (6/7)	10 (5/5)	13 (6/7)	13 (6/7)
RI v/f	9999/0	9999/0	9999/0	9999/0
Panel B. R^f				
$\widehat{\beta}$	1.541	0.645	1.956	1.701
p_{RI}	0.260	0.950	0.185	0.108
CI_{99}	[-1.491, 4.435]	[-7.796, 8.060]	[-1.777, 6.095]	[-0.539, 4.077]
CI_{95}	[-0.664, 3.740]	[-4.792, 6.183]	[-0.723, 5.216]	[-0.002, 3.444]
CI_{90}	[-0.318, 3.434]	[-3.771, 5.107]	[-0.296, 4.604]	[0.284, 3.138]
N_{subj}	134	29	95	258
$N_{cl}(t/c)$	13 (6/7)	10 (5/5)	13 (6/7)	13 (6/7)
RI v/f	9999/0	9999/0	9999/0	9999/0
Panel C. R^{tot}				
$\widehat{\beta}$	1.137	0.847	3.066	2.055
p_{RI}	0.167	0.597	0.039	0.022
CI_{99}	[-1.070, 3.404]	[-3.677, 7.525]	[0.501, 6.080]	[0.584, 3.634]
CI_{95}	[-0.480, 2.866]	[-2.747, 5.302]	[1.100, 5.308]	[0.910, 3.315]
CI_{90}	[-0.271, 2.606]	[-2.010, 4.476]	[1.448, 4.993]	[1.096, 3.120]
N_{subj}	134	29	95	258
$N_{cl}(t/c)$	13 (6/7)	10 (5/5)	13 (6/7)	13 (6/7)
RI v/f	9999/0	9999/0	9999/0	9999/0

Notes. Panel rows report T1–T2 covariate-adjusted ANCOVA contrasts (Prog. − Lin.). Within each outcome panel, $\widehat{\beta}$ is the effect estimate. p_{RI} is the session-day randomization-inference p-value with fixed treated-cluster count and $B_{RI} = 9999$ attempted draws per test. CI_{99}, CI_{95}, and CI_{90} are subject-bootstrap percentile confidence intervals. $N_{cl}(t/c)$ reports total session-day clusters with treated/control support. RI v/f reports valid/failed RI draws.

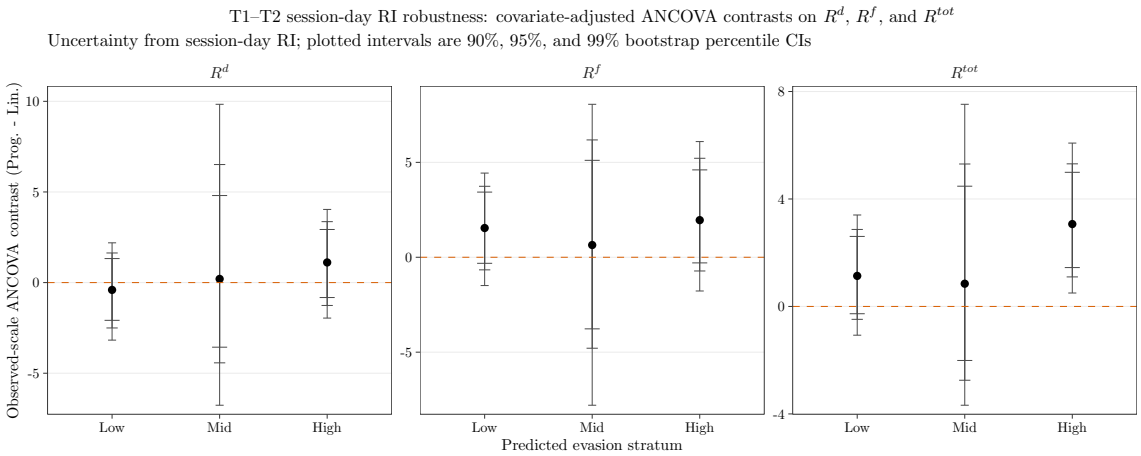

Figure 7: T1–T2 grouped RI robustness with session-day clustering.

Table 5 and Figure 7 show that RI evidence is concentrated in High R^{tot}. In Panel C, $\widehat{\beta}_{\text{High},R^{tot}} = 3.066$, $CI_{95} = [1.100, 5.308]$, with cellwise $p_{RI} = 0.039$. Low and Mid R^{tot} are smaller and imprecise ($\widehat{\beta}_{\text{Low},R^{tot}} = 1.137$, $p_{RI} = 0.167$; $\widehat{\beta}_{\text{Mid},R^{tot}} = 0.847$, $p_{RI} = 0.597$). In Panels A and B, High R^d and High R^f remain positive but imprecise ($\widehat{\beta}_{\text{High},R^d} = 1.111$, $p_{RI} = 0.479$; $\widehat{\beta}_{\text{High},R^f} = 1.956$, $p_{RI} = 0.185$). All grouped cells complete 9,999 valid RI draws with zero RI failures.

For multiplicity-aware interpretation, the High-R^{tot} cell does not survive Holm/BH adjustment across the grouped-cell family ($p_{RI}^{\text{Holm}} = 0.350$, $p_{RI}^{\text{BH}} = 0.350$). In addition, the Mid stratum has limited treated support in T1–T2 ($N_{T1} = 9$, $N_{T2} = 20$; threshold flag), which is consistent with wide intervals in Mid cells. The implication is therefore targeted: under assignment-concordant RI, the only economically large and individually informative T1–T2 signal is High-stratum total revenue, while broader component-level or across-strata claims are not supported.

6.2.2 T3–T4 temporal-control ladder robustness

For T3/T4, identification is within-subject, and the central inferential risk is temporal confounding around the regime switch. The baseline main-text pooled T3/T4 estimator uses round fixed effects, subject-level covariates, and random intercepts (Subsection 4.2). The temporal-control ladder keeps outcomes, grouping, censoring, and covariates fixed and perturbs only the time-control component.

Temporal-control ladder construction. The sample uses T3 and T4 only. Outcomes are R^d, R^f, and R^{tot}, with the usual censoring bounds. Grouping follows the baseline external-ML partition. Define

$$Post_{it} = \mathbf{1}\{round \geq 11\}, \qquad \mathbf{1}\{T3_i\} \times Post_{it} \text{ (sequence-by-post interaction).}$$

Let $\mathbf{Z}_i$ collect standard subject-level covariates. For each outcome and group, we estimate

$$y_{it}^* = \eta_{it}^{(S)} + u_i + \varepsilon_{it}, \qquad u_i \sim \mathcal{N}(0, \tau_u^2),\ \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2),$$

with Tobit censoring at outcome-specific bounds. The seven specifications are:

$$
\begin{aligned}
\text{S1:}\quad & \eta_{it}^{(1)} = \alpha + \beta_1 \mathbf{1}\{T3_i\} + \beta_2 Post_{it} + \beta_3\big(\mathbf{1}\{T3_i\} \times Post_{it}\big) + \boldsymbol{\gamma}'\mathbf{Z}_i, \\
\text{S2:}\quad & \eta_{it}^{(2)} = \eta_{it}^{(1)} + \lambda\, xround_{it}, \\
\text{S3:}\quad & \eta_{it}^{(3)} = \eta_{it}^{(2)} + \kappa_1(xround_{it} \times Post_{it}) + \kappa_2\big(xround_{it} \times \mathbf{1}\{T3_i\}\big), \\
\text{S4:}\quad & \eta_{it}^{(4)} = \alpha + \beta_1 \mathbf{1}\{T3_i\} + \beta_3\big(\mathbf{1}\{T3_i\} \times Post_{it}\big) + \sum_r \delta_r \mathbf{1}\{round = r\} + \boldsymbol{\gamma}'\mathbf{Z}_i, \\
\text{S5:}\quad & \eta_{it}^{(5)} = \eta_{it}^{(4)} \text{ estimated on rounds } r \in \{8, \ldots, 13\}, \\
\text{S6:}\quad & \eta_{it}^{(6)} = \alpha_i + \beta_2 Post_{it} + \beta_3\big(\mathbf{1}\{T3_i\} \times Post_{it}\big) + \lambda\, xround_{it}, \\
\text{S7:}\quad & \eta_{it}^{(7)} = \alpha_i + \beta_3\big(\mathbf{1}\{T3_i\} \times Post_{it}\big) + \sum_r \delta_r \mathbf{1}\{round = r\},
\end{aligned}
$$

where α_i denotes an individual (subject) fixed effect. We refer to the seven steps as: S1 (Block DiD), S2 (+ Linear Trend), S3 (+ Trend Interactions), S4 (Baseline (Round FE)), S5 (Local Round FE (r8–13)), S6 (Individual FE + Linear Time), and S7 (Baseline + Individual FE). S1 removes round fixed effects and uses only block structure; S2 adds a common linear trend; S3 adds flexible linear drift interactions by sequence and post period. S4 matches the main pooled T3/T4 temporal-control specification. S5 (Local Round FE (r8–13)) is the same round-FE structure as S4, estimated only on rounds 8–13, so DiD identification rests entirely on within-window variation near the switch. S5 targets the main temporal-confounding concern directly—learning or fatigue that accumulates far from the switch—at the cost of lower precision. S6 (Individual FE + Linear Time) combines individual fixed effects with a parsimonious linear time trend: α_i absorbs all time-invariant subject heterogeneity (subsuming $\mathbf{Z}_i$ and $\mathbf{1}\{T3_i\}$), while the linear trend captures smooth time dynamics without saturating the round dimension. Because S6 does not include round fixed effects, both pooled and sequence-specific contrasts remain identified, placing it in the same estimand class as S1–S3 but under individual-FE subject controls. S7 (Baseline + Individual FE) replaces the random intercept and observed subject-level covariates with individual fixed effects plus round fixed effects, absorbing all time-invariant subject heterogeneity non-parametrically. Because round fixed effects are present in S4, S5, and S7, standalone sequence-level contrasts are not identified in those steps. S7 uses the full sample (unlike S5) and therefore complements the local-window test with a full-horizon robustness check under the strictest available FE control structure.

As in Subsection 4.2, the reported observed-scale object depends on the time controls. In S1–S3 and S6, we report the pooled contrast $0.5[(Prog - Lin)_{T3} + (Prog - Lin)_{T4}]$. In S4, S5, and S7, round fixed effects absorb sequence-level contrasts, so the identified

object is the T3/T4 DiD, $(Prog - Lin)_{T3} + (Prog - Lin)_{T4}$. Standalone T3 and T4 levels are therefore not identified in S4, S5, and S7.

The main grouped temporal-ladder results are reported in Figure 8, with cell-level details in Table 27. The pattern is stable. High- and Low-stratum R^{tot} effects remain positive across the ladder, whereas Mid stays near zero and uninformative. Among the non-round-FE specifications, S6 is the key comparison because it replaces the baseline random intercept with individual fixed effects while relying only on a linear time control. Even under that stricter subject-control structure, the pooled R^{tot} contrast stays positive and distinguishable from zero for High (1.874, $CI_{95} = [0.210, 3.695]$) and Low (1.606, $CI_{95} = [0.148, 3.148]$).

Under round fixed effects, the estimand is the DiD contrast. High R^{tot} remains positive and distinguishable from zero in S4 (Baseline; 3.984, $CI_{95} = [0.429, 7.648]$), S5 (Local, rounds 8–13; 8.029, $CI_{95} = [0.869, 15.252]$), and S7 (Baseline + Individual FE; 3.913, $CI_{95} = [0.409, 7.797]$). Low R^{tot} shows the same pattern in S4 (3.310, $CI_{95} = [0.394, 6.534]$) and S7 (3.278, $CI_{95} = [0.301, 6.461]$); in S5 the estimate remains positive (5.087) but the restricted window is less precise ($CI_{95} = [-0.453, 10.991]$). At the component level, the R^d/R^f decomposition remains imprecise across the ladder, so the robust result concerns total revenue rather than its internal allocation.

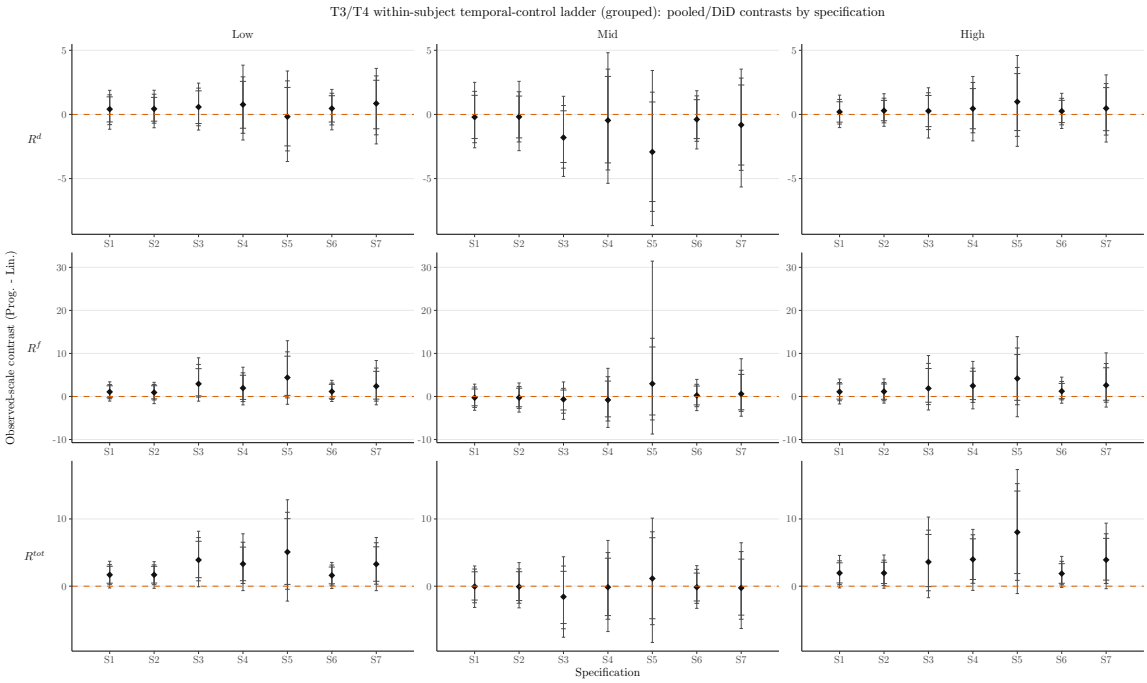

Figure 8: T3/T4 within-subject grouped temporal-control ladder (main robustness estimand). S1–S3 and S6 report pooled contrasts (Prog. − Lin.); S4, S5, and S7 report the round-FE identified DiD contrast. Error bars show nested 90%, 95%, and 99% confidence intervals.

Two comparisons matter most for interpretation. First, S7 is stricter than the baseline S4 because it replaces the random intercept and observed covariates with subject fixed effects. The fact that S7 closely tracks S4 therefore strengthens the baseline result. Second, S6 shows that the positive pooled pattern survives even when round fixed effects are replaced by a much less saturated linear time control. By contrast, S5 is best read as a local-window stress test: its larger High estimate is suggestive, but the overlap with S4 and S7 is too wide to support a separate near-switch effect-size claim.

Figure 9, with details in Table 28, leads to the same sequence-specific reading as in Section 5. The pooled signal continues to load mainly on T4. For High R^{tot}, the T4 contrast remains directionally positive across S1–S3 and S6, whereas T3 is weaker and changes sign across specifications. For Low R^{tot}, the positive evidence again comes mainly from T4, with weaker precision in some steps. The ladder therefore does not overturn the baseline sequence asymmetry; it reinforces the view that the pooled High- and Low-group results are driven mainly by T4, while single-sequence estimates remain too imprecise for standalone claims.

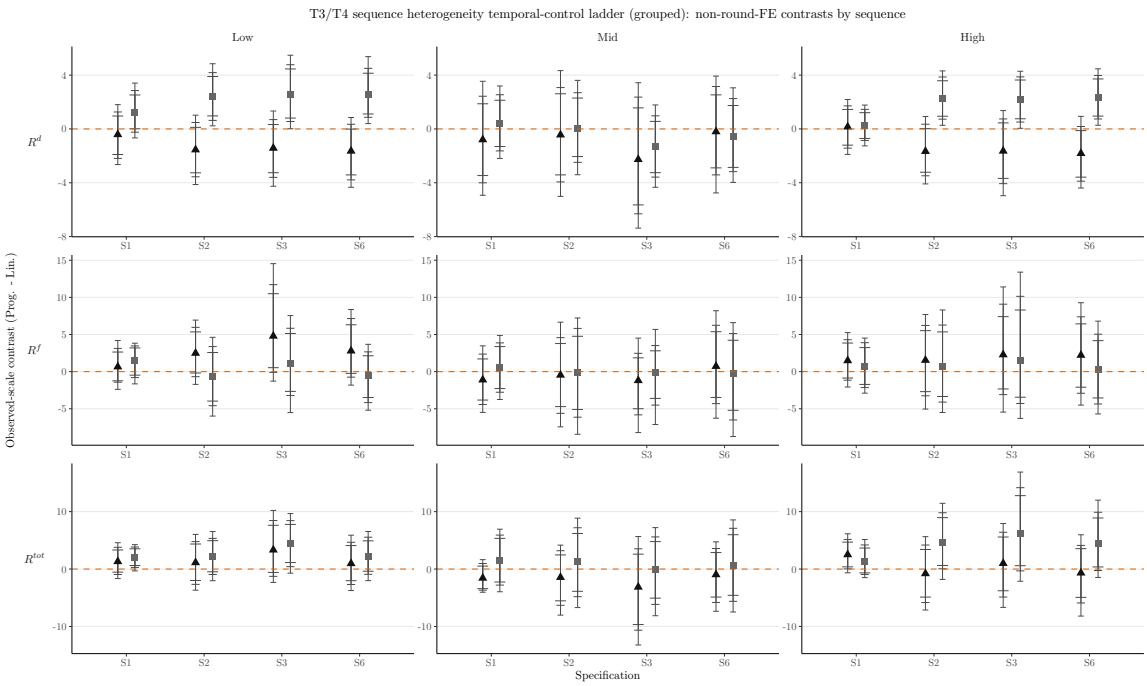

Figure 9: T3/T4 sequence-specific grouped temporal-control ladder (heterogeneity diagnostic). Reported series are T3 and T4 sequence contrasts for S1–S3 and S6 (Prog. − Lin.). Under S4, S5, and S7 round fixed effects, standalone T3/T4 level contrasts are not identified and are therefore omitted. Error bars show nested 90%, 95%, and 99% confidence intervals.

Aggregate results point in the same direction. The aggregate pooled/DiD R^{tot} contrast remains positive across the temporal ladder, and the aggregate sequence diagnostics again

indicate a stronger T4 than T3 contribution. Full aggregate tables and figures are reported in Appendix Subsection A.5.5.

Overall, richer time controls do not overturn the main T3/T4 result. High R^{tot} remains positive throughout the ladder; Low R^{tot} also remains positive and is distinguishable from zero in the baseline and in the stricter S6 and S7 specifications; Mid is uninformative throughout. The temporal ladder also leaves the baseline sequence-specific reading intact: the pooled signal is still primarily associated with T4, but sequence-specific estimates remain too imprecise to carry standalone causal weight. Support constraints are satisfied across all reported cells (minimum 48 observations and 16 subjects per Seq×Post cell in every reported specification), and S4, S5, and S7 sequence-specific levels are not identified by construction.

7 Conclusion

Progressive sanctions are common in real-world deterrence systems, but evidence on the shape of the sanction schedule remains limited. This paper combines a simple model with experimental evidence to study that question in a tax-evasion setting. The model implies that moving from a linear to a progressive fine schedule should weakly increase declared-tax and total revenue for high-evasion types, while effects for low- and mid-evasion types remain ambiguous because they depend on bracket location and threshold effects.

The empirical evidence supports that central upper-tail prediction. Using externally predicted evasion strata, we find that progressive fines raise total revenue most clearly among predicted high evaders, and this result appears in both the between-subject comparison and the pooled within-subject design. Aggregate total revenue is also higher under progressivity. In the pooled within-subject design, we also observe a positive total-revenue estimate for predicted low evaders, but this pattern is more sequence-sensitive and is not confirmed in the between-subject comparison. By contrast, the separate declaration- and fine-revenue components are less precisely estimated, so the most robust conclusion concerns total revenue rather than its internal decomposition.

These findings remain stable across a broad set of robustness checks. The main grouped results survive alternative first-stage partitions, assignment-uncertainty perturbations, session-day randomization inference for T1–T2, and richer temporal controls for T3–T4. The temporal-control ladder also leaves the baseline within-subject interpretation intact: the pooled signal remains concentrated in T4, while sequence-specific estimates are too imprecise to carry standalone causal weight.

As with any laboratory evidence, external validity requires care. The experiment is conducted with student subjects, moderate stakes, and bounded losses, so the magnitudes documented here should be interpreted as evidence on the direction and structure of the response to progressivity rather than as field-ready policy elasticities. More broadly, the paper is most persuasive as evidence on sanction-shape design and the differential incidence of revenue effects across the evasion distribution, rather than as a complete account of the behavioral mechanisms at work**[Note:** Antoine has any ideas on this?**]**. The extent to which these effects carry over to actual tax environments remains an open question, because the salience of punishment, the size of liabilities, and the composition of taxpayers are likely to shape both compliance responses and revenue effects.

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A.1 Experiment

A.1.1 Instructions

GENERAL

This is an experiment in which you will earn "real" money. The total amount you will earn depends on your decisions.

The experiment consists of 2 independent parts. You will receive the proposition for part 2 after the end of part 1. At the end of the experiment, your earning will be the sum of what you earn in part 1 and part 2. All your decisions will be anonymous.

Thank you for your participation.

PART 1

This part consists of 20 periods with identical course of action. The currency used during the experiment is ECU (Experimental Currency Units). Your earning for this part will be converted into euro at the exchange rate of

25 ECU = 1 euro

At the end of part 1, 2 periods will be randomly drawn and your decision for those two periods will be used for your payments of part 1.

DECISIONS

This experiment is a **tax simulation**.

At the beginning of each period you will receive an initial income equal to 125 ECU. After receiving your income, you will be asked to declare it for tax purposes. Only declared income will be taxed. The tax rate is equal to 20% and will be the same in all the periods.

CALCULATION OF PROFITS

Your profit in each round will be equal to:

Your income − (the declared income × 0,2)

The total amount of tax collected during all the 20 periods of the experiment will be invested in a research fund to run more experiments. You will therefore not be getting anything back from those taxes.

Your *declared income* does not have to be equal your *initial income*. It can be equal or lower.

However, there is a 20% chance that you will be audited. The probability of an audit is fixed and does not change during all the 20 periods. It means that if there are 40 participants, 8 participants will be audited in each period.

If an audit occurs, then the computer will examine your declared income comparing it with your initial income. If there is a difference between the declared income and the initial income, you will pay a fine on the unpaid tax.

So, there are three possibilities:

- *No audit.*
  A message will inform you that you were not audited in that period.

- *An audit occurs and there is no difference between the declared income and the initial income.*
  A message will inform you that the audit did not entail any fine.

- *An audit occurs and there is a difference between the declared income and the initial income.*
  A message will inform you that the income declared is not equal to the initial income of that period. A fine will be applied to your earnings.

The fine is determined in the following way:

$$\textbf{Fine} = \text{q} \times \text{ unpaid tax}$$

q is the fine rate and can vary between 0 to 5 times the unpaid tax. [Linear: *It is equal to 3.25 and is fixed in each period.* Progressive-Linear: *Between periods 1 to 10, the fine rate varies between 0 and 5 times according to the income declared. Between periods 11 to 20, the fine is equal to 3.25.* Linear-Progressive: *Between periods 1 to 10, the fine rate is equal to q=3.25. Between periods 11 to 20, the fine rate varies between 0 and 5 times, according to the income declared.*]

Declared Income	**Level of *q***	**Fine**
125	q=0	0
124 - 100	q=1 [=*3.25*]	1 [*3.25*] ×unpaid tax
99 - 75	q=2 [=*3.25*]	2 [*3.25*] ×unpaid tax
74 - 50	q=3 [=*3.25*]	3 [*3.25*] ×unpaid tax
49 - 25	q=4 [=*3.25*]	4 [*3.25*] ×unpaid tax
24 - 0	q=5 [=*3.25*]	5 [*3.25*] ×unpaid tax

In this case your profit will be:

Your income − (the declared income × 0,2) − (q × unpaid tax)

Please complete the following questionnaire to verify the comprehension of the instructions of part 1 :

1. The income that I declare will be taxed at the rate of 20%.
2. In each period I will receive an income equal to 125 ECU.
3. The audit probability is fixed at 20%.
4. The fine can vary between 0 and 5 times the unpaid tax. [Linear: *The fine is equal to 3.25 and is fixed in each period.* Progressive-Linear: *Between periods 1 to 10, the fine varies between 0 and 5 times the unpaid tax. Between periods 11 to 20, the fine is equal to 3.25 the unpaid tax.* Linear-Progressive: *Between periods 1 to 10, the fine is equal to 3.25 the unpaid tax. Between periods 11 to 20, the fine varies between 0 and 5 times the unpaid tax.*]

PART 2

You have an endowment of 10 euros. You need to decide how much you invest in a risky option. You can choose how much to invest (in whole number) between 0 and 10, i.e. 0, 1, 2,..., 9, or 10.

The risky option is a coin toss game. The risky option will bring you 3 times the amount invested if heads is drawn and 0 if tails is drawn.

Your final gain is equal to the number of euros kept + the gain of your investment.

Examples

1. You invest 5 euros. Your final win will be 20 euros if heads are drawn (5 euros kept + 5×3 euros of your 5 euros investment in the risky option) and 5 euros if the pile is drawn (5 euros kept + 5×0 euro of your investment in the risky option).

2. You invest 0 euros, your gain will be 10 euros regardless of the outcome of the coin toss game.

3. You invest 10 euros. Your final win will be 30 euros if heads are drawn (0 euros kept + 10×3 euros of your investment in the risky option) and 0 euros if stack is drawn (0 euros kept + 10×0 euros of your investment in the risky option).

To avoid calculations, the table below directly reports the final gain according to the amount invested and the result of the draw.

Amount invested	Gain	
	if heads	if tails
0	10	10
1	12	9
2	14	8
3	16	7
4	18	6
5	20	5
6	22	4
7	24	3
8	26	2
9	28	1
10	30	0

Questions

1. Age
2. Gender
3. Year of study
4. Nationality
5. Field of study
6. Is this your first experience in a lab experiment?
7. From 1 (completely unwilling to take risks) to 10 (completely willing to take risks), are you generally a person who is completely willing to take risks, or completely unwilling?
8. How much do you spend each month (on average) in leisure/recreational activities? (Examples of expenses in leisure/recreational activities are: subscriptions such as Netflix and Spotify, money spent for your hobbies and in fitness/sport, money spent at coffee shops, at restaurants etc.)
9. How much do you save each month (on average)? (Please consider all the possible forms of savings: cash, savings account, deposits, etc.)

A.2 Model

A.2.1 Primitive Cutoff-Existence Conditions

To keep the main text readable, we collect primitive sufficient conditions for cutoff existence in one block. Let $\rho_L < \rho_H$ be two bounds of the relevant type domain. Require:

$$\begin{aligned} &M(0;\rho_L, f_{avg})\, M(0;\rho_H, f_{avg}) \le 0, \qquad M(Y;\rho_L, f_{avg})\, M(Y;\rho_H, f_{avg}) \le 0, \\ &M(0;\rho_L, f_1)\, M(0;\rho_H, f_1) \le 0, \qquad M(Y;\rho_L, f_5)\, M(Y;\rho_H, f_5) \le 0, \\ &M(B_k;\rho_L, f_k)\, M(B_k;\rho_H, f_k) \le 0, \qquad k = 1, \dots, 4. \end{aligned} \tag{22}$$

By continuity, (22) gives existence of each boundary and interior threshold root. With $M_\rho < 0$, each root is unique.

A.2.2 Local Threshold Attraction under Progressivity

Fix a threshold B_k with $k \in \{1, \dots, 4\}$. Suppose

$$M(B_k^-;\rho, f_k) \ge 0, \qquad M(B_k^+;\rho, f_{k+1}) \le 0.$$

Because utility is strictly concave on each adjacent flat segment, the left inequality implies that utility is weakly increasing up to B_k from the left, whereas the right inequality implies that utility is weakly decreasing immediately to the right. Hence B_k is locally optimal: nearby interior alternatives on either side of the threshold do not dominate it. If the type distribution has positive density on a neighborhood of types for which these inequalities hold, the local threshold selection appears as observable bunching at B_k. This is the local kink argument used in the main text.

A.2.3 Detailed Proof of Proposition 1

(i) Linear schedule. With $f(E) \equiv f_{avg}$, strict concavity follows from (5) evaluated at $\bar{f} = f_{avg}$, so $M_E < 0$ on $[0, Y]$. Hence there is a unique maximizer in $[0, Y]$. On the interior, the implicit-function theorem gives

$$E_\rho = -\frac{M_\rho}{M_E} < 0.$$

For flat-rate comparative statics in the fine level,

$$E_f = -\frac{M_f}{M_E} \le 0.$$

The corner cutoffs ρ^0_{lin} and ρ^Y_{lin} are defined by the boundary equalities $M(0;\rho^0_{lin}, f_{avg}) = 0$ and $M(Y;\rho^Y_{lin}, f_{avg}) = 0$. Since $M_E < 0$, we have $M(0;\rho, f_{avg}) \geq M(Y;\rho, f_{avg})$ for every ρ. Evaluating this inequality at $\rho = \rho^Y_{lin}$ gives $M(0;\rho^Y_{lin}, f_{avg}) \geq 0$, so the root at $E = 0$ must occur weakly later in ρ; hence $\rho^Y_{lin} \leq \rho^0_{lin}$. Because $M_\rho < 0$, the sign of $M(0;\rho, f_{avg})$ and $M(Y;\rho, f_{avg})$ partitions the full-evasion, interior, and full-compliance cases exactly as stated in Proposition 1(i).

(ii) Progressive global selection and local threshold attraction. Within each bracket, (5) with $\bar{f} = f_k$ implies strict concavity, hence at most one interior candidate E^I_k. Since $[0, Y]$ is compact and the schedule has finitely many thresholds, the global solution is obtained by finite comparison over $\mathcal{C}(\rho)$. If $M(B^-_k;\rho, f_k) \geq 0$ and $M(B^+_k;\rho, f_{k+1}) \leq 0$, utility is weakly increasing up to B_k from the left and weakly decreasing to the right, so B_k is locally optimal and can be globally selected. By continuity, these inequalities continue to hold on a neighborhood of such types. Observable bunching then follows once the type distribution places positive density on that neighborhood.

(iii) Type-space mapping under the progressive schedule. For $k = 1, \ldots, 4$, (6) defines the interior-threshold cutoffs $\rho^{B_k}_{prog}$. The sign-bracketing conditions in (22) guarantee existence, and $M_\rho < 0$ implies uniqueness. The same argument gives the corner cutoffs ρ^0_{prog} and ρ^Y_{prog}.

To order the cutoffs, define

$$\begin{aligned} H_0(\rho) &:= M(0;\rho, f_1), \\ H_k(\rho) &:= M(B_k;\rho, f_k), \quad k = 1, \ldots, 4, \\ H_5(\rho) &:= M(Y;\rho, f_5). \end{aligned}$$

Because $M_E < 0$ and $M_f \leq 0$ on flat segments, and because both thresholds and fine levels increase across brackets, we have

$$H_0(\rho) \geq H_1(\rho) \geq H_2(\rho) \geq H_3(\rho) \geq H_4(\rho) \geq H_5(\rho) \quad \text{for every } \rho.$$

Since each H_j is decreasing in ρ, their unique zeros must satisfy

$$\rho^0_{prog} \geq \rho^{B_1}_{prog} \geq \rho^{B_2}_{prog} \geq \rho^{B_3}_{prog} \geq \rho^{B_4}_{prog} \geq \rho^Y_{prog}.$$

Now fix ρ. If $\rho > \rho^{B_1}_{prog}$, then $M(B_1;\rho, f_1) < 0$, so the optimum cannot lie above B_1; hence $E^{prog}(\rho) \in [0, B_1]$. If, in addition, $\rho \geq \rho^0_{prog}$, then $M(0;\rho, f_1) \leq 0$ and the corner $E = 0$ is selected. For $k = 2, 3, 4$, if $\rho^{B_k}_{prog} < \rho \leq \rho^{B_{k-1}}_{prog}$, then

$$M(B_{k-1};\rho, f_{k-1}) \geq 0, \qquad M(B_k;\rho, f_k) < 0.$$

The first inequality implies that the optimum does not stop below B_{k-1}; the second implies that it does not cross above B_k. Hence $E^{prog}(\rho) \in (B_{k-1}, B_k]$. Finally, if $\rho \leq \rho_{prog}^{B_4}$, then $E^{prog}(\rho) \in (B_4, Y]$; and if $\rho \leq \rho_{prog}^{Y}$, then $M(Y; \rho, f_5) \geq 0$, so the corner $E = Y$ is selected. This proves the partition stated in Proposition 1(iii).

Upper-tail admissibility. The bounded upper-tail statement used in the main text is therefore immediate: the top corner $E = Y$ is admissible whenever $\rho \leq \rho_{prog}^{Y}$, that is, whenever the boundary condition $M(Y; \rho, f_5) \geq 0$ holds for sufficiently low-ρ types. ■

A.3 Empirical strategy and econometric implementation

Table 6: Hyperparameter search spaces for ML stratification (main specification)

Model family	Search space (paper-facing summary)	Number of configurations
Penalized multinomial logit	Penalty mix from ridge to lasso ($\alpha \in \{0, 0.025, \ldots, 1\}$); number of penalty values ($nlambda \in \{200, 400, 800\}$); minimum penalty ratio ($\lambda_{\min_ratio} \in \{10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}\}$); covariate standardization on/off	1230
Feed-forward neural network	Hidden-layer width ($size \in \{4, 8, 12, 16, 20, 24, 28, 32, 40, 48\}$); weight decay ($decay \in \{0, 10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 5 \cdot 10^{-3}, 10^{-2}, 5 \cdot 10^{-2}\}$); training iterations ($maxit \in \{400, 800, 1200, 1600\}$); skip connection on/off; initial-weight scale ($rang \in \{0.5, 0.7\}$)	1280
Probability forest	Candidate split-variable counts ($mtry$) set adaptively as $\{\lfloor\sqrt{p}\rfloor, \lfloor p/4\rfloor, \lfloor p/2\rfloor, \lfloor 3p/4\rfloor, p\}$ (deduplicated); minimum node size ($\{1, 2, 3, 5, 8, 12\}$); number of trees ($\{300, 500, 800, 1200, 1600, 2200\}$); sample fraction ($\{0.5, 0.65, 0.8, 0.9, 1.0\}$); split rule (`gini` or `extratrees`)	720 for the socio-demographic exogenous and Risk sets; 1440 for the Risk plus socio-demographic exogenous set
Gradient-boosted trees	Tree depth ($\{2, 3, 4, 5\}$); learning rate ($\eta \in \{0.03, 0.05, 0.10\}$); minimum child weight ($\{1, 3, 5\}$); row and column subsampling ($\{0.70, 0.85\}$, $\{0.70, 0.90\}$); split loss penalty ($\gamma \in \{0, 0.5\}$); boosting rounds ($\{200, 400\}$); L_2 and L_1 penalties ($\lambda = \{5\}$, $\alpha = \{0, 0.5\}$)	1152

Notes. The table reports pre-specified search spaces used to construct ML-based strata in the main specification. For each model family, all grid points were evaluated and ranked by out-of-fold classification criteria. The forest grid varies across feature sets only through p, the number of available predictors.

A.4 Results

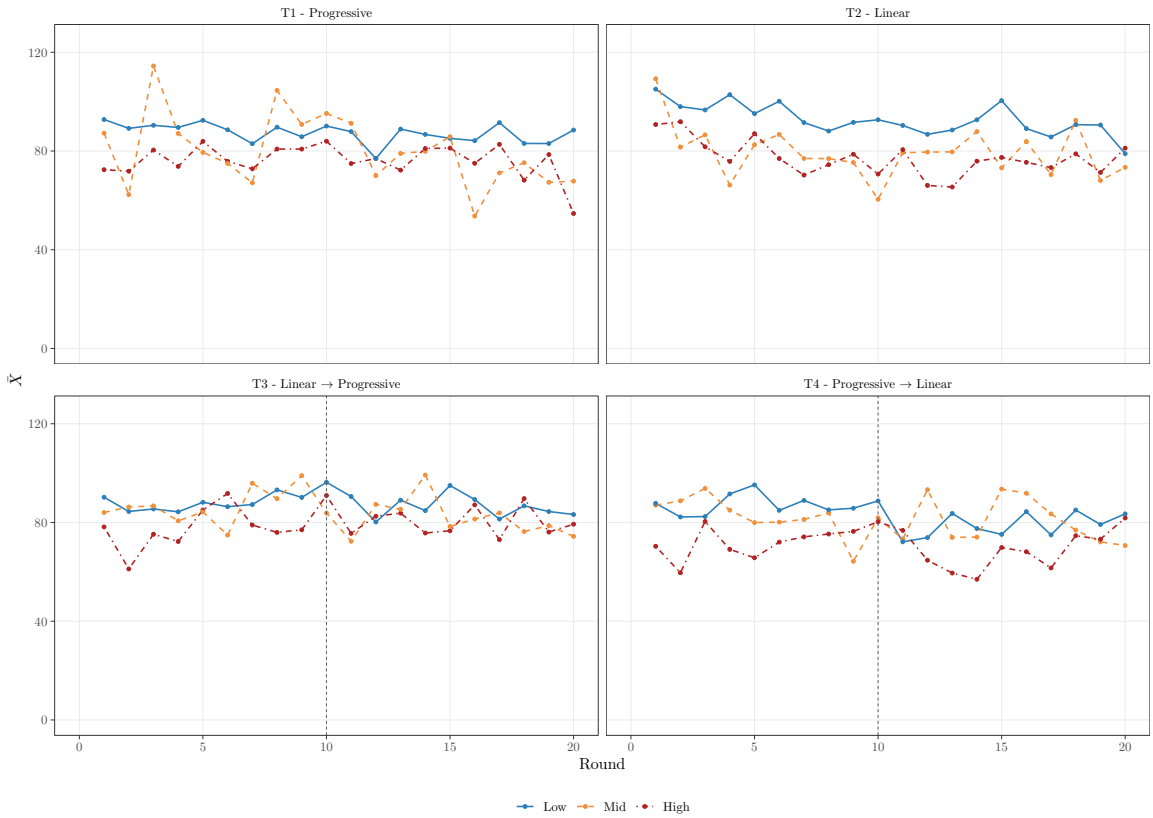

Figure 10: Average declaration by round, treatment, and predicted evasion stratum (group means).

Table 7: RE-Tobit latent coefficients for the T3/T4 within-subject specification (S4 round fixed effects), with standard covariates.

	R^d	R^f	R^{tot}
DiD (Mid baseline)	-0.232	0.040	0.597
	(1.587)	(0.261)	(2.346)
Sequence (T3 vs T4)	2.148	-12.479*	-0.968
	(1.713)	(7.196)	(1.041)
High	-1.625	-0.971	-0.845
	(2.410)	(5.029)	(1.275)
DiD × High	2.070	23.300**	4.066
	(1.782)	(11.556)	(2.489)
Low	1.392	-1.763	0.419
	(2.417)	(6.010)	(1.277)
DiD × Low	0.624	13.569	2.857
	(1.716)	(11.239)	(2.435)
Age	2.335	4.445	1.909
	(2.475)	(8.458)	(1.207)
Female	3.211*	-1.169	1.819**
	(1.683)	(6.064)	(0.818)
French	-9.891**	-3.563	-4.848**
	(4.792)	(16.087)	(2.270)
Study year	-1.124	2.296	-0.648
	(0.940)	(3.393)	(0.458)
Economics student	-1.978	3.151	-0.240
	(1.738)	(6.283)	(0.845)
Experience	-3.955**	1.299	-1.488*
	(1.807)	(6.474)	(0.875)
Constant	28.649***	-128.618***	22.791***
	(7.458)	(23.589)	(3.881)
sigma_u	11.714***	24.540***	3.225***
	(0.656)	(3.950)	(0.537)
sigma_e	12.513***	102.320***	21.961***
	(0.207)	(3.957)	(0.252)
N	4540	4540	4540
Round FE	Yes	Yes	Yes

Notes. Entries are latent-index coefficients from a pooled T3/T4 RE-Tobit model with S4 round fixed effects, group interactions, and standard covariates. Coefficients are reported with standard errors in parentheses. In this parametrization, `SeqPost` is the DiD coefficient for the Mid baseline group; interaction terms report differential DiD for Low/High. Round FE indicates whether round fixed effects are included in the regression. Inference uses two-sided normal-approximation p-values from reported standard errors; stars follow * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group assignment source: ml_main. Covariates included when available: expe, xage, female, study_year, econ, french.

Table 8: RE-Tobit latent coefficients by predicted evasion stratum: T3 (Lin. → Prog.) and T4 (Prog. → Lin.), contrast *Prog. - Lin.*

	T3 (Lin. → Prog.)			T4 (Prog. → Lin.)		
	R^d	R^f	R^{tot}	R^d	R^f	R^{tot}
	(1)	(2)	(3)	(4)	(5)	(6)
Progressive regime	-4.510**	-0.709	-1.936	2.882*	-0.784	5.286**
	(1.904)	(21.277)	(2.966)	(1.629)	(19.536)	(2.693)
High	0.205	-6.065	-1.627	-3.596	-1.578	-0.611
	(3.786)	(17.402)	(2.109)	(3.173)	(14.758)	(2.132)
Prog. × High	1.835	31.478	4.975*	0.160	3.493	-0.291
	(1.873)	(21.640)	(2.907)	(1.562)	(18.610)	(2.580)
Low	0.576	-3.922	-0.186	0.189	-3.267	0.185
	(3.846)	(17.468)	(2.105)	(3.187)	(14.908)	(2.147)
Prog. × Low	0.558	22.560	3.407	1.915	11.527	0.485
	(1.804)	(21.169)	(2.824)	(1.586)	(18.874)	(2.614)
Round	3.068***	-16.848	0.101	2.365**	-2.704	3.264**
	(1.107)	(12.477)	(1.671)	(0.964)	(11.266)	(1.580)
Age	0.339	11.709	1.775	2.658	-41.728*	1.847
	(2.856)	(9.609)	(1.199)	(5.444)	(24.314)	(3.004)
Female	5.589**	-2.841	3.288***	1.020	-3.039	0.545
	(2.491)	(9.189)	(1.035)	(2.224)	(8.156)	(1.229)
French	-11.222*	-13.783	-4.670*	-7.826	-21.747	-3.695
	(6.528)	(23.007)	(2.599)	(7.007)	(24.754)	(3.829)
Study year	0.622	0.185	-0.288	-2.759**	9.820*	-0.849
	(1.444)	(5.286)	(0.604)	(1.290)	(5.207)	(0.711)
Economics student	-5.303*	13.876	-0.317	1.779	-6.234	0.073
	(2.721)	(10.069)	(1.130)	(2.315)	(8.444)	(1.279)
Experience	-3.562	-5.839	-1.869	-5.067**	6.741	-1.317
	(2.820)	(10.305)	(1.168)	(2.292)	(8.412)	(1.262)
Constant	29.993***	-130.845***	22.574***	26.657**	-34.507	17.415**
	(10.117)	(38.446)	(4.383)	(13.187)	(56.069)	(7.711)
sigma_u	11.697***	24.921***	1.287	11.215***	22.532***	4.037***
	(0.980)	(6.198)	(2.060)	(0.848)	(5.334)	(0.686)
sigma_e	12.838***	103.413***	21.784***	12.311***	101.762***	22.169***
	(0.319)	(6.004)	(0.371)	(0.272)	(5.286)	(0.348)
N	2100	2100	2100	2440	2440	2440

Notes. Entries are latent-index coefficients from covariate-adjusted random-intercept Tobit (RE-Tobit) models for outcomes R^d, R^f, and R^{tot}. Each coefficient row reports a point estimate and the following row reports the standard error in parentheses. Mid is the omitted category in interaction specifications. Inference uses two-sided normal-approximation p-values from reported standard errors; stars follow $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Grouped observed-scale contrasts with covariates. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject DiD contrast; Panels C and D report sequence-specific T3 and T4 diagnostics. Contrasts are reported as *Prog. - Lin.*.

	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
Panel A: T1 Progressive, T2 Linear — Contrast: Prog. - Lin.									
Shared column layout for Panels A–D: outcomes (R^d, R^f, R^{tot}) by group (Low, Mid, High).									
$\widehat{\beta}$	-0.404	0.202	1.111	1.541	0.645	1.956	1.137	0.847	3.066***
SE	(1.040)	(2.670)	(1.205)	(1.159)	(2.836)	(1.475)	(0.877)	(1.976)	(1.114)
p	0.658	0.926	0.369	0.171	0.796	0.163	0.202	0.606	0.005
CI_{99}	[-3.038, 2.303]	[-6.139, 9.543]	[-2.105, 4.065]	[-1.300, 4.488]	[-7.346, 8.450]	[-1.562, 6.255]	[-1.019, 3.412]	[-3.877, 6.644]	[0.318, 5.994]
CI_{95}	[-2.431, 1.611]	[-4.065, 6.122]	[-1.249, 3.448]	[-0.662, 3.897]	[-4.873, 6.261]	[-0.759, 4.997]	[-0.588, 2.866]	[-2.680, 5.068]	[0.952, 5.314]
CI_{90}	[-2.087, 1.308]	[-3.382, 4.818]	[-0.898, 3.066]	[-0.351, 3.509]	[-3.866, 5.178]	[-0.372, 4.495]	[-0.324, 2.570]	[-1.928, 4.422]	[1.309, 4.896]
Panel B: Pooled T3/T4 within-subject DiD contrast — Contrast: Prog. - Lin.									
$\widehat{\Delta}$	0.765	-0.465	0.454	1.952	-0.805	2.465	3.310**	-0.151	3.984**
SE	(1.118)	(2.030)	(0.970)	(1.699)	(2.570)	(2.053)	(1.545)	(2.617)	(1.845)
p	0.495	0.805	0.638	0.242	0.716	0.214	0.026	0.941	0.027
CI_{99}	[-1.993, 3.840]	[-5.373, 4.812]	[-2.062, 2.959]	[-1.958, 6.797]	[-7.190, 6.535]	[-2.874, 8.147]	[-0.659, 7.790]	[-6.746, 6.806]	[-0.625, 8.446]
CI_{95}	[-1.470, 2.929]	[-4.329, 3.532]	[-1.434, 2.491]	[-1.191, 5.490]	[-5.704, 4.590]	[-1.375, 6.575]	[0.394, 6.534]	[-4.916, 5.000]	[0.429, 7.648]
CI_{90}	[-1.078, 2.577]	[-3.777, 2.960]	[-1.117, 2.013]	[-0.683, 4.926]	[-4.735, 3.575]	[-0.723, 5.863]	[0.818, 5.813]	[-4.372, 4.163]	[0.997, 7.043]
Panel C: T3 Linear → Progressive — Contrast: Prog. - Lin.									
$\widehat{\Delta}$	-1.831*	-2.208	-1.282	2.273	-0.074	3.599*	1.242	-1.595	2.527
SE	(1.051)	(1.644)	(0.995)	(1.894)	(2.444)	(2.069)	(2.010)	(1.884)	(2.043)
p	0.085	0.187	0.187	0.138	0.794	0.072	0.537	0.412	0.222
CI_{99}	[-4.613, 0.800]	[-6.745, 2.224]	[-3.843, 1.112]	[-1.856, 8.577]	[-5.382, 7.963]	[-1.185, 9.551]	[-4.026, 6.400]	[-6.087, 3.586]	[-2.982, 7.566]
CI_{95}	[-3.992, 0.228]	[-5.497, 0.971]	[-3.284, 0.697]	[-0.844, 6.449]	[-4.046, 5.604]	[-0.245, 8.065]	[-2.605, 5.217]	[-4.974, 2.224]	[-1.768, 6.359]
CI_{90}	[-3.548, -0.071]	[-4.880, 0.459]	[-2.936, 0.332]	[-0.323, 5.905]	[-3.331, 4.529]	[0.464, 7.320]	[-2.038, 4.553]	[-4.514, 1.587]	[-1.003, 5.810]
Panel D: T4 Progressive → Linear — Contrast: Prog. - Lin.									
$\widehat{\Delta}$	2.384***	1.455	1.620*	1.257	-0.088	0.318	4.755***	4.335	4.040*
SE	(0.875)	(1.157)	(0.816)	(1.936)	(2.572)	(1.994)	(1.867)	(2.886)	(2.305)
p	0.008	0.220	0.054	0.467	0.944	0.776	0.010	0.158	0.061
CI_{99}	[0.077, 4.761]	[-1.401, 4.718]	[-0.374, 3.656]	[-3.967, 6.383]	[-6.673, 7.030]	[-4.239, 5.943]	[0.108, 9.644]	[-3.241, 11.462]	[-1.756, 10.076]
CI_{95}	[0.699, 4.129]	[-0.766, 3.794]	[-0.022, 3.234]	[-2.593, 5.044]	[-4.722, 5.475]	[-2.994, 4.599]	[1.102, 8.402]	[-1.769, 9.857]	[-0.149, 8.882]
CI_{90}	[0.973, 3.888]	[-0.430, 3.370]	[0.289, 2.966]	[-1.898, 4.411]	[-3.869, 4.670]	[-2.487, 3.993]	[1.720, 7.926]	[-0.634, 8.844]	[0.403, 7.848]

Notes. Within each panel and outcome, rows report the effect estimate, SE, p-value, and percentile confidence intervals (CI_{99}, CI_{95}, CI_{90}). Significance stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For Panel A, bootstrap SE and plus-one p-values are recomputed from the saved T1–T2 bootstrap array. For Panel B, T3/T4 SE and p-values are read from ladder bootstrap summaries. For Panels C and D, sequence-specific T3/T4 plus-one p-values are recomputed from saved T3 and T4 bootstrap arrays.

Table 10: Grouped observed-scale treatment contrasts by predicted evasion stratum. Panel A reports T1–T2 covariate-adjusted ANCOVA contrasts; Panels B and C report sequence-specific T3 and T4 contrasts from RE-Tobit used for sequence-level heterogeneity diagnostics. All panels report the contrast *Prog. - Lin.*.

Panel A: T1 Progressive, T2 Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\beta}$	-0.404	0.202	1.111	1.541	0.645	1.956	1.137	0.847	3.066***
SE	(1.040)	(2.670)	(1.205)	(1.159)	(2.836)	(1.475)	(0.877)	(1.976)	(1.114)
p	0.670	0.917	0.375	0.174	0.799	0.165	0.202	0.586	0.005
CI_{99}	[-3.038, 2.303]	[-6.139, 9.543]	[-2.105, 4.065]	[-1.300, 4.488]	[-7.346, 8.450]	[-1.562, 6.255]	[-1.019, 3.412]	[-3.877, 6.644]	[0.318, 5.994]
CI_{95}	[-2.431, 1.611]	[-4.065, 6.122]	[-1.249, 3.448]	[-0.662, 3.897]	[-4.873, 6.261]	[-0.759, 4.997]	[-0.588, 2.866]	[-2.680, 5.068]	[0.952, 5.314]
CI_{90}	[-2.087, 1.308]	[-3.382, 4.818]	[-0.898, 3.066]	[-0.351, 3.509]	[-3.866, 5.178]	[-0.372, 4.495]	[-0.324, 2.570]	[-1.928, 4.422]	[1.309, 4.896]
Panel B: T3 Linear → Progressive — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	-1.831*	-2.208	-1.282	2.273	-0.074	3.599*	1.242	-1.595	2.527
SE	(1.051)	(1.644)	(0.995)	(1.894)	(2.444)	(2.069)	(2.010)	(1.884)	(2.043)
p	0.085	0.187	0.187	0.138	0.794	0.072	0.537	0.412	0.222
CI_{99}	[-4.613, 0.800]	[-6.745, 2.224]	[-3.843, 1.112]	[-1.856, 8.577]	[-5.382, 7.963]	[-1.185, 9.551]	[-4.026, 6.400]	[-6.087, 3.586]	[-2.982, 7.566]
CI_{95}	[-3.992, 0.228]	[-5.497, 0.971]	[-3.284, 0.697]	[-0.844, 6.449]	[-4.046, 5.604]	[-0.245, 8.065]	[-2.605, 5.217]	[-4.974, 2.224]	[-1.768, 6.359]
CI_{90}	[-3.548, -0.071]	[-4.880, 0.459]	[-2.936, 0.332]	[-0.323, 5.905]	[-3.331, 4.529]	[0.464, 7.320]	[-2.038, 4.553]	[-4.514, 1.587]	[-1.003, 5.810]
Panel C: T4 Progressive → Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	2.384***	1.455	1.620*	1.257	-0.088	0.318	4.755***	4.335	4.040*
SE	(0.875)	(1.157)	(0.816)	(1.936)	(2.572)	(1.994)	(1.867)	(2.886)	(2.305)
p	0.008	0.220	0.054	0.467	0.944	0.776	0.010	0.158	0.061
CI_{99}	[0.077, 4.761]	[-1.401, 4.718]	[-0.374, 3.656]	[-3.967, 6.383]	[-6.673, 7.030]	[-4.239, 5.943]	[0.108, 9.644]	[-3.241, 11.462]	[-1.756, 10.076]
CI_{95}	[0.699, 4.129]	[-0.766, 3.794]	[-0.022, 3.234]	[-2.593, 5.044]	[-4.722, 5.475]	[-2.994, 4.599]	[1.102, 8.402]	[-1.769, 9.857]	[-0.149, 8.882]
CI_{90}	[0.973, 3.888]	[-0.430, 3.370]	[0.289, 2.966]	[-1.898, 4.411]	[-3.869, 4.670]	[-2.487, 3.993]	[1.720, 7.926]	[-0.634, 8.844]	[0.403, 7.848]

Notes. Panel A entries are grouped T1–T2 covariate-adjusted ANCOVA contrasts; Panels B and C entries are sequence-specific T3 and T4 contrasts from covariate-adjusted random-intercept Tobit (RE-Tobit) models. Within each panel and outcome, rows report the effect estimate ($\widehat{\beta}$ in Panel A; $\widehat{\Delta}$ in Panels B and C), bootstrap SE, two-sided bootstrap plus-one p-value, and percentile confidence intervals at 99%, 95%, and 90%. Significance stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Covariates are round, age, female, French, study year, economics student, and experience.

Table 11: Aggregate observed-scale contrasts with covariates (baseline specification). Columns report T1–T2 ANCOVA, pooled T3/T4 within-subject DiD, and sequence-specific T3 and T4 diagnostics. Contrasts are reported as *Prog. - Lin.*.

	T1 (Prog.) vs. T2 (Lin.)	Pooled T3/T4 DiD	T3 (Lin. → Prog.)	T4 (Prog. → Lin.)
	(1)	(2)	(3)	(4)
Baseline covariate specification				
R^d	0.355	0.369	-1.701**	1.890***
	(0.732)	(0.708)	(0.794)	(0.633)
	p=0.632	p=0.597	p=0.034	p=0.001
	CI_{99}=[-1.655, 2.150]	CI_{99}=[-1.384, 2.268]	CI_{99}=[-3.699, 0.297]	CI_{99}=[0.347, 3.479]
	CI_{95}=[-1.085, 1.788]	CI_{95}=[-1.036, 1.750]	CI_{95}=[-3.282, -0.151]	CI_{95}=[0.674, 3.113]
	CI_{90}=[-0.840, 1.557]	CI_{90}=[-0.791, 1.522]	CI_{90}=[-3.014, -0.374]	CI_{90}=[0.864, 2.912]
R^f	1.701**	1.778	2.916**	1.011
	(0.836)	(1.170)	(1.573)	(1.699)
	p=0.037	p=0.132	p=0.047	p=0.572
	CI_{99}=[-0.249, 3.869]	CI_{99}=[-1.273, 4.671]	CI_{99}=[-1.145, 7.292]	CI_{99}=[-3.008, 5.883]
	CI_{95}=[0.106, 3.353]	CI_{95}=[-0.510, 4.109]	CI_{95}=[0.042, 6.096]	CI_{95}=[-2.037, 4.301]
	CI_{90}=[0.339, 3.099]	CI_{90}=[-0.140, 3.684]	CI_{90}=[0.482, 5.603]	CI_{90}=[-1.673, 3.849]
R^{tot}	2.055***	3.043***	1.331	4.294**
	(0.622)	(1.138)	(1.682)	(1.885)
	p=0.001	p=0.010	p=0.435	p=0.019
	CI_{99}=[0.491, 3.731]	CI_{99}=[0.009, 6.303]	CI_{99}=[-3.337, 5.438]	CI_{99}=[-0.315, 9.446]
	CI_{95}=[0.847, 3.263]	CI_{95}=[0.932, 5.365]	CI_{95}=[-2.076, 4.562]	CI_{95}=[0.776, 7.929]
	CI_{90}=[1.059, 3.084]	CI_{90}=[1.160, 4.928]	CI_{90}=[-1.600, 4.060]	CI_{90}=[1.255, 7.450]

Notes. For each outcome and treatment contrast, rows report the effect estimate, SE, two-sided p-value, and percentile confidence intervals (CI_{99}, CI_{95}, CI_{90}). Significance stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For the T1–T2 column, bootstrap SE and p-values are read from the baseline aggregate CSV. For the pooled T3/T4 column, SE and p-values are read from ladder bootstrap summaries. For the sequence-specific T3 and T4 columns, SE and p-values are read from the baseline aggregate CSV.

Table 12: Aggregate observed-scale treatment contrasts with covariates. Panel A reports the T1–T2 ANCOVA contrast; Panels B and C report sequence-specific T3 and T4 contrasts used for sequence-level heterogeneity diagnostics. Contrasts are reported as *Prog. - Lin.*.

Panel A: T1 Progressive, T2 Linear — Contrast: Prog. - Lin.			
	R^d	R^f	R^{tot}
$\widehat{\beta}$	0.355	1.701**	2.055***
SE	(0.732)	(0.836)	(0.622)
p	0.632	0.037	0.001
CI_{99}	[-1.655, 2.150]	[-0.249, 3.869]	[0.491, 3.731]
CI_{95}	[-1.085, 1.788]	[0.106, 3.353]	[0.847, 3.263]
CI_{90}	[-0.840, 1.557]	[0.339, 3.099]	[1.059, 3.084]
Panel B: T3 Linear → Progressive — Contrast: Prog. - Lin.			
	R^d	R^f	R^{tot}
$\widehat{\Delta}$	-1.701**	2.916**	1.331
SE	(0.794)	(1.573)	(1.682)
p	0.034	0.047	0.435
CI_{99}	[-3.699, 0.297]	[-1.145, 7.292]	[-3.337, 5.438]
CI_{95}	[-3.282, -0.151]	[0.042, 6.096]	[-2.076, 4.562]
CI_{90}	[-3.014, -0.374]	[0.482, 5.603]	[-1.600, 4.060]
Panel C: T4 Progressive → Linear — Contrast: Prog. - Lin.			
	R^d	R^f	R^{tot}
$\widehat{\Delta}$	1.890***	1.011	4.294**
SE	(0.633)	(1.699)	(1.885)
p	0.001	0.572	0.019
CI_{99}	[0.347, 3.479]	[-3.008, 5.883]	[-0.315, 9.446]
CI_{95}	[0.674, 3.113]	[-2.037, 4.301]	[0.776, 7.929]
CI_{90}	[0.864, 2.912]	[-1.673, 3.849]	[1.255, 7.450]

Notes. Panel A entries are aggregate T1–T2 covariate-adjusted ANCOVA contrasts; Panels B and C entries are sequence-specific T3 and T4 contrasts from covariate-adjusted random-intercept Tobit models (round, age, female, French, study year, economics student, experience). Within each treatment contrast panel, rows report the effect estimate ($\widehat{\beta}$ in Panel A; $\widehat{\Delta}$ in Panels B and C), SE, two-sided bootstrap p-value, and percentile intervals CI_{99}, CI_{95}, CI_{90}. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Covariate-ladder robustness (grouped OS-APEs): pooled T3/T4 within-subject DiD contrast for R^d. Contrast: Prog. - Lin. Columns are ladder rungs grouped by covariate-affinity batches.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Low	0.832	0.833	0.833	0.763	0.770	0.766	0.764	0.769	0.769	0.773
SE	(1.139)	(1.137)	(1.139)	(1.144)	(1.150)	(1.149)	(1.152)	(1.153)	(1.153)	(1.156)
p	0.462	0.460	0.459	0.501	0.498	0.497	0.500	0.498	0.502	0.497
CI_{99}	[-2.156, 3.544]	[-2.136, 3.525]	[-2.130, 3.535]	[-2.358, 3.592]	[-2.384, 3.601]	[-2.354, 3.582]	[-2.388, 3.557]	[-2.392, 3.581]	[-2.405, 3.579]	[-2.415, 3.605]
CI_{95}	[-1.536, 3.004]	[-1.527, 3.005]	[-1.539, 3.005]	[-1.590, 2.931]	[-1.602, 2.938]	[-1.609, 2.942]	[-1.589, 2.970]	[-1.598, 2.961]	[-1.597, 2.961]	[-1.592, 2.966]
CI_{90}	[-1.086, 2.618]	[-1.100, 2.614]	[-1.089, 2.630]	[-1.173, 2.586]	[-1.166, 2.622]	[-1.200, 2.605]	[-1.189, 2.600]	[-1.174, 2.611]	[-1.173, 2.613]	[-1.190, 2.630]
Mid	-0.812	-0.805	-0.816	-0.437	-0.441	-0.443	-0.437	-0.440	-0.439	-0.420
SE	(1.878)	(1.867)	(1.891)	(1.987)	(1.989)	(1.991)	(1.993)	(1.989)	(1.985)	(1.964)
p	0.682	0.681	0.681	0.806	0.803	0.808	0.809	0.805	0.806	0.805
CI_{99}	[-5.477, 3.733]	[-5.417, 3.698]	[-5.455, 3.813]	[-5.329, 4.900]	[-5.366, 4.854]	[-5.440, 4.818]	[-5.409, 4.937]	[-5.370, 4.940]	[-5.352, 4.897]	[-5.213, 5.004]
CI_{95}	[-4.351, 2.843]	[-4.325, 2.813]	[-4.311, 2.820]	[-4.398, 3.580]	[-4.441, 3.568]	[-4.480, 3.590]	[-4.421, 3.579]	[-4.392, 3.587]	[-4.391, 3.581]	[-4.200, 3.568]
CI_{90}	[-3.913, 2.282]	[-3.862, 2.272]	[-3.926, 2.300]	[-3.704, 2.901]	[-3.756, 2.905]	[-3.782, 2.886]	[-3.780, 2.911]	[-3.769, 2.899]	[-3.772, 2.871]	[-3.668, 2.920]
High	0.421	0.419	0.423	0.370	0.371	0.370	0.375	0.376	0.376	0.378
SE	(0.953)	(0.946)	(0.949)	(0.961)	(0.960)	(0.961)	(0.964)	(0.968)	(0.968)	(0.973)
p	0.653	0.658	0.648	0.703	0.699	0.704	0.696	0.697	0.694	0.697
CI_{99}	[-1.963, 3.036]	[-1.941, 2.954]	[-1.924, 3.033]	[-1.989, 2.927]	[-1.984, 2.919]	[-2.005, 2.915]	[-2.007, 2.931]	[-2.023, 2.962]	[-2.007, 2.965]	[-2.018, 2.978]
CI_{95}	[-1.452, 2.208]	[-1.450, 2.207]	[-1.466, 2.201]	[-1.491, 2.309]	[-1.499, 2.280]	[-1.488, 2.317]	[-1.501, 2.319]	[-1.497, 2.310]	[-1.499, 2.322]	[-1.498, 2.322]
CI_{90}	[-1.105, 1.962]	[-1.101, 1.938]	[-1.126, 1.935]	[-1.189, 1.921]	[-1.177, 1.909]	[-1.188, 1.907]	[-1.200, 1.931]	[-1.205, 1.947]	[-1.212, 1.939]	[-1.196, 1.945]

Notes. Entries are OS-APEs on observed outcomes from pooled T3/T4 within-subject S4 DiD estimates by predicted evasion stratum. Within each stratum block, rows report the effect estimate, bootstrap SE (in parentheses), two-sided bootstrap plus-one p-value, and percentile confidence intervals CI_{99}, CI_{95}, and CI_{90}. Stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (1)–(10) follow the manifest rung order.

Table 14: Covariate-ladder robustness (grouped OS-APEs): pooled T3/T4 within-subject DiD contrast for R^f. Contrast: Prog. - Lin. Columns are ladder rungs grouped by covariate-affinity batches.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Low	1.790	1.768	1.667	1.932	1.969	1.973	2.036	2.049	2.069	2.057
SE	(1.813)	(1.794)	(1.721)	(1.702)	(1.718)	(1.709)	(1.685)	(1.695)	(1.699)	(1.691)
p	0.317	0.320	0.333	0.240	0.239	0.229	0.204	0.195	0.201	0.199
CI_{99}	[-2.505, 7.025]	[-2.521, 6.832]	[-2.331, 6.536]	[-1.998, 6.944]	[-1.986, 7.000]	[-1.832, 6.921]	[-1.965, 6.850]	[-1.969, 6.938]	[-1.956, 7.015]	[-1.931, 6.809]
CI_{95}	[-1.609, 5.464]	[-1.595, 5.364]	[-1.527, 5.218]	[-1.171, 5.546]	[-1.158, 5.651]	[-1.151, 5.633]	[-1.170, 5.503]	[-1.199, 5.602]	[-1.105, 5.641]	[-1.113, 5.722]
CI_{90}	[-1.193, 4.882]	[-1.119, 4.831]	[-1.021, 4.626]	[-0.687, 4.922]	[-0.670, 4.974]	[-0.667, 4.993]	[-0.625, 4.962]	[-0.605, 5.004]	[-0.607, 5.041]	[-0.580, 4.958]
Mid	0.043	0.039	0.010	-0.937	-0.958	-0.922	-0.799	-0.818	-0.817	-0.822
SE	(2.379)	(2.404)	(2.384)	(2.334)	(2.424)	(2.466)	(2.468)	(2.486)	(2.487)	(2.408)
p	0.980	0.971	0.975	0.674	0.679	0.692	0.740	0.730	0.723	0.727
CI_{99}	[-6.253, 6.699]	[-6.092, 6.688]	[-6.168, 6.613]	[-6.808, 5.306]	[-7.516, 5.634]	[-7.455, 5.761]	[-7.056, 6.302]	[-7.170, 5.803]	[-7.430, 6.048]	[-7.017, 5.928]
CI_{95}	[-4.567, 4.713]	[-4.514, 4.858]	[-4.451, 4.663]	[-5.367, 3.859]	[-5.477, 3.963]	[-5.580, 4.137]	[-5.518, 4.286]	[-5.635, 4.208]	[-5.641, 4.259]	[-5.535, 3.928]
CI_{90}	[-3.850, 3.948]	[-3.914, 4.003]	[-3.937, 3.854]	[-4.722, 2.846]	[-4.955, 3.071]	[-4.953, 3.186]	[-4.786, 3.283]	[-4.947, 3.282]	[-4.874, 3.257]	[-4.855, 3.105]
High	1.972	1.980	2.226	2.383	2.403	2.423	2.532	2.498	2.504	2.537
SE	(1.721)	(1.740)	(1.912)	(1.981)	(1.996)	(2.008)	(2.055)	(2.085)	(2.089)	(2.117)
p	0.223	0.228	0.221	0.219	0.215	0.218	0.212	0.218	0.213	0.213
CI_{99}	[-2.257, 7.143]	[-2.306, 7.099]	[-2.851, 7.489]	[-2.718, 8.316]	[-2.624, 7.949]	[-2.775, 7.883]	[-2.685, 7.998]	[-2.892, 8.426]	[-2.893, 8.218]	[-2.996, 8.213]
CI_{95}	[-1.280, 5.691]	[-1.280, 5.740]	[-1.370, 6.239]	[-1.229, 6.436]	[-1.346, 6.453]	[-1.364, 6.538]	[-1.365, 6.758]	[-1.447, 6.760]	[-1.491, 6.741]	[-1.466, 6.856]
CI_{90}	[-0.700, 4.918]	[-0.797, 4.947]	[-0.828, 5.497]	[-0.705, 5.871]	[-0.820, 5.882]	[-0.715, 5.890]	[-0.770, 6.130]	[-0.772, 6.102]	[-0.768, 6.070]	[-0.812, 6.107]

Notes. Entries are OS-APEs on observed outcomes from pooled T3/T4 within-subject S4 DiD estimates by predicted evasion stratum. Within each stratum block, rows report the effect estimate, bootstrap SE (in parentheses), two-sided bootstrap plus-one p-value, and percentile confidence intervals CI_{99}, CI_{95}, and CI_{90}. Stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (1)–(10) follow the manifest rung order.

Table 15: Covariate-ladder robustness (grouped OS-APEs): pooled T3/T4 within-subject DiD contrast for R^{tot}. Contrast: Prog. - Lin. Columns are ladder rungs grouped by covariate-affinity batches.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Low	3.134**	3.136**	3.126**	3.334**	3.331**	3.339**	3.335**	3.336**	3.333**	3.331**
SE	(1.517)	(1.522)	(1.517)	(1.526)	(1.526)	(1.525)	(1.529)	(1.519)	(1.521)	(1.524)
p	0.038	0.035	0.036	0.026	0.024	0.024	0.023	0.024	0.023	0.025
CI_{99}	[-0.716, 7.083]	[-0.685, 7.069]	[-0.749, 7.051]	[-0.669, 7.431]	[-0.627, 7.427]	[-0.607, 7.468]	[-0.493, 7.365]	[-0.543, 7.429]	[-0.575, 7.416]	[-0.734, 7.377]
CI_{95}	[0.213, 6.252]	[0.247, 6.253]	[0.209, 6.217]	[0.458, 6.400]	[0.464, 6.418]	[0.474, 6.430]	[0.488, 6.448]	[0.483, 6.394]	[0.462, 6.404]	[0.471, 6.407]
CI_{90}	[0.668, 5.658]	[0.629, 5.669]	[0.606, 5.631]	[0.811, 5.912]	[0.832, 5.928]	[0.830, 5.925]	[0.799, 5.935]	[0.812, 5.871]	[0.816, 5.876]	[0.819, 5.889]
Mid	-0.343	-0.336	-0.332	-0.112	-0.106	-0.101	-0.077	-0.085	-0.090	-0.126
SE	(2.520)	(2.523)	(2.518)	(2.567)	(2.566)	(2.565)	(2.575)	(2.580)	(2.581)	(2.579)
p	0.887	0.892	0.895	0.936	0.936	0.933	0.950	0.946	0.940	0.934
CI_{99}	[-6.520, 6.099]	[-6.461, 6.001]	[-6.519, 6.080]	[-6.150, 6.136]	[-6.186, 6.133]	[-6.202, 6.182]	[-6.137, 6.197]	[-6.177, 6.328]	[-6.142, 6.213]	[-6.226, 6.274]
CI_{95}	[-4.924, 4.892]	[-4.967, 4.901]	[-5.000, 4.889]	[-4.890, 5.139]	[-4.865, 5.130]	[-4.850, 5.169]	[-4.877, 5.241]	[-4.866, 5.244]	[-4.858, 5.260]	[-4.894, 5.125]
CI_{90}	[-4.376, 3.829]	[-4.370, 3.841]	[-4.325, 3.834]	[-4.141, 4.236]	[-4.179, 4.225]	[-4.137, 4.248]	[-4.115, 4.280]	[-4.119, 4.261]	[-4.116, 4.256]	[-4.130, 4.219]
High	3.867**	3.895**	3.884**	3.934**	3.931**	3.933**	3.901**	3.893**	3.895**	3.890**
SE	(1.843)	(1.842)	(1.829)	(1.824)	(1.825)	(1.820)	(1.825)	(1.823)	(1.822)	(1.823)
p	0.026	0.022	0.026	0.032	0.031	0.030	0.029	0.029	0.028	0.028
CI_{99}	[-0.398, 8.996]	[-0.277, 9.077]	[-0.292, 9.069]	[-0.469, 9.155]	[-0.414, 9.139]	[-0.390, 9.214]	[-0.406, 9.160]	[-0.402, 9.245]	[-0.397, 9.248]	[-0.418, 9.227]
CI_{95}	[0.401, 7.581]	[0.413, 7.577]	[0.456, 7.608]	[0.408, 7.603]	[0.406, 7.593]	[0.425, 7.587]	[0.429, 7.560]	[0.462, 7.512]	[0.440, 7.559]	[0.456, 7.541]
CI_{90}	[0.860, 7.004]	[0.898, 7.026]	[0.917, 6.922]	[0.971, 6.926]	[0.970, 6.921]	[0.999, 6.929]	[0.955, 6.940]	[0.994, 6.925]	[0.968, 6.929]	[0.985, 6.906]

Notes. Entries are OS-APEs on observed outcomes from pooled T3/T4 within-subject S4 DiD estimates by predicted evasion stratum. Within each stratum block, rows report the effect estimate, bootstrap SE (in parentheses), two-sided bootstrap plus-one p-value, and percentile confidence intervals CI_{99}, CI_{95}, and CI_{90}. Stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (1)–(10) follow the manifest rung order.

Table 16: T3 (Lin. → Prog.). Contrast: Prog. - Lin. Predicted evasion strata are based on the baseline ML prediction. Panels report R^d, R^f, and R^{tot}. Columns are cumulative covariate-ladder rungs.

	B0 Baseline	B1 Demographics			B2 Human Capital / Exposure			B3 Risk		B4 Financial Behavior	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A: R^d											
Low	-0.360	-1.586	-1.603	-1.570	-1.842*	-1.838*	-1.831*	-1.813*	-1.806*	-1.827*	-1.826*
SE	(0.852)	(1.031)	(1.030)	(1.033)	(1.036)	(1.036)	(1.038)	(1.038)	(1.038)	(1.039)	(1.039)
p	0.679	0.133	0.133	0.136	0.076	0.075	0.076	0.074	0.076	0.077	0.077
CI_{99}	[-2.624, 1.773]	[-4.246, 1.029]	[-4.240, 1.042]	[-4.283, 1.053]	[-4.611, 0.756]	[-4.610, 0.762]	[-4.611, 0.753]	[-4.600, 0.773]	[-4.597, 0.741]	[-4.597, 0.757]	[-4.589, 0.769]
CI_{95}	[-2.066, 1.290]	[-3.655, 0.483]	[-3.622, 0.464]	[-3.639, 0.477]	[-3.949, 0.182]	[-3.945, 0.178]	[-3.938, 0.184]	[-3.951, 0.178]	[-3.960, 0.180]	[-3.971, 0.192]	[-3.964, 0.183]
CI_{90}	[-1.809, 1.069]	[-3.271, 0.106]	[-3.257, 0.103]	[-3.263, 0.103]	[-3.524, -0.125]	[-3.525, -0.123]	[-3.527, -0.116]	[-3.525, -0.119]	[-3.531, -0.117]	[-3.532, -0.121]	[-3.534, -0.111]
Mid	-1.177	-2.456	-2.456	-2.481	-2.212	-2.225	-2.208	-2.242	-2.210	-2.212	-2.207
SE	(1.444)	(1.502)	(1.501)	(1.504)	(1.668)	(1.673)	(1.663)	(1.677)	(1.683)	(1.682)	(1.685)
p	0.410	0.105	0.107	0.106	0.199	0.196	0.195	0.200	0.200	0.200	0.198
CI_{99}	[-5.160, 2.754]	[-6.695, 1.400]	[-6.711, 1.398]	[-6.663, 1.397]	[-6.778, 2.225]	[-6.776, 2.214]	[-6.745, 2.224]	[-6.783, 2.215]	[-6.762, 2.213]	[-6.782, 2.212]	[-6.829, 2.239]
CI_{95}	[-4.154, 1.625]	[-5.512, 0.512]	[-5.501, 0.529]	[-5.534, 0.538]	[-5.580, 1.012]	[-5.560, 0.986]	[-5.525, 0.997]	[-5.560, 1.052]	[-5.582, 1.016]	[-5.582, 1.006]	[-5.553, 0.991]
CI_{90}	[-3.513, 1.244]	[-4.924, 0.008]	[-4.921, 0.045]	[-4.931, 0.041]	[-4.916, 0.489]	[-4.951, 0.483]	[-4.919, 0.491]	[-4.975, 0.476]	[-4.995, 0.495]	[-4.986, 0.498]	[-4.986, 0.499]
High	0.223	-1.061	-1.080	-1.036	-1.293	-1.284	-1.282	-1.263	-1.254	-1.280	-1.281
SE	(0.838)	(0.988)	(0.987)	(0.988)	(0.984)	(0.982)	(0.984)	(0.970)	(0.970)	(0.970)	(0.970)
p	0.797	0.281	0.284	0.282	0.185	0.186	0.186	0.187	0.187	0.187	0.184
CI_{99}	[-1.987, 2.382]	[-3.467, 1.443]	[-3.458, 1.426]	[-3.473, 1.435]	[-3.734, 1.131]	[-3.728, 1.115]	[-3.748, 1.112]	[-3.717, 1.092]	[-3.723, 1.086]	[-3.722, 1.087]	[-3.723, 1.082]
CI_{95}	[-1.499, 1.856]	[-2.936, 0.937]	[-2.934, 0.941]	[-2.954, 0.948]	[-3.213, 0.690]	[-3.217, 0.691]	[-3.227, 0.689]	[-3.184, 0.682]	[-3.205, 0.689]	[-3.205, 0.685]	[-3.207, 0.686]
CI_{90}	[-1.220, 1.563]	[-2.690, 0.608]	[-2.677, 0.621]	[-2.673, 0.608]	[-2.902, 0.310]	[-2.905, 0.318]	[-2.912, 0.330]	[-2.880, 0.310]	[-2.881, 0.315]	[-2.885, 0.320]	[-2.880, 0.315]
Panel B: R^f											
Low	0.373	2.461	2.283	2.368	2.405	2.540	2.273	2.367	2.620	2.592	2.595
SE	(1.210)	(1.807)	(1.786)	(1.856)	(1.899)	(1.879)	(1.885)	(1.886)	(1.844)	(1.838)	(1.823)
p	0.769	0.153	0.144	0.166	0.144	0.139	0.136	0.146	0.134	0.132	0.128
CI_{99}	[-2.546, 3.817]	[-1.974, 7.617]	[-1.888, 7.535]	[-2.050, 7.601]	[-2.098, 7.983]	[-2.178, 7.974]	[-2.224, 7.842]	[-2.201, 8.018]	[-2.055, 7.829]	[-2.052, 7.856]	[-2.057, 7.867]
CI_{95}	[-1.955, 2.748]	[-0.875, 6.146]	[-0.815, 6.099]	[-1.013, 6.248]	[-0.852, 6.485]	[-0.865, 6.441]	[-0.853, 6.368]	[-0.837, 6.386]	[-0.807, 6.398]	[-0.771, 6.398]	[-0.718, 6.315]
CI_{90}	[-1.540, 2.346]	[-0.393, 5.535]	[-0.277, 5.551]	[-0.484, 5.576]	[-0.267, 5.924]	[-0.272, 5.892]	[-0.311, 5.900]	[-0.304, 5.824]	[-0.204, 5.771]	[-0.216, 5.737]	[-0.211, 5.714]
Mid	-1.051	1.301	1.280	1.524	0.223	0.062	-0.074	-0.848	0.620	0.629	0.635
SE	(1.597)	(2.097)	(2.114)	(2.403)	(2.353)	(2.430)	(2.422)	(2.613)	(2.581)	(2.612)	(2.613)
p	0.462	0.446	0.434	0.429	0.715	0.776	0.780	0.892	0.804	0.794	0.787
CI_{99}	[-3.631, 5.657]	[-3.273, 8.347]	[-3.251, 8.645]	[-3.895, 9.213]	[-4.966, 7.841]	[-5.337, 8.037]	[-5.362, 8.251]	[-5.916, 8.228]	[-5.873, 8.111]	[-5.883, 8.186]	[-5.900, 8.283]
CI_{95}	[-2.609, 3.638]	[-2.067, 6.218]	[-2.050, 6.248]	[-2.769, 7.032]	[-3.692, 5.544]	[-3.881, 5.694]	[-3.907, 5.594]	[-4.369, 5.732]	[-4.037, 6.144]	[-4.049, 6.141]	[-4.073, 6.096]
CI_{90}	[-2.357, 2.764]	[-1.607, 5.228]	[-1.560, 5.302]	[-1.982, 5.923]	[-2.998, 4.651]	[-3.165, 4.635]	[-3.206, 4.529]	[-3.763, 4.623]	[-3.354, 4.905]	[-3.409, 5.113]	[-3.437, 5.135]
High	1.616	3.695*	3.900**	3.527*	3.866*	3.504*	3.599*	3.909*	3.968**	3.947**	3.953**
SE	(1.302)	(1.970)	(1.912)	(2.022)	(2.083)	(2.064)	(2.059)	(2.010)	(1.978)	(1.963)	(1.965)
p	0.281	0.055	0.043	0.069	0.061	0.067	0.068	0.052	0.045	0.044	0.046
CI_{99}	[-1.942, 4.869]	[-0.905, 9.123]	[-0.759, 9.083]	[-0.981, 9.342]	[-1.143, 9.631]	[-1.100, 9.588]	[-1.144, 9.498]	[-0.853, 9.564]	[-0.860, 9.580]	[-0.827, 9.512]	[-0.883, 9.505]
CI_{95}	[-1.095, 4.051]	[-0.026, 7.791]	[0.116, 7.702]	[-0.225, 7.821]	[-0.225, 8.210]	[-0.222, 8.145]	[-0.183, 8.080]	[-0.031, 8.041]	[0.055, 7.932]	[0.170, 7.876]	[0.169, 7.851]
CI_{90}	[-0.755, 3.437]	[0.539, 7.128]	[0.761, 6.996]	[0.372, 7.145]	[0.470, 7.372]	[0.509, 7.325]	[0.517, 7.320]	[0.672, 7.224]	[0.739, 7.180]	[0.858, 7.195]	[0.860, 7.197]
Panel C: R^{tot}											
Low	1.069	1.404	1.286	1.388	1.189	1.127	1.242	1.237	1.247	1.191	1.189
SE	(1.204)	(1.971)	(1.973)	(1.961)	(2.002)	(2.002)	(2.003)	(2.003)	(2.018)	(2.015)	(2.014)
p	0.351	0.486	0.493	0.492	0.541	0.546	0.535	0.538	0.542	0.546	0.538
CI_{99}	[-1.892, 4.624]	[-3.989, 6.354]	[-4.045, 6.379]	[-4.047, 6.262]	[-4.243, 6.230]	[-4.238, 6.145]	[-4.273, 6.242]	[-4.183, 6.229]	[-4.260, 6.286]	[-4.182, 6.298]	[-4.175, 6.238]
CI_{95}	[-1.094, 3.477]	[-2.450, 5.157]	[-2.459, 5.165]	[-2.416, 5.138]	[-2.649, 5.208]	[-2.664, 5.212]	[-2.641, 5.217]	[-2.683, 5.179]	[-2.686, 5.238]	[-2.696, 5.230]	[-2.702, 5.249]
CI_{90}	[-0.776, 3.071]	[-1.774, 4.620]	[-1.806, 4.601]	[-1.806, 4.572]	[-1.998, 4.485]	[-1.980, 4.501]	[-1.986, 4.523]	[-2.009, 4.517]	[-2.021, 4.531]	[-2.037, 4.498]	[-2.056, 4.518]
Mid	-1.648	-1.206	-1.627	-0.962	-1.821	-1.882	-1.595	-1.243	-1.363	-1.486	-1.450
SE	(1.076)	(1.759)	(1.767)	(1.747)	(1.885)	(1.879)	(1.887)	(1.914)	(1.945)	(1.857)	(1.855)
p	0.137	0.415	0.401	0.430	0.427	0.424	0.422	0.417	0.431	0.397	0.392
CI_{99}	[-4.014, 1.507]	[-5.738, 3.277]	[-5.857, 3.371]	[-5.843, 3.207]	[-6.092, 3.621]	[-6.220, 3.553]	[-6.087, 3.620]	[-6.234, 3.676]	[-6.287, 3.977]	[-6.250, 3.563]	[-6.251, 3.558]
CI_{95}	[-3.530, 0.677]	[-4.754, 1.995]	[-4.842, 2.021]	[-4.705, 2.017]	[-5.021, 2.258]	[-5.000, 2.260]	[-4.951, 2.293]	[-5.116, 2.235]	[-5.148, 2.404]	[-5.146, 2.011]	[-5.075, 2.048]
CI_{90}	[-3.255, 0.175]	[-4.255, 1.440]	[-4.315, 1.395]	[-4.250, 1.384]	[-4.509, 1.542]	[-4.519, 1.599]	[-4.513, 1.637]	[-4.540, 1.569]	[-4.564, 1.671]	[-4.553, 1.421]	[-4.529, 1.405]
High	2.558*	2.815	2.728	2.851	2.503	2.406	2.527	2.561	2.571	2.603	2.597
SE	(1.326)	(1.994)	(1.994)	(1.983)	(2.036)	(2.037)	(2.039)	(2.048)	(2.052)	(2.042)	(2.040)
p	0.051	0.169	0.170	0.172	0.219	0.216	0.214	0.220	0.220	0.218	0.217
CI_{99}	[-0.761, 5.889]	[-2.499, 7.467]	[-2.491, 7.473]	[-2.461, 7.455]	[-2.908, 7.512]	[-2.891, 7.489]	[-2.857, 7.487]	[-2.943, 7.388]	[-2.980, 7.499]	[-2.903, 7.408]	[-2.898, 7.397]
CI_{95}	[0.001, 5.200]	[-1.372, 6.499]	[-1.360, 6.491]	[-1.329, 6.523]	[-1.692, 6.422]	[-1.708, 6.408]	[-1.724, 6.359]	[-1.708, 6.419]	[-1.777, 6.391]	[-1.705, 6.400]	[-1.693, 6.429]
CI_{90}	[0.336, 4.662]	[-0.735, 6.017]	[-0.725, 5.988]	[-0.702, 5.954]	[-0.979, 5.805]	[-0.959, 5.821]	[-0.946, 5.810]	[-1.011, 5.821]	[-1.014, 5.875]	[-1.012, 5.855]	[-1.013, 5.857]

Notes. Entries are OS-APEs on observed outcomes R^d, R^f, and R^{tot} from RE-Tobit margins. Within each stratum block, rows report the effect estimate, bootstrap SE (in parentheses), two-sided bootstrap plus-one p-value, and percentile confidence intervals CI_{99}, CI_{95}, and CI_{90}. Stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (1)–(11): (1) no covariates; (2)–(4) demographics; (5)–(7) human-capital/exposure; (8)–(9) risk controls; (10)–(11) financial-behavior controls.

Table 17: T4 (Prog. → Lin.). Contrast: Prog. - Lin. Predicted evasion strata are based on the baseline ML prediction. Panels report R^d, R^f, and R^{tot}. Columns are cumulative covariate-ladder rungs.

	B0 Baseline	B1 Demographics			B2 Human Capital / Exposure			B3 Risk		B4 Financial Behavior	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A: R^d											
Low	1.223	2.420***	2.429***	2.421***	2.373***	2.371***	2.384***	2.367***	2.353***	2.353***	2.352***
SE	(0.795)	(0.867)	(0.869)	(0.870)	(0.869)	(0.870)	(0.871)	(0.868)	(0.865)	(0.865)	(0.865)
p	0.110	0.006	0.006	0.006	0.008	0.008	0.008	0.008	0.008	0.008	0.008
CI_{99}	[-0.768, 3.373]	[0.173, 4.759]	[0.161, 4.767]	[0.153, 4.767]	[0.086, 4.762]	[0.094, 4.763]	[0.094, 4.747]	[0.090, 4.702]	[0.078, 4.690]	[0.078, 4.712]	[0.078, 4.718]
CI_{95}	[-0.294, 2.894]	[0.759, 4.154]	[0.740, 4.152]	[0.731, 4.157]	[0.707, 4.115]	[0.706, 4.113]	[0.701, 4.129]	[0.696, 4.086]	[0.692, 4.078]	[0.692, 4.078]	[0.687, 4.080]
CI_{90}	[-0.032, 2.581]	[1.033, 3.906]	[1.029, 3.906]	[1.031, 3.900]	[0.989, 3.868]	[0.996, 3.859]	[0.995, 3.874]	[0.983, 3.853]	[0.985, 3.820]	[0.984, 3.820]	[0.982, 3.818]
Mid	0.326	1.495	1.503	1.500	1.424	1.440	1.455	1.496	1.491	1.492	1.489
SE	(1.058)	(1.149)	(1.149)	(1.148)	(1.149)	(1.150)	(1.148)	(1.162)	(1.167)	(1.167)	(1.167)
p	0.806	0.194	0.194	0.194	0.201	0.206	0.207	0.205	0.206	0.207	0.210
CI_{99}	[-2.418, 3.252]	[-1.285, 4.775]	[-1.291, 4.784]	[-1.299, 4.649]	[-1.356, 4.693]	[-1.358, 4.649]	[-1.375, 4.718]	[-1.401, 4.716]	[-1.337, 4.799]	[-1.341, 4.815]	[-1.338, 4.803]
CI_{95}	[-1.688, 2.355]	[-0.711, 3.846]	[-0.716, 3.840]	[-0.707, 3.848]	[-0.738, 3.838]	[-0.741, 3.816]	[-0.746, 3.796]	[-0.739, 3.868]	[-0.756, 3.888]	[-0.757, 3.884]	[-0.761, 3.870]
CI_{90}	[-1.443, 2.000]	[-0.310, 3.378]	[-0.327, 3.389]	[-0.315, 3.361]	[-0.357, 3.364]	[-0.367, 3.367]	[-0.363, 3.351]	[-0.384, 3.413]	[-0.373, 3.424]	[-0.374, 3.423]	[-0.373, 3.402]
High	0.301	1.605*	1.617*	1.604*	1.589*	1.590*	1.620*	1.605*	1.601*	1.601*	1.599*
SE	(0.630)	(0.800)	(0.802)	(0.802)	(0.814)	(0.815)	(0.814)	(0.815)	(0.815)	(0.815)	(0.814)
p	0.616	0.053	0.054	0.053	0.056	0.057	0.057	0.055	0.056	0.055	0.055
CI_{99}	[-1.262, 2.086]	[-0.365, 3.636]	[-0.371, 3.641]	[-0.370, 3.640]	[-0.408, 3.653]	[-0.411, 3.656]	[-0.419, 3.656]	[-0.408, 3.655]	[-0.397, 3.645]	[-0.397, 3.646]	[-0.398, 3.645]
CI_{95}	[-0.932, 1.509]	[-0.025, 3.183]	[-0.021, 3.194]	[-0.040, 3.190]	[-0.049, 3.236]	[-0.058, 3.227]	[-0.064, 3.224]	[-0.044, 3.249]	[-0.039, 3.236]	[-0.037, 3.236]	[-0.044, 3.232]
CI_{90}	[-0.739, 1.335]	[0.299, 2.932]	[0.297, 2.939]	[0.291, 2.937]	[0.283, 2.955]	[0.284, 2.961]	[0.284, 2.951]	[0.290, 2.965]	[0.284, 2.953]	[0.285, 2.953]	[0.284, 2.966]
Panel B: R^f											
Low	1.699	1.466	1.390	1.067	1.185	1.167	1.257	1.018	1.023	1.025	1.017
SE	(1.108)	(1.981)	(1.992)	(1.938)	(1.949)	(1.944)	(1.922)	(1.929)	(1.936)	(1.933)	(1.940)
p	0.178	0.565	0.568	0.544	0.475	0.474	0.468	0.502	0.509	0.501	0.481
CI_{99}	[-1.326, 4.099]	[-3.867, 6.286]	[-3.835, 6.220]	[-3.976, 5.997]	[-3.991, 6.334]	[-3.869, 6.419]	[-3.677, 6.248]	[-3.758, 6.473]	[-3.489, 6.376]	[-3.480, 6.379]	[-3.485, 6.441]
CI_{95}	[-0.816, 3.636]	[-2.812, 4.843]	[-2.908, 4.839]	[-2.799, 4.674]	[-2.485, 4.916]	[-2.417, 4.941]	[-2.450, 4.910]	[-2.537, 4.844]	[-2.588, 4.907]	[-2.571, 4.918]	[-2.554, 4.961]
CI_{90}	[-0.451, 3.286]	[-2.151, 4.307]	[-2.262, 4.266]	[-2.156, 4.221]	[-1.947, 4.446]	[-1.953, 4.405]	[-1.909, 4.389]	[-1.906, 4.393]	[-1.938, 4.373]	[-1.942, 4.400]	[-1.895, 4.430]
Mid	-0.769	-0.599	-0.528	-0.185	-0.151	-0.161	-0.088	-0.629	-0.517	-0.518	-0.533
SE	(1.841)	(2.838)	(2.853)	(2.720)	(2.551)	(2.547)	(2.567)	(2.703)	(2.699)	(2.703)	(2.729)
p	0.746	0.835	0.855	0.931	0.958	0.950	0.950	0.980	0.952	0.944	0.936
CI_{99}	[-3.057, 5.171]	[-7.017, 7.374]	[-7.033, 7.146]	[-7.239, 6.730]	[-6.492, 6.893]	[-6.486, 6.892]	[-6.702, 6.810]	[-6.759, 7.206]	[-6.833, 7.067]	[-6.832, 6.990]	[-7.029, 7.044]
CI_{95}	[-2.437, 4.207]	[-5.837, 5.516]	[-5.553, 5.573]	[-5.273, 5.261]	[-4.852, 5.442]	[-4.697, 5.430]	[-4.710, 5.436]	[-5.355, 5.394]	[-5.283, 5.329]	[-5.191, 5.306]	[-5.173, 5.450]
CI_{90}	[-2.162, 3.670]	[-4.846, 4.629]	[-4.727, 4.773]	[-4.532, 4.536]	[-3.963, 4.633]	[-3.910, 4.624]	[-3.870, 4.602]	[-4.360, 4.645]	[-4.309, 4.635]	[-4.328, 4.637]	[-4.321, 4.662]
High	0.689	0.426	1.041	0.308	0.232	0.215	0.318	0.000	-0.032	-0.031	-0.041
SE	(1.328)	(2.031)	(2.037)	(1.996)	(1.998)	(2.003)	(1.984)	(1.990)	(1.994)	(1.992)	(2.005)
p	0.609	0.867	0.871	0.866	0.765	0.768	0.776	0.802	0.817	0.809	0.792
CI_{99}	[-2.654, 4.259]	[-4.592, 5.978]	[-4.671, 6.066]	[-4.630, 5.650]	[-4.349, 5.830]	[-4.167, 5.973]	[-4.282, 5.936]	[-4.452, 5.952]	[-4.313, 5.884]	[-4.415, 5.958]	[-4.398, 5.986]
CI_{95}	[-1.730, 3.459]	[-3.300, 4.472]	[-3.331, 4.469]	[-3.314, 4.322]	[-3.090, 4.481]	[-3.023, 4.609]	[-2.963, 4.532]	[-3.079, 4.612]	[-3.072, 4.533]	[-3.085, 4.569]	[-3.064, 4.687]
CI_{90}	[-1.418, 2.952]	[-2.835, 3.818]	[-2.841, 3.839]	[-2.805, 3.780]	[-2.558, 4.003]	[-2.496, 4.001]	[-2.487, 3.987]	[-2.582, 3.847]	[-2.547, 3.924]	[-2.520, 3.951]	[-2.527, 3.986]
Panel C: R^{tot}											
Low	2.112**	4.616**	4.480**	4.549**	4.779***	4.725***	4.755***	4.833***	4.703***	4.699***	4.693***
SE	(0.860)	(1.874)	(1.867)	(1.861)	(1.863)	(1.861)	(1.861)	(1.858)	(1.860)	(1.864)	(1.867)
p	0.016	0.013	0.013	0.011	0.008	0.008	0.008	0.008	0.009	0.008	0.008
CI_{99}	[-0.124, 4.446]	[-0.110, 9.617]	[-0.153, 9.618]	[0.044, 9.570]	[0.141, 9.639]	[0.142, 9.664]	[0.174, 9.628]	[0.148, 9.650]	[0.097, 9.610]	[0.093, 9.650]	[0.101, 9.660]
CI_{95}	[0.466, 3.791]	[0.994, 8.263]	[1.009, 8.303]	[1.026, 8.305]	[1.138, 8.374]	[1.159, 8.397]	[1.128, 8.418]	[1.139, 8.389]	[1.142, 8.370]	[1.128, 8.383]	[1.151, 8.384]
CI_{90}	[0.676, 3.549]	[1.587, 7.757]	[1.608, 7.745]	[1.578, 7.770]	[1.746, 7.909]	[1.709, 7.929]	[1.720, 7.923]	[1.738, 7.919]	[1.706, 7.943]	[1.709, 7.950]	[1.692, 7.945]
Mid	1.437	3.944	3.947	3.989	4.340	4.365	4.335	4.338	4.072	4.053	4.030
SE	(2.336)	(2.856)	(2.858)	(2.874)	(2.861)	(2.860)	(2.862)	(2.826)	(2.829)	(2.826)	(2.817)
p	0.529	0.167	0.166	0.169	0.161	0.162	0.158	0.154	0.150	0.155	0.156
CI_{99}	[-4.305, 7.565]	[-3.386, 11.155]	[-3.345, 11.091]	[-3.304, 11.296]	[-3.186, 11.257]	[-2.956, 11.345]	[-3.152, 11.396]	[-3.064, 11.354]	[-3.185, 11.127]	[-3.195, 11.119]	[-3.146, 11.089]
CI_{95}	[-3.058, 6.005]	[-1.746, 9.668]	[-1.758, 9.691]	[-1.712, 9.721]	[-1.621, 9.847]	[-1.661, 9.786]	[-1.734, 9.815]	[-1.628, 9.695]	[-1.644, 9.706]	[-1.620, 9.688]	[-1.640, 9.689]
CI_{90}	[-2.448, 5.125]	[-0.650, 8.683]	[-0.696, 8.693]	[-0.710, 8.734]	[-0.516, 8.793]	[-0.528, 8.789]	[-0.605, 8.820]	[-0.496, 8.722]	[-0.529, 8.741]	[-0.571, 8.733]	[-0.536, 8.702]
High	1.372	3.859*	3.747*	3.816*	4.052*	4.010*	4.040*	4.086*	4.055*	4.055*	4.051*
SE	(1.346)	(2.292)	(2.288)	(2.279)	(2.299)	(2.296)	(2.299)	(2.294)	(2.296)	(2.299)	(2.302)
p	0.306	0.070	0.070	0.066	0.059	0.059	0.059	0.059	0.059	0.059	0.058
CI_{99}	[-1.609, 5.054]	[-1.828, 9.873]	[-1.923, 9.756]	[-1.907, 9.835]	[-1.689, 10.082]	[-1.692, 10.074]	[-1.678, 10.076]	[-1.682, 10.156]	[-1.685, 9.933]	[-1.688, 9.933]	[-1.688, 10.002]
CI_{95}	[-1.047, 4.147]	[-0.282, 8.626]	[-0.286, 8.603]	[-0.231, 8.605]	[-0.132, 8.849]	[-0.148, 8.856]	[-0.122, 8.852]	[-0.125, 8.818]	[-0.157, 8.882]	[-0.107, 8.877]	[-0.107, 8.876]
CI_{90}	[-0.693, 3.726]	[0.331, 7.684]	[0.332, 7.666]	[0.373, 7.688]	[0.441, 7.804]	[0.438, 7.831]	[0.445, 7.823]	[0.466, 7.825]	[0.487, 7.875]	[0.468, 7.866]	[0.430, 7.863]

Notes. Entries are OS-APEs on observed outcomes R^d, R^f, and R^{tot} from RE-Tobit margins. Within each stratum block, rows report the effect estimate, bootstrap SE (in parentheses), two-sided bootstrap plus-one p-value, and percentile confidence intervals CI_{99}, CI_{95}, and CI_{90}. Stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (1)–(11): (1) no covariates; (2)–(4) demographics; (5)–(7) human-capital/exposure; (8)–(9) risk controls; (10)–(11) financial-behavior controls.

Table 18: T1 (Prog.) vs. T2 (Lin.). Contrast: Prog. - Lin. Predicted evasion strata are based on the baseline ML prediction. Panels report R^d, R^f, and R^{tot}. Columns are cumulative covariate-ladder rungs.

	B0 Baseline	B1 Demographics			B2 Human Capital / Exposure			B3 Risk		B4 Financial Behavior	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A: R^d											
Low	-1.080	-0.849	-0.848	-0.765	-1.032	-1.013	-0.404	-0.368	-0.451	-0.278	-0.536
SE	(0.936)	(0.939)	(0.946)	(0.949)	(0.977)	(0.983)	(1.038)	(1.005)	(0.996)	(1.014)	(0.993)
p	0.251	0.349	0.358	0.389	0.260	0.280	0.658	0.680	0.631	0.767	0.557
CI_{99}	[-3.568, 1.213]	[-3.312, 1.497]	[-3.337, 1.510]	[-3.297, 1.578]	[-3.695, 1.323]	[-3.691, 1.348]	[-3.043, 2.275]	[-2.926, 2.171]	[-2.983, 2.142]	[-2.821, 2.373]	[-3.179, 1.977]
CI_{95}	[-2.971, 0.667]	[-2.692, 0.933]	[-2.724, 0.977]	[-2.714, 1.002]	[-2.937, 0.827]	[-2.923, 0.845]	[-2.431, 1.574]	[-2.293, 1.521]	[-2.353, 1.431]	[-2.200, 1.728]	[-2.506, 1.383]
CI_{90}	[-2.625, 0.429]	[-2.357, 0.675]	[-2.373, 0.674]	[-2.347, 0.730]	[-2.656, 0.547]	[-2.657, 0.576]	[-2.109, 1.299]	[-2.026, 1.227]	[-2.032, 1.181]	[-1.869, 1.342]	[-2.134, 1.038]
Mid	0.140	0.173	0.312	0.312	0.206	0.172	0.202	0.043	0.229	1.848	1.882
SE	(1.901)	(2.004)	(2.151)	(2.151)	(2.248)	(2.348)	(2.644)	(3.061)	(3.212)	(6.059)	(5.930)
p	0.970	0.969	0.954	0.954	0.949	0.979	0.926	0.982	0.986	0.577	0.584
CI_{99}	[-4.291, 5.645]	[-4.457, 6.648]	[-5.486, 7.399]	[-5.486, 7.399]	[-5.280, 7.909]	[-5.623, 8.098]	[-6.139, 8.757]	[-7.770, 10.542]	[-9.270, 11.151]	[-8.812, 13.399]	[-9.605, 14.867]
CI_{95}	[-3.367, 4.220]	[-3.508, 4.129]	[-3.651, 4.691]	[-3.651, 4.691]	[-3.733, 4.940]	[-3.801, 5.328]	[-4.188, 6.122]	[-5.455, 6.678]	[-5.860, 6.917]	[-4.963, 9.828]	[-5.735, 10.527]
CI_{90}	[-2.863, 3.353]	[-2.962, 3.509]	[-2.994, 3.783]	[-2.994, 3.783]	[-3.065, 4.184]	[-3.238, 4.217]	[-3.382, 4.835]	[-4.383, 5.230]	[-4.843, 5.201]	[-3.826, 7.688]	[-4.239, 8.507]
High	-0.216	0.328	0.235	0.454	0.320	0.425	1.111	1.349	1.353	1.347	1.330
SE	(1.181)	(1.167)	(1.193)	(1.198)	(1.161)	(1.175)	(1.206)	(1.200)	(1.208)	(1.220)	(1.240)
p	0.884	0.774	0.850	0.706	0.784	0.734	0.369	0.275	0.284	0.273	0.286
CI_{99}	[-3.188, 2.732]	[-2.580, 3.296]	[-2.666, 3.303]	[-2.512, 3.511]	[-2.562, 3.181]	[-2.434, 3.315]	[-2.105, 4.062]	[-1.775, 4.357]	[-1.796, 4.346]	[-1.910, 4.456]	[-1.991, 4.501]
CI_{95}	[-2.509, 2.057]	[-1.946, 2.558]	[-2.147, 2.530]	[-1.889, 2.766]	[-1.954, 2.587]	[-1.856, 2.685]	[-1.235, 3.463]	[-0.942, 3.734]	[-0.978, 3.730]	[-0.942, 3.818]	[-1.048, 3.800]
CI_{90}	[-2.186, 1.682]	[-1.585, 2.215]	[-1.714, 2.178]	[-1.501, 2.409]	[-1.574, 2.237]	[-1.472, 2.367]	[-0.898, 3.100]	[-0.625, 3.318]	[-0.648, 3.321]	[-0.676, 3.328]	[-0.699, 3.324]
Panel B: R^f											
Low	2.781**	2.614**	2.638**	2.535**	2.826**	2.708**	1.541	1.512	1.565	1.333	1.469
SE	(1.197)	(1.160)	(1.162)	(1.164)	(1.194)	(1.196)	(1.161)	(1.173)	(1.180)	(1.190)	(1.207)
p	0.019	0.020	0.017	0.022	0.015	0.019	0.171	0.185	0.187	0.271	0.200
CI_{99}	[-0.205, 5.901]	[-0.278, 5.593]	[-0.183, 5.653]	[-0.231, 5.600]	[-0.101, 6.026]	[-0.247, 5.773]	[-1.300, 4.488]	[-1.339, 4.623]	[-1.462, 4.687]	[-1.656, 4.449]	[-1.456, 4.771]
CI_{95}	[0.544, 5.073]	[0.486, 4.960]	[0.451, 4.942]	[0.397, 4.900]	[0.583, 5.229]	[0.470, 5.144]	[-0.662, 3.890]	[-0.688, 3.839]	[-0.681, 3.889]	[-0.997, 3.656]	[-0.850, 3.977]
CI_{90}	[0.856, 4.736]	[0.776, 4.598]	[0.805, 4.609]	[0.731, 4.531]	[0.961, 4.891]	[0.814, 4.780]	[-0.351, 3.517]	[-0.339, 3.501]	[-0.343, 3.499]	[-0.603, 3.294]	[-0.429, 3.513]
Mid	1.236	1.206	1.227	1.227	1.116	1.119	0.645	0.449	0.312	-0.599	-0.696
SE	(2.027)	(2.215)	(2.287)	(2.287)	(2.388)	(2.482)	(2.837)	(3.170)	(3.380)	(4.899)	(6.867)
p	0.550	0.522	0.535	0.535	0.594	0.604	0.796	0.854	0.891	0.860	0.902
CI_{99}	[-4.141, 5.984]	[-4.335, 7.744]	[-4.601, 7.643]	[-4.601, 7.643]	[-5.128, 8.221]	[-5.388, 8.226]	[-7.397, 8.489]	[-8.425, 10.199]	[-8.824, 11.299]	[-10.850, 11.844]	[-12.441, 15.376]
CI_{95}	[-2.835, 5.064]	[-3.051, 5.906]	[-3.162, 6.021]	[-3.162, 6.021]	[-3.533, 5.920]	[-3.745, 6.276]	[-4.867, 6.311]	[-5.543, 7.050]	[-5.966, 7.348]	[-8.095, 7.535]	[-8.453, 8.489]
CI_{90}	[-2.081, 4.488]	[-2.285, 5.044]	[-2.388, 5.094]	[-2.388, 5.094]	[-2.728, 5.114]	[-2.846, 5.381]	[-3.935, 5.176]	[-4.322, 5.752]	[-4.560, 5.999]	[-6.353, 5.929]	[-6.601, 6.655]
High	3.604**	3.075**	3.111**	3.045**	3.193**	2.845**	1.956	1.654	1.740	1.724	1.736
SE	(1.471)	(1.475)	(1.519)	(1.602)	(1.544)	(1.502)	(1.484)	(1.416)	(1.413)	(1.413)	(1.440)
p	0.011	0.029	0.032	0.040	0.026	0.043	0.163	0.225	0.192	0.196	0.192
CI_{99}	[0.007, 7.527]	[-0.604, 7.194]	[-0.596, 7.309]	[-0.853, 7.447]	[-0.368, 7.544]	[-0.735, 6.967]	[-1.561, 6.148]	[-2.006, 5.620]	[-1.936, 5.732]	[-1.712, 6.014]	[-1.739, 5.982]
CI_{95}	[0.892, 6.716]	[0.383, 5.979]	[0.325, 6.198]	[0.147, 6.368]	[0.442, 6.355]	[0.078, 5.932]	[-0.770, 5.045]	[-1.029, 4.560]	[-0.958, 4.708]	[-0.894, 4.690]	[-0.873, 4.787]
CI_{90}	[1.253, 6.224]	[0.739, 5.566]	[0.735, 5.677]	[0.565, 5.720]	[0.805, 5.806]	[0.545, 5.412]	[-0.386, 4.560]	[-0.606, 4.116]	[-0.470, 4.183]	[-0.482, 4.165]	[-0.427, 4.240]
Panel C: R^{tot}											
Low	1.701**	1.765**	1.790**	1.770**	1.794**	1.695**	1.137	1.145	1.114	1.055	0.933
SE	(0.810)	(0.835)	(0.823)	(0.828)	(0.831)	(0.843)	(0.877)	(0.871)	(0.875)	(0.887)	(0.899)
p	0.038	0.026	0.024	0.032	0.024	0.038	0.202	0.196	0.202	0.237	0.285
CI_{99}	[-0.509, 3.716]	[-0.178, 3.987]	[-0.118, 4.073]	[-0.202, 4.098]	[-0.203, 4.011]	[-0.313, 3.991]	[-1.038, 3.372]	[-1.035, 3.380]	[-1.070, 3.279]	[-1.072, 3.218]	[-1.189, 3.192]
CI_{95}	[0.096, 3.252]	[0.212, 3.408]	[0.256, 3.414]	[0.204, 3.418]	[0.241, 3.431]	[0.081, 3.362]	[-0.555, 2.857]	[-0.566, 2.834]	[-0.621, 2.842]	[-0.653, 2.771]	[-0.745, 2.673]
CI_{90}	[0.332, 3.018]	[0.437, 3.195]	[0.474, 3.223]	[0.458, 3.208]	[0.448, 3.193]	[0.311, 3.115]	[-0.323, 2.572]	[-0.269, 2.586]	[-0.314, 2.578]	[-0.364, 2.555]	[-0.493, 2.471]
Mid	1.376	1.379	1.539	1.539	1.322	1.291	0.847	0.492	0.541	1.249	1.187
SE	(1.592)	(1.838)	(1.775)	(1.775)	(1.770)	(1.839)	(1.985)	(2.090)	(2.248)	(4.052)	(4.822)
p	0.410	0.367	0.339	0.339	0.370	0.390	0.606	0.715	0.790	0.579	0.572
CI_{99}	[-2.499, 5.586]	[-3.170, 6.959]	[-2.802, 7.055]	[-2.802, 7.055]	[-2.884, 6.845]	[-2.913, 7.438]	[-3.877, 6.644]	[-4.891, 6.553]	[-4.932, 7.832]	[-6.128, 9.129]	[-9.204, 10.858]
CI_{95}	[-1.596, 4.498]	[-2.020, 5.632]	[-1.743, 5.436]	[-1.743, 5.436]	[-1.779, 5.324]	[-1.941, 5.388]	[-2.705, 5.022]	[-3.131, 5.266]	[-3.306, 5.678]	[-3.608, 7.022]	[-4.084, 7.722]
CI_{90}	[-1.173, 4.030]	[-1.213, 4.521]	[-1.200, 4.577]	[-1.200, 4.577]	[-1.270, 4.615]	[-1.343, 4.661]	[-1.928, 4.451]	[-2.402, 4.401]	[-2.564, 4.636]	[-2.677, 5.879]	[-3.073, 6.716]
High	3.388***	3.404***	3.346***	3.499***	3.513***	3.269***	3.066***	3.003***	3.093***	3.071***	3.065***
SE	(1.115)	(1.077)	(1.095)	(1.149)	(1.152)	(1.086)	(1.115)	(1.111)	(1.082)	(1.091)	(1.112)
p	< 0.001	0.003	0.003	0.003	0.002	0.002	0.005	0.004	0.003	0.004	0.003
CI_{99}	[0.703, 6.370]	[0.637, 6.270]	[0.636, 6.275]	[0.760, 6.776]	[0.693, 6.601]	[0.545, 6.173]	[0.280, 5.957]	[0.238, 5.856]	[0.426, 6.132]	[0.415, 6.121]	[0.385, 6.128]
CI_{95}	[1.299, 5.599]	[1.360, 5.546]	[1.268, 5.559]	[1.337, 5.802]	[1.296, 5.792]	[1.215, 5.457]	[0.918, 5.302]	[0.844, 5.234]	[0.969, 5.281]	[1.006, 5.320]	[0.984, 5.451]
CI_{90}	[1.718, 5.326]	[1.668, 5.137]	[1.628, 5.135]	[1.716, 5.405]	[1.663, 5.413]	[1.529, 5.077]	[1.301, 4.886]	[1.220, 4.855]	[1.345, 4.933]	[1.367, 4.933]	[1.319, 5.000]

Notes. Entries are covariate-adjusted T1–T2 ANCOVA contrasts on observed outcomes R^d, R^f, and R^{tot}. Within each stratum block, rows report the effect estimate, bootstrap SE (in parentheses), two-sided bootstrap plus-one p-value, and percentile confidence intervals CI_{99}, CI_{95}, and CI_{90}. Stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (1)–(11): (1) no covariates; (2)–(4) demographics; (5)–(7) human-capital/exposure; (8)–(9) risk controls; (10)–(11) financial-behavior controls.

A.5 Robustness

A.5.1 Alternative first-stage specifications

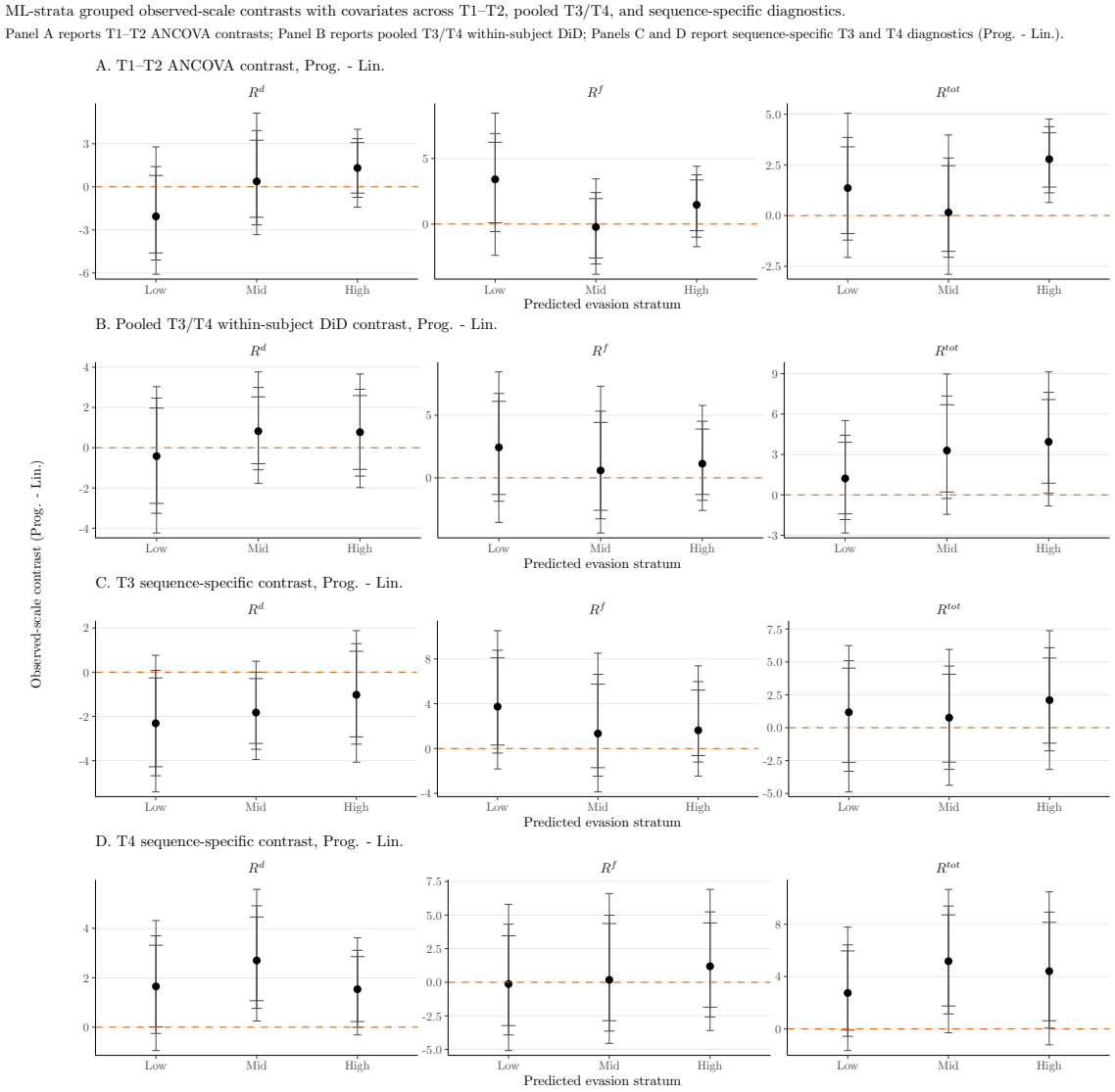

Figure 11: Robustness grouped observed-scale contrasts with covariates using the socio-demographic exogenous set (age and female indicator) under the baseline specification. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panels C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.). Error bars show nested 90%, 95%, and 99% confidence intervals.

Table 19: Robustness grouped observed-scale contrasts with covariates using the socio-demographic exogenous set (age and female indicator) under the baseline specification. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panels C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.).

Panel A: T1 Progressive, T2 Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\beta}$	-2.057	0.376	1.313	3.411*	-0.224	1.467	1.354	0.152	2.780***
SE	(1.662)	(1.636)	(1.047)	(1.900)	(1.372)	(1.193)	(1.308)	(1.284)	(0.814)
p	0.235	0.830	0.214	0.092	0.858	0.217	0.337	0.931	< 0.001
CI_{99}	[-6.090, 2.782]	[-3.334, 5.129]	[-1.418, 4.008]	[-2.385, 8.454]	[-3.831, 3.444]	[-1.727, 4.408]	[-2.069, 5.051]	[-2.897, 3.977]	[0.644, 4.760]
CI_{95}	[-5.097, 1.408]	[-2.649, 3.918]	[-0.731, 3.357]	[-0.576, 6.904]	[-3.042, 2.396]	[-0.997, 3.754]	[-1.210, 3.853]	[-2.059, 2.838]	[1.120, 4.388]
CI_{90}	[-4.605, 0.789]	[-2.120, 3.244]	[-0.438, 3.076]	[0.094, 6.232]	[-2.590, 1.936]	[-0.508, 3.360]	[-0.888, 3.391]	[-1.764, 2.458]	[1.404, 4.083]
Panel B: Pooled T3/T4 within-subject DiD contrast — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	-0.417	0.822	0.771	2.433	0.588	1.130	1.219	3.293*	3.941**
SE	(1.437)	(1.029)	(1.104)	(2.233)	(2.167)	(1.611)	(1.626)	(1.957)	(1.894)
p	0.761	0.424	0.462	0.281	0.848	0.502	0.460	0.079	0.044
CI_{99}	[-4.240, 3.034]	[-1.771, 3.767]	[-1.976, 3.664]	[-3.558, 8.459]	[-4.407, 7.316]	[-2.600, 5.782]	[-2.837, 5.522]	[-1.449, 8.977]	[-0.825, 9.132]
CI_{95}	[-3.258, 2.463]	[-1.099, 2.995]	[-1.411, 2.908]	[-1.871, 6.739]	[-3.271, 5.323]	[-1.796, 4.518]	[-1.835, 4.426]	[-0.270, 7.339]	[0.112, 7.616]
CI_{90}	[-2.767, 1.972]	[-0.795, 2.527]	[-1.075, 2.592]	[-1.326, 6.110]	[-2.590, 4.424]	[-1.320, 3.893]	[-1.412, 3.901]	[0.209, 6.691]	[0.866, 7.079]
Panel C: T3 Linear → Progressive — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	-2.313*	-1.824*	-1.024	3.737*	1.337	1.625	1.170	0.757	2.095
SE	(1.237)	(0.885)	(1.176)	(2.354)	(2.276)	(1.806)	(2.189)	(2.039)	(1.973)
p	0.058	0.052	0.397	0.079	0.400	0.203	0.642	0.756	0.304
CI_{99}	[-5.410, 0.765]	[-3.949, 0.495]	[-4.071, 1.879]	[-1.823, 10.521]	[-3.861, 8.514]	[-2.456, 7.380]	[-4.874, 6.237]	[-4.378, 5.948]	[-3.177, 7.376]
CI_{95}	[-4.685, 0.078]	[-3.488, 0.003]	[-3.256, 1.291]	[-0.409, 8.773]	[-2.468, 6.624]	[-1.209, 5.970]	[-3.317, 5.092]	[-3.168, 4.685]	[-1.766, 6.076]
CI_{90}	[-4.278, -0.260]	[-3.223, -0.292]	[-2.925, 0.946]	[0.306, 8.096]	[-1.710, 5.747]	[-0.633, 5.217]	[-2.635, 4.526]	[-2.621, 4.063]	[-1.178, 5.293]
Panel D: T4 Progressive → Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	1.650*	2.699***	1.535*	-0.127	0.180	1.189	2.746	5.172**	4.409**
SE	(1.007)	(1.037)	(0.791)	(2.063)	(2.199)	(1.956)	(1.817)	(2.152)	(2.262)
p	0.096	0.003	0.052	0.955	0.774	0.529	0.109	0.017	0.047
CI_{99}	[-0.943, 4.312]	[0.252, 5.574]	[-0.311, 3.616]	[-5.079, 5.802]	[-4.544, 6.603]	[-3.588, 6.914]	[-1.652, 7.781]	[-0.297, 10.665]	[-1.217, 10.476]
CI_{95}	[-0.253, 3.698]	[0.762, 4.910]	[-0.002, 3.110]	[-3.916, 4.319]	[-3.628, 4.988]	[-2.581, 5.241]	[-0.565, 6.424]	[1.149, 9.389]	[0.066, 8.927]
CI_{90}	[0.015, 3.316]	[1.072, 4.460]	[0.228, 2.856]	[-3.223, 3.460]	[-2.858, 4.374]	[-1.851, 4.415]	[-0.090, 5.964]	[1.754, 8.716]	[0.625, 8.146]

Notes. Within each panel and outcome, rows report the effect estimate, SE, p-value, and percentile confidence intervals (CI_{99}, CI_{95}, CI_{90}). Significance stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For Panel A, bootstrap plus-one p-values are recomputed from the saved T1–T2 bootstrap array. For Panel B, T3/T4 SE and p-values are read from ladder bootstrap summaries. For Panels C and D, sequence-specific T3/T4 plus-one p-values are recomputed from saved T3 and T4 bootstrap arrays.

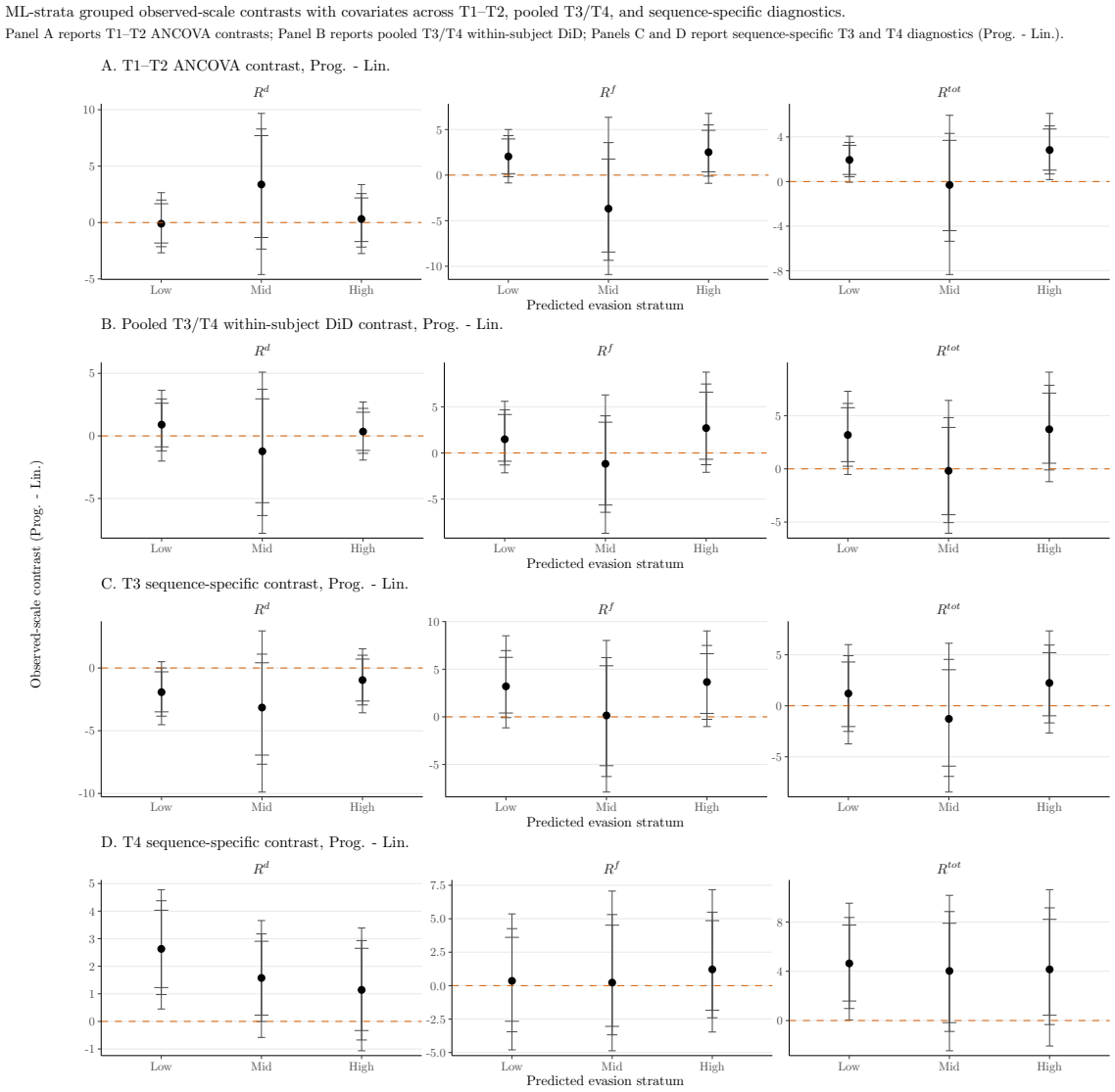

Figure 12: Robustness grouped observed-scale contrasts with covariates using the Risk plus socio-demographic exogenous set under the baseline specification. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panels C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.). Error bars show nested 90%, 95%, and 99% confidence intervals.

Table 20: Robustness grouped observed-scale contrasts with covariates using the Risk plus socio-demographic exogenous set under the baseline specification. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panels C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.).

Panel A: T1 Progressive, T2 Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\beta}$	-0.104	3.374	0.315	2.045*	-3.678	2.514*	1.941**	-0.304	2.829***
SE	(1.064)	(2.746)	(1.190)	(1.142)	(3.195)	(1.424)	(0.806)	(2.445)	(1.120)
p	0.919	0.213	0.845	0.076	0.232	0.060	0.019	0.913	0.009
CI_{99}	[-2.683, 2.647]	[-4.611, 9.673]	[-2.749, 3.368]	[-0.859, 5.001]	[-10.909, 6.343]	[-0.900, 6.768]	[-0.062, 4.065]	[-8.343, 5.945]	[0.174, 6.111]
CI_{95}	[-2.141, 1.995]	[-2.351, 8.293]	[-2.172, 2.568]	[-0.179, 4.328]	[-9.343, 3.564]	[-0.116, 5.514]	[0.430, 3.511]	[-5.355, 4.318]	[0.679, 4.991]
CI_{90}	[-1.817, 1.667]	[-1.326, 7.705]	[-1.685, 2.180]	[0.159, 3.967]	[-8.438, 1.765]	[0.360, 4.914]	[0.634, 3.248]	[-4.408, 3.695]	[1.024, 4.711]
Panel B: Pooled T3/T4 within-subject DiD contrast — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	0.898	-1.231	0.347	1.483	-1.177	2.693	3.184**	-0.185	3.715*
SE	(1.081)	(2.544)	(0.927)	(1.550)	(2.687)	(2.198)	(1.520)	(2.543)	(2.020)
p	0.413	0.648	0.731	0.336	0.620	0.209	0.037	0.961	0.058
CI_{99}	[-2.013, 3.641]	[-7.780, 5.093]	[-1.930, 2.710]	[-2.155, 5.591]	[-8.703, 6.284]	[-2.098, 8.751]	[-0.528, 7.289]	[-6.055, 6.428]	[-1.207, 9.085]
CI_{95}	[-1.193, 2.955]	[-6.363, 3.731]	[-1.392, 2.198]	[-1.304, 4.680]	[-6.430, 4.018]	[-1.276, 7.466]	[0.242, 6.147]	[-5.054, 4.814]	[-0.103, 7.854]
CI_{90}	[-0.872, 2.612]	[-5.336, 2.962]	[-1.157, 1.889]	[-0.895, 4.163]	[-5.639, 3.327]	[-0.686, 6.589]	[0.679, 5.743]	[-4.310, 3.892]	[0.526, 7.117]
Panel C: T3 Linear → Progressive — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	-1.925*	-3.152	-0.961	3.202*	0.145	3.656*	1.206	-1.280	2.241
SE	(0.974)	(2.311)	(1.004)	(1.790)	(3.230)	(1.912)	(1.920)	(2.926)	(1.914)
p	0.052	0.160	0.338	0.056	0.905	0.064	0.545	0.684	0.263
CI_{99}	[-4.519, 0.502]	[-9.887, 2.959]	[-3.571, 1.533]	[-1.151, 8.511]	[-7.890, 8.018]	[-1.019, 9.021]	[-3.733, 5.998]	[-8.454, 6.129]	[-2.663, 7.342]
CI_{95}	[-3.838, 0.005]	[-7.668, 1.105]	[-2.945, 1.019]	[-0.082, 6.956]	[-6.263, 6.219]	[-0.258, 7.493]	[-2.514, 4.911]	[-6.922, 4.550]	[-1.675, 5.955]
CI_{90}	[-3.495, -0.304]	[-6.943, 0.417]	[-2.631, 0.710]	[0.404, 6.244]	[-5.124, 5.359]	[0.361, 6.625]	[-2.035, 4.292]	[-5.914, 3.537]	[-0.979, 5.196]
Panel D: T4 Progressive → Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	2.630***	1.576*	1.142	0.355	0.225	1.212	4.636**	4.026	4.150*
SE	(0.856)	(0.817)	(0.888)	(1.933)	(2.272)	(2.068)	(1.881)	(2.496)	(2.415)
p	0.006	0.054	0.203	0.764	0.838	0.495	0.011	0.122	0.062
CI_{99}	[0.450, 4.779]	[-0.583, 3.657]	[-1.066, 3.387]	[-4.800, 5.361]	[-4.868, 7.066]	[-3.465, 7.174]	[0.050, 9.525]	[-2.455, 10.160]	[-2.070, 10.624]
CI_{95}	[0.972, 4.376]	[-0.004, 3.174]	[-0.676, 2.930]	[-3.449, 4.258]	[-3.677, 5.312]	[-2.402, 5.492]	[0.977, 8.361]	[-0.886, 8.844]	[-0.329, 9.147]
CI_{90}	[1.226, 4.031]	[0.225, 2.906]	[-0.332, 2.652]	[-2.660, 3.609]	[-3.045, 4.518]	[-1.844, 4.859]	[1.583, 7.753]	[-0.171, 7.895]	[0.431, 8.214]

Notes. Within each panel and outcome, rows report the effect estimate, SE, p-value, and percentile confidence intervals (CI_{99}, CI_{95}, CI_{90}). Significance stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For Panel A, bootstrap plus-one p-values are recomputed from the saved T1–T2 bootstrap array. For Panel B, T3/T4 SE and p-values are read from ladder bootstrap summaries. For Panels C and D, sequence-specific T3/T4 plus-one p-values are recomputed from saved T3 and T4 bootstrap arrays.

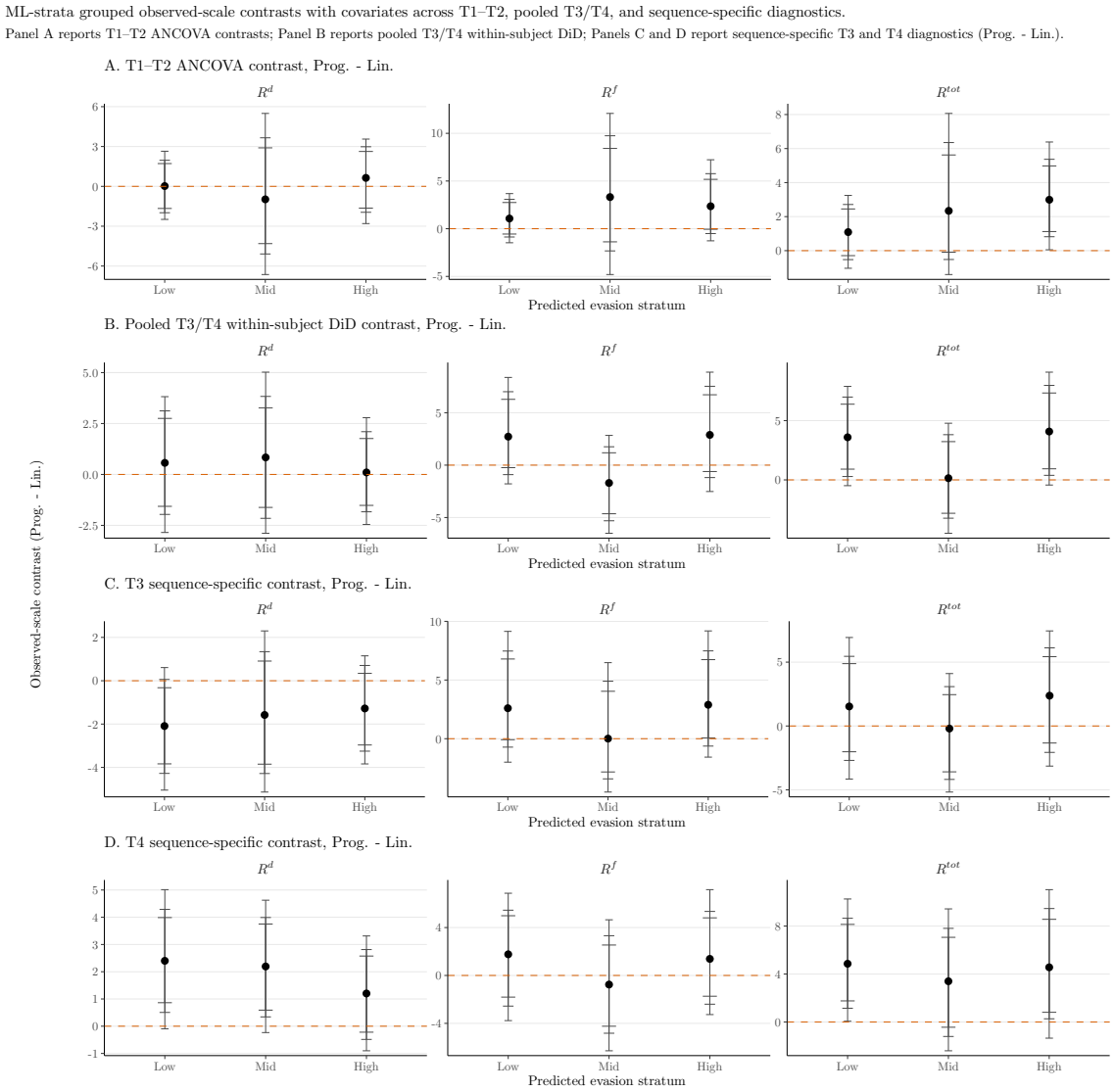

Figure 13: Robustness grouped observed-scale contrasts with covariates using the Risk set under the extended depurated specification. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panels C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.). Error bars show nested 90%, 95%, and 99% confidence intervals.

Table 21: Robustness grouped observed-scale contrasts with covariates using the Risk set under the extended depurated specification. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panel C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.).

Panel A: T1 Progressive, T2 Linear — Contrast: Prog. - Lin.

	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\beta}$	0.021	-0.977	0.637	1.070	3.317	2.354	1.090	2.340	2.992***
SE	(1.023)	(2.286)	(1.270)	(1.003)	(3.105)	(1.611)	(0.840)	(1.742)	(1.185)
p	0.989	0.729	0.654	0.272	0.264	0.111	0.193	0.110	0.010
CI_{99}	[-2.489, 2.645]	[-6.642, 5.492]	[-2.804, 3.561]	[-1.471, 3.681]	[-4.804, 12.086]	[-1.269, 7.229]	[-1.036, 3.247]	[-1.405, 8.066]	[0.052, 6.387]
CI_{95}	[-1.987, 1.964]	[-5.091, 3.657]	[-1.953, 2.984]	[-0.869, 3.077]	[-2.325, 9.745]	[-0.508, 5.771]	[-0.527, 2.713]	[-0.520, 6.353]	[0.818, 5.370]
CI_{90}	[-1.664, 1.716]	[-4.312, 2.901]	[-1.640, 2.628]	[-0.558, 2.748]	[-1.375, 8.416]	[-0.057, 5.176]	[-0.293, 2.450]	[-0.091, 5.623]	[1.122, 4.980]

Panel B: Pooled T3/T4 within-subject DiD contrast — Contrast: Prog. - Lin.

	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	0.569	0.837	0.100	2.721	-1.701	2.888	3.592**	0.144	4.079**
SE	(1.306)	(1.518)	(0.987)	(2.010)	(1.779)	(2.215)	(1.673)	(1.839)	(1.926)
p	0.664	0.599	0.933	0.139	0.332	0.182	0.037	0.944	0.027
CI_{99}	[-2.848, 3.819]	[-2.888, 5.022]	[-2.456, 2.787]	[-1.787, 8.372]	[-6.505, 2.840]	[-2.516, 8.875]	[-0.486, 7.885]	[-4.501, 4.778]	[-0.432, 9.083]
CI_{95}	[-1.961, 3.122]	[-2.154, 3.832]	[-1.833, 2.100]	[-0.905, 7.010]	[-5.311, 1.755]	[-1.187, 7.525]	[0.285, 6.983]	[-3.229, 3.812]	[0.385, 7.968]
CI_{90}	[-1.565, 2.752]	[-1.618, 3.271]	[-1.518, 1.764]	[-0.238, 6.282]	[-4.640, 1.178]	[-0.610, 6.717]	[0.906, 6.383]	[-2.814, 3.228]	[0.935, 7.327]

Panel C: T3 Linear → Progressive — Contrast: Prog. - Lin.

	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	-2.091*	-1.579	-1.279	2.597	0.009	2.889*	1.529	-0.202	2.374
SE	(1.091)	(1.458)	(0.999)	(2.077)	(2.095)	(2.040)	(2.097)	(1.809)	(2.044)
p	0.061	0.294	0.194	0.107	0.756	0.098	0.506	0.749	0.281
CI_{99}	[-5.046, 0.607]	[-5.136, 2.299]	[-3.839, 1.149]	[-1.998, 9.144]	[-4.544, 6.480]	[-1.569, 9.185]	[-4.156, 6.931]	[-5.173, 4.104]	[-3.141, 7.447]
CI_{95}	[-4.276, 0.059]	[-4.281, 1.340]	[-3.250, 0.705]	[-0.717, 7.468]	[-3.421, 4.900]	[-0.628, 7.484]	[-2.696, 5.455]	[-4.185, 3.082]	[-2.069, 6.119]
CI_{90}	[-3.837, -0.323]	[-3.850, 0.907]	[-2.965, 0.340]	[-0.085, 6.791]	[-2.834, 4.041]	[0.070, 6.737]	[-2.019, 4.883]	[-3.591, 2.454]	[-1.333, 5.429]

Panel D: T4 Progressive → Linear — Contrast: Prog. - Lin.

	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	2.400**	2.193**	1.201	1.766	-0.759	1.384	4.859**	3.401	4.562**
SE	(0.951)	(0.949)	(0.840)	(2.059)	(2.045)	(1.994)	(1.935)	(2.273)	(2.384)
p	0.013	0.021	0.165	0.386	0.680	0.478	0.011	0.148	0.042
CI_{99}	[-0.095, 5.013]	[-0.234, 4.626]	[-0.902, 3.314]	[-3.770, 6.866]	[-6.290, 4.648]	[-3.266, 7.166]	[0.069, 10.258]	[-2.395, 9.433]	[-1.343, 11.037]
CI_{95}	[0.506, 4.289]	[0.342, 3.991]	[-0.485, 2.814]	[-2.558, 5.452]	[-4.825, 3.319]	[-2.400, 5.348]	[1.151, 8.652]	[-1.203, 7.810]	[0.270, 9.461]
CI_{90}	[0.863, 3.988]	[0.590, 3.750]	[-0.219, 2.576]	[-1.809, 4.988]	[-4.239, 2.541]	[-1.734, 4.812]	[1.764, 8.141]	[-0.433, 7.062]	[0.813, 8.565]

Notes. Within each panel and outcome, rows report the effect estimate, SE, p-value, and percentile confidence intervals (CI_{99}, CI_{95}, CI_{90}). Significance stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For Panel A, bootstrap plus-one p-values are recomputed from the saved T1–T2 bootstrap array. For Panel B, T3/T4 SE and p-values are read from ladder bootstrap summaries. For Panels C and D, sequence-specific T3/T4 plus-one p-values are recomputed from saved T3 and T4 bootstrap arrays.

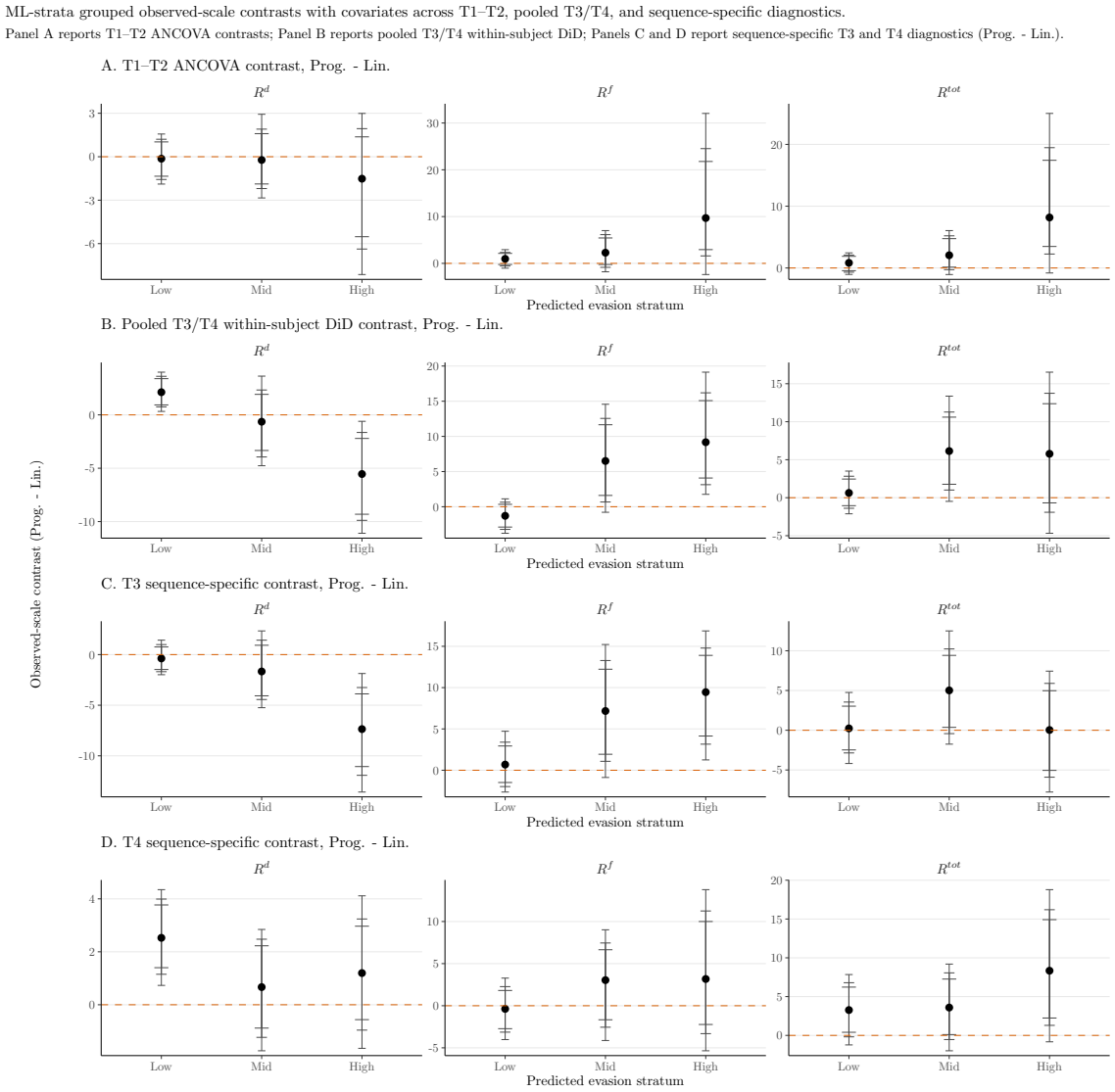

Figure 14: Grouped observed-scale contrasts with covariates under the average-observed-evasion benchmark grouping. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panels C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.). Error bars show nested 90%, 95%, and 99% confidence intervals.

Table 22: Grouped observed-scale contrasts with covariates under the average-observed-evasion benchmark grouping. Panel A reports T1–T2 ANCOVA contrasts; Panel B reports the pooled T3/T4 within-subject contrast; Panels C and D report sequence-specific T3 and T4 contrasts (Prog. − Lin.).

Panel A: T1 Progressive, T2 Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\beta}$	-0.137	-0.225	-1.518	0.966	2.284	9.691**	0.830	2.059*	8.173**
SE	(0.716)	(1.075)	(2.097)	(0.733)	(1.718)	(5.925)	(0.705)	(1.389)	(4.467)
p	0.820	0.718	0.333	0.215	0.135	0.027	0.306	0.077	0.018
CI_{99}	[-1.888, 1.568]	[-2.846, 2.931]	[-8.135, 2.989]	[-1.026, 2.930]	[-1.798, 7.006]	[-2.389, 32.048]	[-1.042, 2.446]	[-1.063, 6.050]	[-0.807, 25.015]
CI_{95}	[-1.570, 1.207]	[-2.198, 1.904]	[-6.374, 1.937]	[-0.559, 2.344]	[-0.845, 6.142]	[1.595, 24.527]	[-0.686, 2.093]	[-0.274, 5.208]	[2.259, 19.473]
CI_{90}	[-1.347, 1.024]	[-1.881, 1.593]	[-5.522, 1.371]	[-0.241, 2.135]	[-0.242, 5.432]	[2.943, 21.781]	[-0.440, 1.887]	[0.136, 4.749]	[3.481, 17.429]
Panel B: Pooled T3/T4 within-subject DiD contrast — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	2.124***	-0.642	-5.543***	-1.293	6.505**	9.172***	0.630	6.139**	5.778
SE	(0.725)	(1.619)	(2.117)	(0.973)	(3.059)	(3.367)	(1.057)	(2.692)	(4.035)
p	0.003	0.712	0.006	0.197	0.022	< 0.001	0.546	0.020	0.139
CI_{99}	[0.323, 4.001]	[-4.765, 3.631]	[-11.097, -0.604]	[-3.777, 1.120]	[-0.789, 14.571]	[1.763, 19.131]	[-2.119, 3.504]	[-0.482, 13.369]	[-4.684, 16.522]
CI_{95}	[0.749, 3.613]	[-3.942, 2.330]	[-9.882, -1.650]	[-3.221, 0.641]	[0.677, 12.557]	[3.135, 16.194]	[-1.392, 2.825]	[1.008, 11.273]	[-1.917, 13.742]
CI_{90}	[0.933, 3.393]	[-3.337, 1.922]	[-9.311, -2.209]	[-2.892, 0.340]	[1.605, 11.658]	[4.081, 15.084]	[-1.051, 2.455]	[1.769, 10.603]	[-0.682, 12.371]
Panel C: T3 Linear → Progressive — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	-0.380	-1.666	-7.359***	0.683	7.171**	9.452***	0.234	5.014*	0.031
SE	(0.675)	(1.500)	(2.188)	(1.366)	(3.072)	(2.938)	(1.654)	(2.758)	(3.025)
p	0.591	0.285	< 0.001	0.628	0.023	0.004	0.859	0.077	0.987
CI_{99}	[-1.985, 1.425]	[-5.245, 2.340]	[-13.576, -1.869]	[-2.639, 4.721]	[-0.877, 15.218]	[1.251, 16.865]	[-4.184, 4.754]	[-1.749, 12.499]	[-7.774, 7.437]
CI_{95}	[-1.688, 0.987]	[-4.437, 1.424]	[-11.925, -3.255]	[-1.979, 3.398]	[1.080, 13.281]	[3.156, 14.788]	[-2.855, 3.547]	[-0.424, 10.232]	[-5.908, 5.895]
CI_{90}	[-1.472, 0.757]	[-4.073, 0.921]	[-11.061, -3.880]	[-1.484, 2.936]	[1.934, 12.225]	[4.140, 13.903]	[-2.474, 3.030]	[0.365, 9.419]	[-5.052, 4.976]
Panel D: T4 Progressive → Linear — Contrast: Prog. - Lin.									
	R^d			R^f			R^{tot}		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
$\widehat{\Delta}$	2.528***	0.667	1.196	-0.377	3.041	3.173	3.267*	3.588*	8.337**
SE	(0.730)	(0.947)	(1.085)	(1.398)	(2.569)	(3.677)	(1.744)	(2.165)	(3.815)
p	< 0.001	0.500	0.280	0.830	0.339	0.306	0.058	0.094	0.022
CI_{99}	[0.730, 4.348]	[-1.739, 2.844]	[-1.647, 4.114]	[-4.021, 3.280]	[-4.132, 9.017]	[-5.350, 13.785]	[-1.226, 7.841]	[-1.982, 9.191]	[-0.825, 18.786]
CI_{95}	[1.151, 3.992]	[-1.233, 2.476]	[-0.959, 3.240]	[-3.137, 2.269]	[-2.554, 7.453]	[-3.315, 11.252]	[-0.187, 6.789]	[-0.535, 8.060]	[1.293, 16.198]
CI_{90}	[1.400, 3.770]	[-0.882, 2.228]	[-0.563, 2.967]	[-2.726, 1.812]	[-1.668, 6.649]	[-2.227, 10.000]	[0.403, 6.224]	[0.105, 7.260]	[2.240, 14.919]

Notes. Within each panel and outcome, rows report the effect estimate, SE, p-value, and percentile confidence intervals (CI_{99}, CI_{95}, CI_{90}). Significance stars use these p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For Panel A, bootstrap p-values are read from the baseline grouped margins CSV. For Panel B, T3/T4 SE and p-values are read from ladder bootstrap summaries. For Panels C and D, sequence-specific T3/T4 SE and p-values are read from grouped bootstrap outputs.

A.5.2 Alternative hyperparameter selection

Table 23: Alternative hyperparameter selection for T1–T2 ANCOVA and pooled T3/T4 within-subject DiD under the Risk set (baseline specification)

	T1–T2 ANCOVA						Pooled T3/T4 within-subject DiD					
Group	$\widehat{\beta}$	CI_{95}	Median	IQR	Range	Matched baseline sign	$\widehat{\Delta}$	CI_{95}	Median	IQR	Range	Matched baseline sign
Panel A. $\mathbf{R}^d$												
Low	-0.40	[-2.43, 1.61]	0.11	[-0.23, 0.16]	[-0.40, 0.32]	40%	0.76	[-1.47, 2.93]	0.43	[0.37, 0.71]	[0.24, 0.92]	100%
Mid	0.20	[-4.06, 6.12]	-0.31	[-1.47, 0.19]	[-3.42, 2.08]	50%	-0.47	[-4.33, 3.53]	0.25	[-0.08, 0.62]	[-1.09, 1.55]	30%
High	1.11	[-1.25, 3.45]	0.92	[0.82, 0.98]	[0.74, 1.44]	100%	0.45	[-1.43, 2.49]	0.47	[0.36, 0.57]	[0.09, 0.64]	100%
Panel B. $\mathbf{R}^f$												
Low	1.54	[-0.66, 3.90]	1.28	[1.08, 1.52]	[0.86, 1.70]	100%	1.95	[-1.19, 5.49]	2.14	[1.73, 2.32]	[0.97, 2.61]	100%
Mid	0.64	[-4.87, 6.26]	1.17	[0.14, 4.02]	[-2.53, 8.90]	70%	-0.81	[-5.70, 4.59]	-0.76	[-1.78, 0.30]	[-2.82, 3.68]	70%
High	1.96	[-0.76, 5.00]	2.06	[1.77, 2.24]	[1.70, 2.59]	100%	2.46	[-1.38, 6.58]	2.43	[2.01, 2.80]	[1.54, 3.21]	100%
Panel C. $\mathbf{R}^{tot}$												
Low	1.14	[-0.59, 2.87]	1.29	[1.17, 1.34]	[0.96, 1.54]	100%	3.31	[0.39, 6.53]	3.26	[2.97, 3.37]	[1.96, 4.00]	100%
Mid	0.85	[-2.68, 5.07]	1.11	[0.14, 1.57]	[-0.74, 5.48]	80%	-0.15	[-4.92, 5.00]	0.10	[-0.32, 1.28]	[-1.25, 2.42]	40%
High	3.07	[0.95, 5.31]	3.05	[2.96, 3.13]	[2.59, 3.39]	100%	3.98	[0.43, 7.65]	4.01	[3.73, 4.31]	[3.19, 4.66]	100%

Notes. The left block reports T1–T2 ANCOVA results and the right block reports pooled T3/T4 within-subject DiD results. In each block, the table reports the baseline estimate, its 95% confidence interval, and the distribution across the 10 alternative hyperparameter configurations; all 10 satisfy the support requirements. "Matched baseline sign" is the share of alternatives with the same sign as the baseline estimate. Median partition similarity across alternatives is ARI = 0.77.

Table 24: Selected configurations for alternative hyperparameter selection under the Risk set (baseline specification)

Rank	Configuration	Macro-F1	α	*nlambda*	λ_{min_ratio}	Standardize
1	1205	0.403	1.000	200	1e-03	No
2	1159	0.403	0.950	400	1e-02	No
3	1213	0.402	1.000	400	3e-04	No
4	1155	0.402	0.950	400	1e-03	No
5	1117	0.402	0.925	200	3e-03	No
6	1217	0.402	1.000	400	3e-03	No
7	1139	0.402	0.925	800	1e-02	No
8	167	0.402	0.125	400	3e-03	No
9	685	0.399	0.550	800	1e-03	No
10	1017	0.398	0.825	800	3e-03	No

Notes. All 10 selected configurations satisfy the support requirements and use penalized multinomial logit. Covariate standardization is disabled in every selected configuration. Rows are ordered by the cached out-of-fold macro-F1 ranking used in the main inference run; multiclass log-loss and classification error break ties. The full pre-specified search spaces are reported in Table 6.

A.5.3 Assignment uncertainty

Table 25: Assignment-uncertainty robustness effects for T1–T2 (between-subject ANCOVA) under the Risk set (baseline specification)

Group	Stat	Deterministic checks					Posterior-draw summary			
		Baseline	Excl. 10%	Excl. 20%	Reassign 10%	Reassign 20%	Median	IQR	Range	Match baseline sign
Panel A. $\mathbf{R}^d$										
Low	$\widehat{\beta}$	-0.40	-0.52	-0.61	-0.18	-0.24	0.69	[-0.17, 1.54]	[-1.61, 2.40]	32%
	CI_{95}	[-2.43, 1.57]	[-2.70, 1.63]	[-2.98, 1.74]	[-2.31, 1.92]	[-2.35, 2.00]				
Mid	$\widehat{\beta}$	0.20	2.02	2.55	0.60	0.90	-0.01	[-0.74, 0.76]	[-2.07, 2.27]	50%
	CI_{95}	[-4.19, 6.12]	[-5.12, 11.59]	[-11.37, 17.82]	[-3.24, 4.99]	[-2.41, 5.37]				
High	$\widehat{\beta}$	1.11	1.16	1.07	0.93	0.75	0.60	[-0.23, 1.32]	[-2.25, 3.13]	64%
	CI_{95}	[-1.23, 3.46]	[-1.21, 3.55]	[-1.51, 3.50]	[-1.36, 3.14]	[-1.65, 3.04]				
Panel B. $\mathbf{R}^f$										
Low	$\widehat{\beta}$	1.54	1.63	1.82	1.61	1.96	1.55	[0.86, 2.01]	[-1.61, 4.90]	86%
	CI_{95}	[-0.66, 3.89]	[-0.75, 4.13]	[-0.72, 4.44]	[-0.74, 4.03]	[-0.43, 4.31]				
Mid	$\widehat{\beta}$	0.64	-3.06	-6.14	-1.17	1.24	1.82	[1.21, 2.97]	[-0.51, 4.63]	94%
	CI_{95}	[-4.87, 6.31]	[-10.26, 4.22]	[-22.56, 13.02]	[-5.69, 2.39]	[-3.57, 4.79]				
High	$\widehat{\beta}$	1.96	1.94	2.07	2.04	1.93	1.61	[0.79, 2.35]	[-2.44, 4.68]	82%
	CI_{95}	[-0.77, 5.04]	[-0.76, 5.09]	[-0.78, 5.36]	[-0.58, 4.93]	[-0.77, 5.04]				
Panel C. $\mathbf{R}^{tot}$										
Low	$\widehat{\beta}$	1.14	1.11	1.20	1.43	1.73^{**}	2.18	[1.57, 2.45]	[-0.29, 3.87]	98%
	CI_{95}	[-0.55, 2.86]	[-0.60, 2.96]	[-0.41, 3.00]	[-0.20, 3.18]	[0.31, 3.32]				
Mid	$\widehat{\beta}$	0.85	-1.04	-3.59	-0.58	2.14	2.09	[1.58, 2.71]	[-0.15, 4.85]	98%
	CI_{95}	[-2.70, 5.02]	[-6.42, 7.69]	[-33.93, 30.84]	[-4.08, 2.58]	[-1.72, 5.45]				
High	$\widehat{\beta}$	3.07^{**}	3.09^{**}	3.14^{**}	2.97^{**}	2.69^{**}	2.20	[1.51, 2.71]	[-0.02, 3.60]	98%
	CI_{95}	[0.92, 5.30]	[0.92, 5.33]	[0.96, 5.45]	[0.85, 5.08]	[0.55, 4.91]				

Notes. Each panel reports the treatment contrast under alternative ways of handling uncertain assignments. Baseline uses the main grouping rule. Excl. 10% and Excl. 20% drop the most uncertain observations. Reassign 10% and Reassign 20% deterministically reallocate those observations. The Stat row reports the effect estimate ($\widehat{\beta}$ for T1–T2 and $\widehat{\Delta}$ for the T3/T4 within-subject DiD contrast). The posterior-draw columns summarize stochastic reassignment based on posterior probabilities: median, interquartile range, full range, and the share matching the baseline sign. Coefficients in deterministic columns carry ** when the reported 95% CI excludes zero.

Table 26: Assignment-uncertainty robustness effects for pooled T3/T4 (within-subject DiD) under the Risk set (baseline specification)

Group	Stat	Deterministic checks					Posterior-draw summary			
		Baseline	Excl. 10%	Excl. 20%	Reassign 10%	Reassign 20%	Median	IQR	Range	Match baseline sign
Panel A. $\mathbf{R}^d$										
Low	$\widehat{\Delta}$	0.77	0.62	0.66	0.66	0.42	0.59	[-0.16, 1.38]	[-1.71, 2.71]	70%
	CI_{95}	[-1.47, 2.93]	[-1.79, 3.09]	[-1.90, 3.26]	[-1.67, 2.79]	[-2.05, 2.76]				
Mid	$\widehat{\Delta}$	-0.47	-1.71	-2.43	-0.82	-0.26	0.47	[-0.26, 1.05]	[-1.50, 2.48]	30%
	CI_{95}	[-4.33, 3.53]	[-6.95, 3.66]	[-7.64, 3.65]	[-4.84, 3.31]	[-3.86, 3.41]				
High	$\widehat{\Delta}$	0.45	0.44	0.35	0.64	0.52	0.11	[-0.61, 0.79]	[-2.18, 2.20]	58%
	CI_{95}	[-1.43, 2.49]	[-1.56, 2.37]	[-1.62, 2.25]	[-1.16, 2.40]	[-1.30, 2.41]				
Panel B. $\mathbf{R}^f$										
Low	$\widehat{\Delta}$	1.95	2.93	3.23	2.61	2.98	1.11	[-0.11, 2.79]	[-1.83, 6.12]	70%
	CI_{95}	[-1.19, 5.49]	[-0.39, 6.67]	[-0.38, 7.34]	[-0.54, 6.35]	[-0.37, 6.67]				
Mid	$\widehat{\Delta}$	-0.81	0.91	3.29	-1.44	-0.03	1.47	[0.74, 3.13]	[-1.18, 4.72]	8%
	CI_{95}	[-5.70, 4.59]	[-7.13, 9.13]	[-4.87, 12.25]	[-6.99, 4.53]	[-5.54, 4.83]				
High	$\widehat{\Delta}$	2.46	2.53	2.27	1.91	1.26	1.53	[0.77, 2.50]	[-0.98, 4.85]	94%
	CI_{95}	[-1.38, 6.57]	[-1.32, 6.73]	[-1.88, 6.66]	[-1.63, 5.85]	[-2.23, 5.48]				
Panel C. $\mathbf{R}^{tot}$										
Low	$\widehat{\Delta}$	3.31^{**}	4.02^{**}	4.04^{**}	4.30^{**}	3.85^{**}	2.99	[1.75, 4.22]	[0.32, 7.27]	100%
	CI_{95}	[0.39, 6.53]	[0.87, 7.44]	[0.96, 7.40]	[1.17, 7.50]	[0.88, 7.00]				
Mid	$\widehat{\Delta}$	-0.15	-1.18	0.22	-1.42	1.53	3.21	[2.41, 3.97]	[0.26, 7.12]	0%
	CI_{95}	[-4.92, 5.00]	[-6.29, 4.69]	[-6.28, 7.86]	[-5.70, 2.99]	[-2.77, 6.15]				
High	$\widehat{\Delta}$	3.98^{**}	4.08^{**}	3.77	3.65^{**}	3.00	2.77	[1.70, 3.89]	[-2.37, 6.07]	92%
	CI_{95}	[0.43, 7.65]	[0.50, 7.99]	[-0.15, 7.79]	[0.05, 7.34]	[-0.57, 6.68]				

Notes. Each panel reports the treatment contrast under alternative ways of handling uncertain assignments. Baseline uses the main grouping rule. Excl. 10% and Excl. 20% drop the most uncertain observations. Reassign 10% and Reassign 20% deterministically reallocate those observations. The Stat row reports the effect estimate ($\widehat{\beta}$ for T1–T2 and $\widehat{\Delta}$ for the T3/T4 within-subject DiD contrast). The posterior-draw columns summarize stochastic reassignment based on posterior probabilities: median, interquartile range, full range, and the share matching the baseline sign. Coefficients in deterministic columns carry ** when the reported 95% CI excludes zero.

A.5.4 T3–T4 temporal-control ladder: grouped evidence

Table 27: Grouped T3/T4 within-subject temporal-control ladder (pooled and DiD contrasts). Columns report specifications S1–S7; rows report predicted evasion groups. S1–S3 and S6 are pooled contrasts (Prog. − Lin.), while S4, S5, and S7 are round-FE identified DiD contrasts.

Group	Statistic	S1	S2	S3	S4	S5	S6	S7
	Model	Block DiD	Block DiD + Linear Trend	Block DiD + Trend Interactions	Baseline (Round FE)	Local Round FE (r8–13)	Individual FE + Linear Time	Baseline + Individual FE
	Estimand	Pooled	Pooled	Pooled	DiD	DiD	Pooled	DiD
Panel A. Outcome R^d								
Low	Estimate	0.411	0.433	0.579	0.765	-0.174	0.466	0.853
	CI_{95}	[-0.796, 1.528]	[-0.684, 1.575]	[-0.884, 2.050]	[-1.470, 2.929]	[-2.844, 2.616]	[-0.820, 1.652]	[-1.589, 3.003]
Mid	Estimate	-0.207	-0.188	-1.804	-0.465	-2.924	-0.389	-0.821
	CI_{95}	[-2.205, 1.797]	[-2.145, 1.769]	[-4.190, 0.695]	[-4.329, 3.532]	[-7.560, 1.736]	[-2.096, 1.447]	[-4.359, 2.834]
High	Estimate	0.196	0.292	0.268	0.454	0.987	0.255	0.474
	CI_{95}	[-0.749, 1.170]	[-0.656, 1.267]	[-1.182, 1.689]	[-1.434, 2.491]	[-1.709, 3.658]	[-0.812, 1.254]	[-1.610, 2.404]
Panel B. Outcome R^f								
Low	Estimate	1.062	0.927	2.931*	1.952	4.389*	1.145	2.387
	CI_{95}	[-0.528, 2.755]	[-0.812, 2.794]	[-0.153, 7.448]	[-1.191, 5.490]	[-0.329, 10.372]	[-0.699, 3.122]	[-1.076, 6.604]
Mid	Estimate	-0.282	-0.267	-0.670	-0.805	2.960	0.248	0.591
	CI_{95}	[-2.538, 2.202]	[-2.764, 2.332]	[-3.894, 1.889]	[-5.704, 4.590]	[-5.453, 13.526]	[-2.375, 2.825]	[-3.469, 6.086]
High	Estimate	1.088	1.148	1.878	2.465	4.177	1.225	2.605
	CI_{95}	[-0.990, 3.245]	[-1.003, 3.258]	[-1.897, 7.687]	[-1.375, 6.575]	[-1.924, 11.262]	[-0.785, 3.491]	[-1.335, 7.655]
Panel C. Outcome R^{tot}								
Low	Estimate	1.688**	1.690**	3.898**	3.310**	5.087*	1.606**	3.278**
	CI_{95}	[0.280, 3.205]	[0.222, 3.170]	[0.799, 7.227]	[0.394, 6.534]	[-0.453, 10.991]	[0.148, 3.148]	[0.301, 6.461]
Mid	Estimate	-0.052	-0.070	-1.576	-0.151	1.142	-0.151	-0.257
	CI_{95}	[-2.465, 2.546]	[-2.538, 2.574]	[-6.331, 2.997]	[-4.916, 5.000]	[-5.725, 8.089]	[-2.553, 2.504]	[-4.919, 5.146]
High	Estimate	1.955**	1.953**	3.604	3.984**	8.029**	1.874**	3.913**
	CI_{95}	[0.233, 3.774]	[0.088, 3.873]	[-0.684, 8.336]	[0.429, 7.648]	[0.869, 15.252]	[0.210, 3.695]	[0.409, 7.797]

Notes. Entries are observed-scale integrated mean contrasts by predicted evasion group. Within each outcome panel, each group is reported on two lines (estimate and CI_{95}). S1–S3 and S6 report pooled contrasts, $0.5 \times [(Prog - Lin)_{T3} + (Prog - Lin)_{T4}]$. In round-FE specifications (S4, S5, and S7), sequence-level levels are absorbed and the identified estimand is the DiD contrast $(Prog - Lin)_{T3} + (Prog - Lin)_{T4}$. Significance markers: *** for CI_{99} excluding zero, ** for CI_{95}, and * for CI_{90}.

Table 28: Grouped T3/T4 sequence-specific heterogeneity temporal-control ladder (S1–S3 and S6).

Panel A. Outcome R^d						
	Low		Mid		High	
Specification	T3	T4	T3	T4	T3	T4
S1 Block DiD	-0.416	1.238*	-0.799	0.386	0.154	0.239
CI_{95}	[-2.190, 1.261]	[-0.241, 2.853]	[-4.004, 2.430]	[-1.633, 2.541]	[-1.428, 1.714]	[-0.856, 1.455]
S2 + Linear Trend	-1.547	2.414***	-0.441	0.065	-1.666	2.250***
CI_{95}	[-3.557, 0.476]	[0.634, 4.186]	[-3.936, 3.074]	[-2.483, 2.694]	[-3.482, 0.350]	[0.727, 3.860]
S3 + Trend Interactions	-1.428	2.586***	-2.267	-1.340	-1.642	2.178***
CI_{95}	[-3.600, 0.693]	[0.551, 4.772]	[-6.308, 2.365]	[-3.580, 0.975]	[-4.067, 0.743]	[0.511, 3.869]
S6 Individual FE + Linear Time	-1.645*	2.576***	-0.202	-0.575	-1.818*	2.329***
CI_{95}	[-3.785, 0.346]	[0.865, 4.503]	[-3.416, 3.154]	[-3.175, 2.255]	[-3.883, 0.183]	[0.741, 3.974]
Panel B. Outcome R^f						
	Low		Mid		High	
Specification	T3	T4	T3	T4	T3	T4
S1 Block DiD	0.651	1.473	-1.102	0.537	1.487	0.688
CI_{95}	[-1.421, 3.129]	[-0.802, 3.468]	[-4.421, 2.366]	[-2.774, 3.872]	[-1.149, 4.301]	[-2.145, 3.900]
S2 + Linear Trend	2.485	-0.632	-0.459	-0.075	1.540	0.756
CI_{95}	[-0.680, 5.966]	[-4.586, 3.374]	[-5.610, 4.576]	[-6.132, 5.820]	[-3.252, 6.224]	[-4.099, 6.271]
S3 + Trend Interactions	4.790*	1.071	-1.193	-0.147	2.283	1.474
CI_{95}	[-0.096, 11.717]	[-3.219, 5.841]	[-5.817, 2.469]	[-4.506, 3.513]	[-3.094, 9.083]	[-4.290, 10.142]
S6 Individual FE + Linear Time	2.794	-0.504	0.719	-0.223	2.217	0.232
CI_{95}	[-0.741, 7.147]	[-4.169, 2.666]	[-4.309, 6.243]	[-6.506, 5.129]	[-2.913, 7.387]	[-4.353, 5.043]
Panel C. Outcome R^{tot}						
	Low		Mid		High	
Specification	T3	T4	T3	T4	T3	T4
S1 Block DiD	1.312	2.064**	-1.570	1.467	2.514**	1.396
CI_{95}	[-0.945, 3.790]	[0.286, 3.837]	[-3.675, 0.965]	[-2.793, 5.915]	[0.034, 5.160]	[-0.935, 4.198]
S2 + Linear Trend	1.128	2.253	-1.441	1.300	-0.788	4.694**
CI_{95}	[-2.664, 4.780]	[-0.938, 5.341]	[-6.309, 3.195]	[-4.809, 7.194]	[-5.832, 4.182]	[0.049, 9.809]
S3 + Trend Interactions	3.335	4.460**	-3.131	-0.020	0.994	6.214*
CI_{95}	[-1.270, 8.451]	[0.416, 8.430]	[-10.646, 3.530]	[-6.154, 5.597]	[-4.873, 6.406]	[-0.313, 14.172]
S6 Individual FE + Linear Time	0.965	2.247	-0.967	0.665	-0.672	4.419*
CI_{95}	[-2.687, 4.663]	[-0.912, 5.547]	[-5.813, 3.581]	[-5.617, 7.096]	[-5.903, 4.119]	[-0.249, 9.887]

Notes. Sequence-specific contrasts are identified and reported for non-round-FE specifications (S1–S3 and S6). Under round-FE specifications (S4, S5, and S7), standalone T3/T4 level contrasts are not identified by construction and are therefore excluded from this table. Significance markers: *** for CI_{99} excluding zero, ** for CI_{95}, * for CI_{90}.

A.5.5 T3–T4 temporal-control ladder: aggregate evidence

This subsection collects the aggregate counterparts to Subsection 6.2.2. Table 29 and Figure 15 report the pooled/DiD temporal-control ladder; Table 30 and Figure 16 report the corresponding sequence-specific aggregate diagnostics.

Table 29: Aggregate T3/T4 within-subject temporal-control ladder. S1–S3 and S6 report pooled contrasts (Prog. − Lin.); S4, S5, and S7 report round-FE identified DiD contrasts. Uncertainty is reported using 95% confidence intervals only; a single superscript star (*) marks estimates whose 95% CI excludes zero.

Specification	Estimand	R^d	R^f	R^{tot}
S1 Block DiD	Pooled	0.208	0.770	1.529*
		[-0.522, 0.878]	[-0.332, 1.954]	[0.470, 2.616]
S2 + Linear Trend	Pooled	0.235	0.772	1.489*
		[-0.451, 0.939]	[-0.461, 2.008]	[0.429, 2.571]
S3 + Trend Interactions	Pooled	0.098	1.798	2.890*
		[-0.862, 1.099]	[-0.429, 4.689]	[0.499, 5.552]
S4 Baseline (Round FE)	DiD	0.369	1.778	3.043*
		[-1.036, 1.750]	[-0.510, 4.109]	[0.932, 5.365]
S5 Local Round FE (r8–13)	DiD	-0.139	3.931*	5.924*
		[-1.918, 1.741]	[0.300, 7.889]	[1.616, 10.227]
S6 Individual FE + Linear Time	Pooled	0.193	1.036	1.423*
		[-0.571, 0.967]	[-0.193, 2.310]	[0.402, 2.512]
S7 Baseline + Individual FE	DiD	0.362	2.076	2.920*
		[-1.119, 1.835]	[-0.388, 4.865]	[0.810, 5.212]

Notes. Entries are observed-scale integrated mean contrasts; each cell stacks the point estimate and CI_{95}. A single marker level is used: * if and only if the 95% CI excludes zero. S1–S3 and S6 report pooled contrasts $0.5 \times [(Prog - Lin)_{T3} + (Prog - Lin)_{T4}]$. In round-FE specifications (S4, S5, and S7), sequence-level levels are absorbed and the identified object is the DiD contrast $(Prog - Lin)_{T3} + (Prog - Lin)_{T4}$.

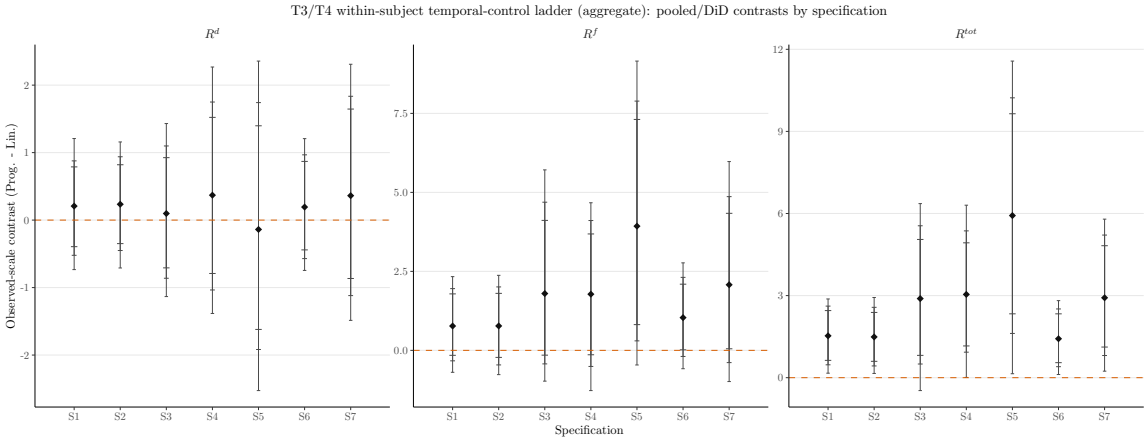

Figure 15: T3/T4 within-subject aggregate temporal-control ladder (main robustness estimand). S1–S3 and S6 report pooled contrasts (Prog. − Lin.); S4, S5, and S7 report the round-FE identified DiD contrast. Error bars show nested 90%, 95%, and 99% confidence intervals.

Table 30: Aggregate T3/T4 sequence-specific heterogeneity temporal-control ladder (S1–S3 and S6).

Panel A. Outcome R^d		
Specification	T3	T4
S1 Block DiD	-0.251	0.666
CI_{95}	[-1.359, 0.863]	[-0.208, 1.553]
S2 + Linear Trend	-1.516	1.986
CI_{95}	[-2.757, -0.258]	[0.909, 3.100]
S3 + Trend Interactions	-1.667	1.862
CI_{95}	[-3.191, -0.062]	[0.663, 3.068]
S6 Individual FE + Linear Time	-1.548	1.934
CI_{95}	[-2.899, -0.138]	[0.830, 3.092]
Panel B. Outcome R^f		
Specification	T3	T4
S1 Block DiD	0.605	0.935
CI_{95}	[-0.977, 2.264]	[-0.616, 2.581]
S2 + Linear Trend	1.544	0.001
CI_{95}	[-0.844, 4.173]	[-2.906, 2.877]
S3 + Trend Interactions	2.694	0.902
CI_{95}	[-0.440, 6.566]	[-2.288, 4.662]
S6 Individual FE + Linear Time	2.190	-0.119
CI_{95}	[-0.521, 5.099]	[-2.575, 2.387]
Panel C. Outcome R^{tot}		
Specification	T3	T4
S1 Block DiD	1.378	1.680
CI_{95}	[-0.102, 3.006]	[0.247, 3.103]
S2 + Linear Trend	-0.118	3.095
CI_{95}	[-2.821, 2.652]	[0.381, 5.930]
S3 + Trend Interactions	1.360	4.421
CI_{95}	[-2.047, 4.694]	[0.857, 8.322]
S6 Individual FE + Linear Time	-0.036	2.881
CI_{95}	[-2.818, 2.748]	[0.295, 5.609]

Notes. Sequence-specific contrasts are reported for non-round-FE specifications (S1–S3 and S6). Under round-FE specifications (S4, S5, and S7), standalone T3/T4 level contrasts are not separately identified by construction.

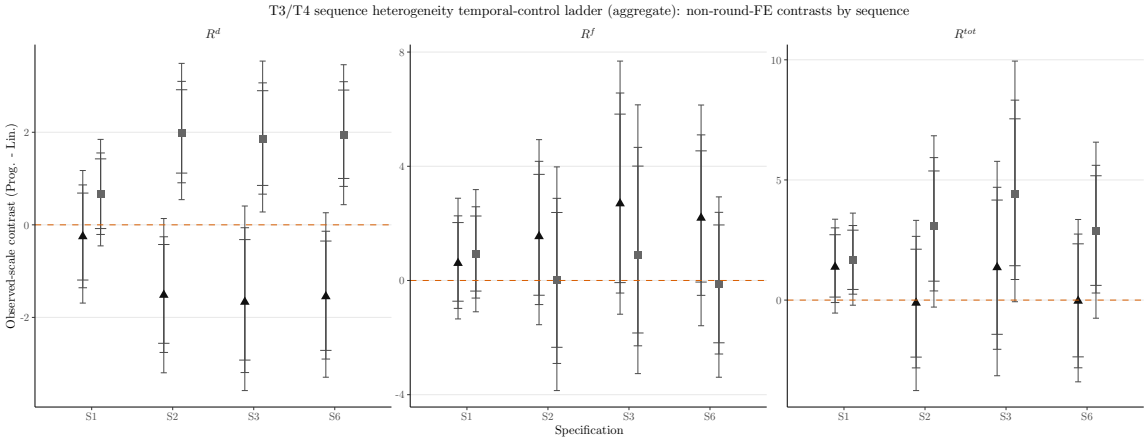

Figure 16: T3/T4 aggregate sequence-specific temporal-control ladder (heterogeneity diagnostic). Reported series are T3 and T4 sequence contrasts for S1–S3 and S6 (Prog. − Lin.). Under S4, S5, and S7 round fixed effects, standalone T3/T4 level contrasts are not identified and are therefore omitted. Error bars show nested 90%, 95%, and 99% confidence intervals.